ISKO ENGINEERING - EXCEL ERP

Excel ERP Rebuild Plan

100,000-Row Resilience Blueprint

Prepared for	Managing Director and department heads (Stores, Production, Dispatch, Finance)
Scope	Four workbooks: 1 Stores, 2 Production, 3 Dispatch & Challan, 4 MIS (2026 files)
Version / date	1.0 - 27 October 2026
Status	Planning blueprint for approval. No production workbook has been rebuilt or overwritten.
Performance status	**Not yet benchmark-tested in Excel.** Targets in Section 17 are provisional.

Current State **Recommendation** **Removed from Phase 1** **Risk** **Decision Required** **Not Yet Tested**

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2. Executive summary

Current State - observed in the attached files. The four 2026 workbooks contain **120 sheets** and **430,095 formula cells** (Stores 91,698; Production 217,775; Dispatch & Challan 119,138; MIS 1,484). Three sheets hold **81%** of all formulas: *2.1 Production Entry* (203,113), *3.3 JWC Lines* (88,661) and *1.6 RM Issue & Return* (57,683). The books hold about six months of history (April-September 2026): 11,406 production lines, 3,023 RM movements, 3,508 JWC lines and 626 DC lines.

Key findings

1. **The present design cannot reach 100,000 rows.** 53,011 formulas sum their own table on every row and 47,369 use expanding "first row to this row" ranges. That work grows with the *square* of the row count: taking *2.1 Production Entry* from 12,538 to 100,000 rows multiplies rows by 8 but this calculation work by about 64. Today the book is usable because it is small.
2. **Record IDs are not permanent.** Issue, production, JWC-line, DC-line, PO-line and consumable IDs are built from `ROW()`; RM lot IDs and batch IDs from running counts. Sorting or inserting a row renumbers them, and every link that quotes them breaks silently.
3. **History can change when a master changes.** Coater rate and basis, customer sale rate, set weight, part weight per piece and dispatch UOM are live lookups on posted rows. Editing a master re-prices or re-weighs old transactions.
4. **About 61% of formulas (≈264,000) depend on functions added in Excel 2019 or later**, mostly Excel 2021 / Microsoft 365 (`XLOOKUP`, `LET`, `FILTER`, `SORTBY`, `UNIQUE`, `SEQUENCE`), and 109 defined names use spill references. The installed Excel version is not recorded in the files and must be confirmed (Decision D-01).
5. **Numbers already disagree between books.** There are three product masters (Stores 249, Production 269, Dispatch 278 rows). MIS was last refreshed on 06-Oct-2026: it shows 2 PO lines where Stores has 3, and reports 442 RM issues with a lot problem where Stores now shows 72.
6. **Controls exist on paper but are weak in practice.** All 217 stock adjustments (net −97,471 kg) have no reason code and no approver; QC status is blank on all 187 lots; validation is bypassed by pasting (form values `SHET`, `SHEE`; 33 DC lines with unit `0`).
7. **The four scope removals touch every book**: purchase requisition (1 sheet, 5 PO columns, 6 names, list columns, MIS import and 3 checks), print (5 print sheets, 15 print areas, 19 printer-settings parts), mail (Gmail links on 10 sheets, 4 contact masters, mail names, the *Mail Coater* and *Mail the team* columns), and physical counts (3 count sheets, 3 MIS imports, count tiles and count-driven adjustments).

Recommendation - target architecture. Keep four Excel books and the same business flow, but change *how* each transaction table works:

- **Entry pad → append-only values register → bounded balance.** Users type into a small entry pad (about 200 rows) that holds all lookups and checks. Posting copies valid rows into the register **as values** with a permanent ID. Registers contain no formulas, so 100,000 rows cost little to recalculate.
- **Balances only for open items** (open lots, open batches, open JWC lines, FG by product), rebuilt on posting with Power Query or bounded `SUMIFS`. Every hard stop reads these balances.
- **Excel 2016 function set** (`INDEX`/`MATCH`, `SUMIFS`, `COUNTIFS`), Excel Tables, data validation, PivotTables and Power Query. No volatile functions, no spill formulas, no sheet protection, no freeze panes, light pastel styling.
- **Controlled transfer:** each book publishes values-only extracts; consumers refresh at defined points; every refresh writes a **Refresh Log** row (file, modified time, rows, distinct IDs, control totals, status).
- **MIS is read-only:** director KPIs from small Power Query summaries, an exception table with owners, and parameter-driven trace queries.
- **FY-wise files** keep each live register inside the tested volume; open balances carry forward.

Decisions required before build (full list in Section 20): (1) Excel version on every PC and whether a signed “Post” macro is allowed (D-01, D-02); (2) how the statutory delivery challan and job-work challan documents will be produced once print is removed (D-03); (3) the stage/route and FG rules that make WIP and FG transaction-calculated, and the approved cut-over opening balances (D-05, D-06, D-08).

Do not build until agreed: the posting mechanism (depends on D-01/D-02), the WIP and FG balances (D-05/D-06), the opening-balance load (D-08) and any document-number sequence for DC/JWC (D-03). Masters, IDs and the Stores pilot can start once D-01 and D-02 are taken.

Not Yet Tested. No Excel benchmark has been run. The 100,000-row figure is a design target to be proven with the test plan in Section 17 on the company’s own PCs.

3. What the current workbooks tell us

All figures in this section were measured from the attached files by reading the workbook XML (every cell, formula, table, defined name, validation and Power Query definition). They describe the files as received.

3.1 Inventory

Workbook	Sheets	Formula cells	File MB	Excel Tables	Defined names	Protected / freeze-pane sheets	Largest formula sheet
1 Stores	28 (27 visible / 1 hidden)	91,698	2.79	18	92 (35 spill)	28 / 18	1.6 RM Issue & Return (57,683)
2 Production	23 (19 visible / 4 hidden)	217,775	7.23	16	48 (19 spill)	20 / 12	2.1 Production Entry (203,113)
3 Dispatch & Challan	23 (20 visible / 3 hidden)	119,138	2.44	15	73 (29 spill)	21 / 12	3.3 JWC Lines (88,661)
4 MIS	46 (14 visible / 32 hidden)	1,484	3.91	32	114 (26 spill)	15 / 2	Lists (582)
Total	**120**	**430,095**	**16.37**	**81**	**327 (109 spill)**	**84 / 44**	

3.2 Where the formulas are

Workbook	Sheet	Formula cells	% of all	Data rows	Table rows (incl. pre-filled)	Formulas per data row	Used range
2 Production	2.1 Production Entry	203,113	47.2%	11,406	12,538	17.8	A1:AK12542
3 Dispatch & Challan	3.3 JWC Lines	88,661	20.6%	3,508	4,030	25.3	A1:AG4034
1 Stores	1.6 RM Issue & Return	57,683	13.4%	3,023	3,393	19.1	A1:AD3397
3 Dispatch & Challan	3.10 DC Lines	21,329	5.0%	626	1,292	34.1	A1:Y1296
1 Stores	1.7 Consumable Issue	10,346	2.4%	128	862	80.8	A1:V866
2 Production	M_Part	6,630	1.5%	122	3,315	54.3	A1:I3319
1 Stores	1.4 RM Receipt GRN	5,918	1.4%	187	348	31.6	A1:AN352
1 Stores	1.2 Purchase Order	5,632	1.3%	3	202	1877.3	A1:AH206
1 Stores	1.5 Consumable Receipt	3,305	0.8%	202	413	16.4	A1:S417
3 Dispatch & Challan	3.9 DC Header	3,159	0.7%	184	451	17.2	A1:Q455
1 Stores	M_Consumable	2,684	0.6%	97	244	27.7	A1:O248
3 Dispatch & Challan	3.2 JWC Header	2,617	0.6%	152	327	17.2	A1:Q331

3.3 Formula patterns that limit scale

Workbook	Formula cells	Post-2016 functions	Dynamic-array functions	Expanding ranges	Self-table aggregates	Position-dependent (ROW())	Volatile	Reads Power Query tables
1 Stores	91,698	60,607	4,111	17,271	15,397	10,455	5,571	0
2 Production	217,775	154,699	40	25,076	37,614	38,077	3	103,080

Workbook	Formula cells	Post-2016 functions	Dynamic-array functions	Expanding ranges	Self-table aggregates	Position-dependent (ROW())	Volatile	Reads Power Query tables
3 Dispatch & Challan	119,138	48,747	163	5,022	0	6,200	3	16,123
4 MIS	1,484	128	126	0	0	0	3	589
Total	**430,095**	**264,181**	**4,440**	**47,369**	**53,011**	**54,732**	**5,580**	**119,792**

Self-table aggregate = a row formula that runs `SUMIFS`/`COUNTIFS` over its own table. *Expanding range* = `INDEX(col,1):[this row]`. *Position-dependent* = formula uses `ROW()` (IDs and running ranges). *Reads imports* = formula reads a Power Query table that refreshes on open and every 5 minutes.

3.4 Growth and calculation load

Table	Data rows now	Period	Rows / month	Rows / year (est.)	Years to 100,000 (est.)	Formula cells now	Formulas per row
2.1 Production Entry	11,406	Apr-Sep 2026	1,901	22,812	4.4	203,113	17.8
1.6 RM Issue & Return	3,023	Apr-Sep 2026	504	6,046	16.5	57,683	19.1
3.3 JWC Lines	3,508	Apr-Sep 2026	585	7,016	14.3	88,661	25.3
3.10 DC Lines	626	Apr-Sep 2026	104	1,252	>50	21,329	34.1
1.4 RM Receipt GRN	187	Apr-Sep 2026	31	374	>50	5,918	31.6
1.7 Consumable Issue	128	Aug-Sep 2026	64	768	>50	10,346	80.8

> **Calculation-load estimate (analytical, not a benchmark).** For a row formula that aggregates its own table, a full recalculation evaluates roughly *rows²* comparisons per aggregate. Moving *2.1* from 12,538 to 100,000 rows multiplies that by (100,000 / 12,538)² ≈ **64**; *1.6* from 3,393 to 100,000 rows by ≈ **869**; *3.3* from 4,030 by ≈ **616**. Whatever the time is today, the same design at 100,000 rows is far slower. This is why the plan removes per-row aggregates instead of simply extending formulas.

3.5 Compatibility, duplication and design issues observed

- **Excel version:** files were last saved by Excel 16.x and contain calculation features for `LET` and dynamic arrays. About 264,000 formula cells use post-2016 functions; 109 names use `ANCHORARRAY` spill references. These books will show `#NAME?` in Excel 2016/2019.
- **Power Query in use:** Production reads Stores (3 queries), Dispatch reads Production (2), MIS reads all three (31 query tables). Production and Dispatch queries refresh on open and every 5 minutes; MIS refreshes on open. The source folder is an absolute path (`D:\Claude code folder\ISKO ERP\...`) held in *Settings*.
- **Duplicated data:** production history is stored in Production and again in MIS (296,551 cells); RM movements in Stores, Production (52,091 cells) and MIS (79,943 cells); JWC lines in Dispatch and MIS (102,388 cells). MIS imports pre-filled blank rows (3,393 rows for 3,023 issues).
- **Duplicated logic:** batch balance is computed in Production (*2.8*) and again inside MIS (*qBatch*); product, people and contact lists exist in several books.
- **Sheet protection and freeze panes:** 84 of 120 sheets are protected and 44 have freeze panes; all four books lock workbook structure. The target removes both, as instructed, and replaces protection with values-only registers and visible checks.
- **No VBA:** all four files are `.xlsx`; no macro project exists. Print and mail features are formula hyperlinks (Gmail compose links built with `ENCODEURL`) and print areas.
- **Migrated history:** most history was loaded from older registers (Source text such as “Part Production sheet”); 3 production rows are balancing lines (“Yield sheet says … not itemised”), and MIS reports 190,137 kg of production not itemised. Shift and QC status are blank on all production history.

3.6 Measured facts versus assumptions

Statement	Basis
Sheet, formula, table, name, validation and query counts	**Observed in the attached files**
Data-row counts and monthly volumes (Apr-Sep 2026)	**Observed** (non-blank date cells in each table)
Growth projections (rows per year, years to 100,000)	**Assumption requiring confirmation** - straight-line from six months
Calculation-load multipliers	**Analytical estimate** - not measured in Excel
Installed Excel version and bitness	**Unknown** - go-live check (D-01)
File size of a 100,000-row values register (14.8 MB)	**Measured** with a synthetic file written by Python/openpyxl, not by Excel
Open/save/recalculation times	**Not Yet Tested**

4. Design principles and scope

4.1 Phase 1 scope

Phase 1 covers **Purchase Order → RM receipt/GRN → RM lot stock → RM issue/return → Cutting → Bending → Fabrication → Rejection/rework → transaction-calculated WIP → Powder coating/job work → Finished goods → Dispatch/challan → Management MIS**, across the four existing books.

Removed from Phase 1 (explicit exclusions)

- **Purchase requisition** - no sheet, table, field, ID, approval queue, KPI or link. A PO is created on its own; purchase need is shown by the MIS RM Requirement report and re-order flags. Removed: *1.1 Requisition*; PO columns *Req No, Req Item, Req Line, Req kg, Req left kg*; names *ReqFrom, TotReqPO, RqItem, RqLen, RqWid, RqLine, ReqFirst, ReqNoList, ReqOpenList, ReqItemByReq, ReqItmId, ReqItmKey*; *Lists* columns AL:AR; MIS *D_Requisition* and query; checks "Requisitions waiting", "PO lines without a requisition", "PO lines ordering more than their requisition"; the requisition step in *4.7 DC Trace* and the START pages.
- **All print functionality** - *1.3 Print PO, 1.6b Print RM Issue Slip, 1.8 Print Issue Slip, 3.4 Print JWC, 3.11 Print DC*; all 15 print areas and 6 print-title names; 19 printer-settings parts; "Copy page" links; print-only settings (company address/email/GST print fields). No replacement templates.
- **All mailing / email-sending functionality** - "Gmail to" and "Open Gmail" links on 1.3, 1.8, 1.12, 1.13, 2.5, 3.4, 3.11, 4.1, 4.2, 4.8; *M_Contacts* in all four books; *MailTo/MailPick/POMailTo/DCMailTo/JWCMailTo* names; mail list columns; *Mail Coater* (3.6) and *Mail the team* (4.10). Supplier, customer and vendor email fields stay only as optional plain reference text.
- **All physical-count and count-reconciliation features** - *1.10 Stock Count, 2.4 WIP Count, 3.8 FG Count*; MIS *D_Count, D_WIPCount, D_FGCount*; "latest physical count" (4.3), "FG counted" (4.1) and "Counted (kg)" (4.8); count-difference adjustment reasons. RM, consumable, WIP and FG balances are calculated from valid transactions only. A necessary correction uses the **authorised stock-adjustment movement** (reason, reference, requester, independent approver, value).
- **Machine log and utilisation** (recommended for Phase 2, Decision D-09).

4.2 Design principles

1. **Enter once, at source.** Each record has one owning book; others import values.
2. **Registers are values.** A posted row never recalculates; corrections are reversal rows.
3. **Permanent IDs** written at posting: prefix + financial year + sequence.
4. **Snapshot what history needs** (rates, weights, UOM, names on documents); look up only what may follow the current master.
5. **Bounded calculation:** formulas in pads and open-item balances only; no whole-column ranges, no volatile functions, no spill formulas.
6. **Controls at entry, exceptions in MIS.** Hard stops in the pad; MIS lists what slipped through and who owns it.
7. **Calm, readable sheets:** light pastel palette, one header row, no merged cells in tables, no freeze panes, no protected sheets.
8. **Version-safe:** Excel 2016 functions unless D-01 approves more.

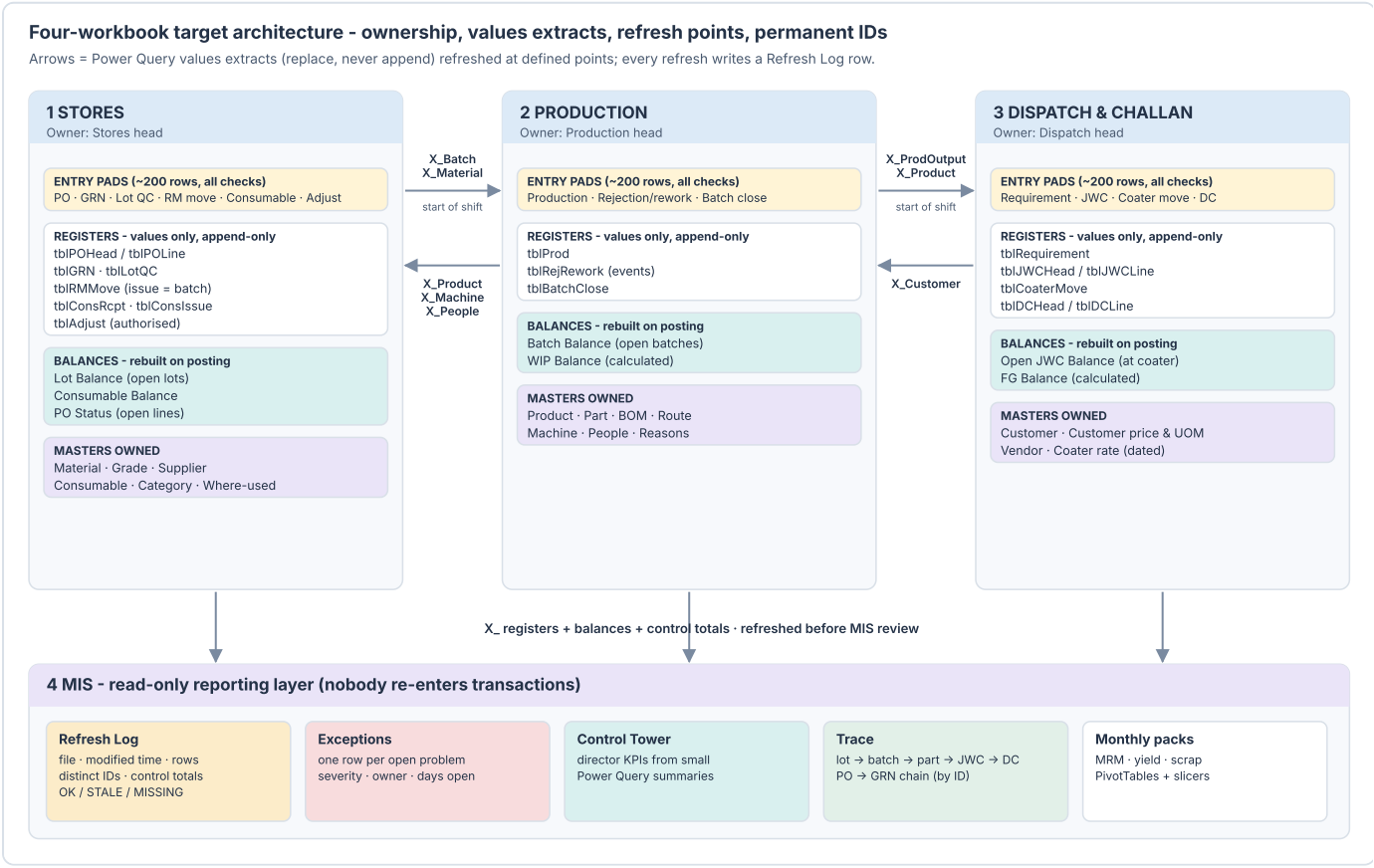
4.3 Assumptions

- Windows desktop Excel, one editor per book at a time, books in one shared folder on a local network or PC.
- Power Query (Get & Transform) is available in the installed Excel - it is already used by three of the books, which is evidence but not proof for every PC.
- Typical office PC: 4-core CPU, 8-16 GB RAM, SSD, 64-bit Excel recommended (go-live check).

4.4 Non-negotiable transaction controls

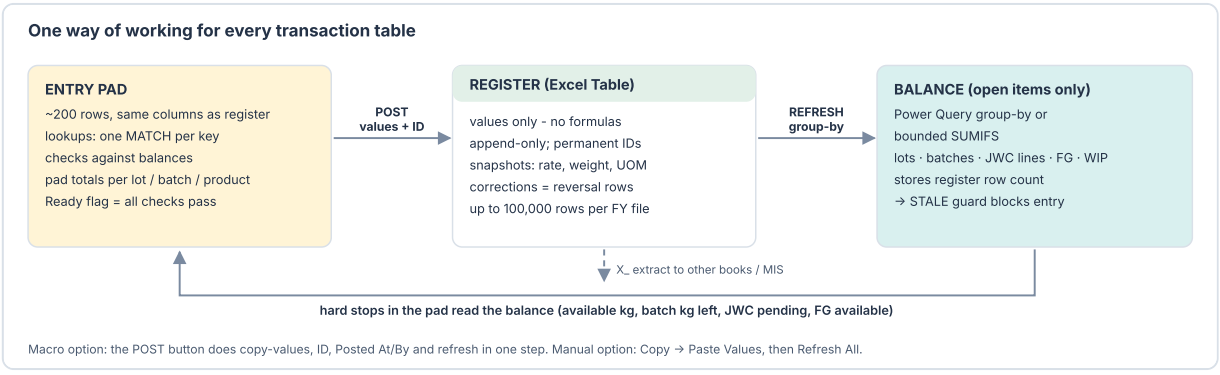
Unique permanent IDs; no negative stock through ordinary issue; QC-accepted lots only; return only against a batch; cutting cannot exceed batch kg; dispatch cannot exceed FG or JWC pending; customer UOM mandatory; no posted-row editing (reversal only); period lock; every adjustment reasoned and independently approved.

5. Four-workbook target architecture



Four-workbook target architecture: ownership, extracts, refresh points and IDs

How a transaction flows. A user fills the entry pad in the owning book. The pad shows lookups, available balances and a *Ready* flag. *Post* copies ready rows into the register as values with a permanent ID, then the book's balances are refreshed. At defined points the downstream book refreshes its import (X_ tables) and MIS refreshes everything, writing the Refresh Log.

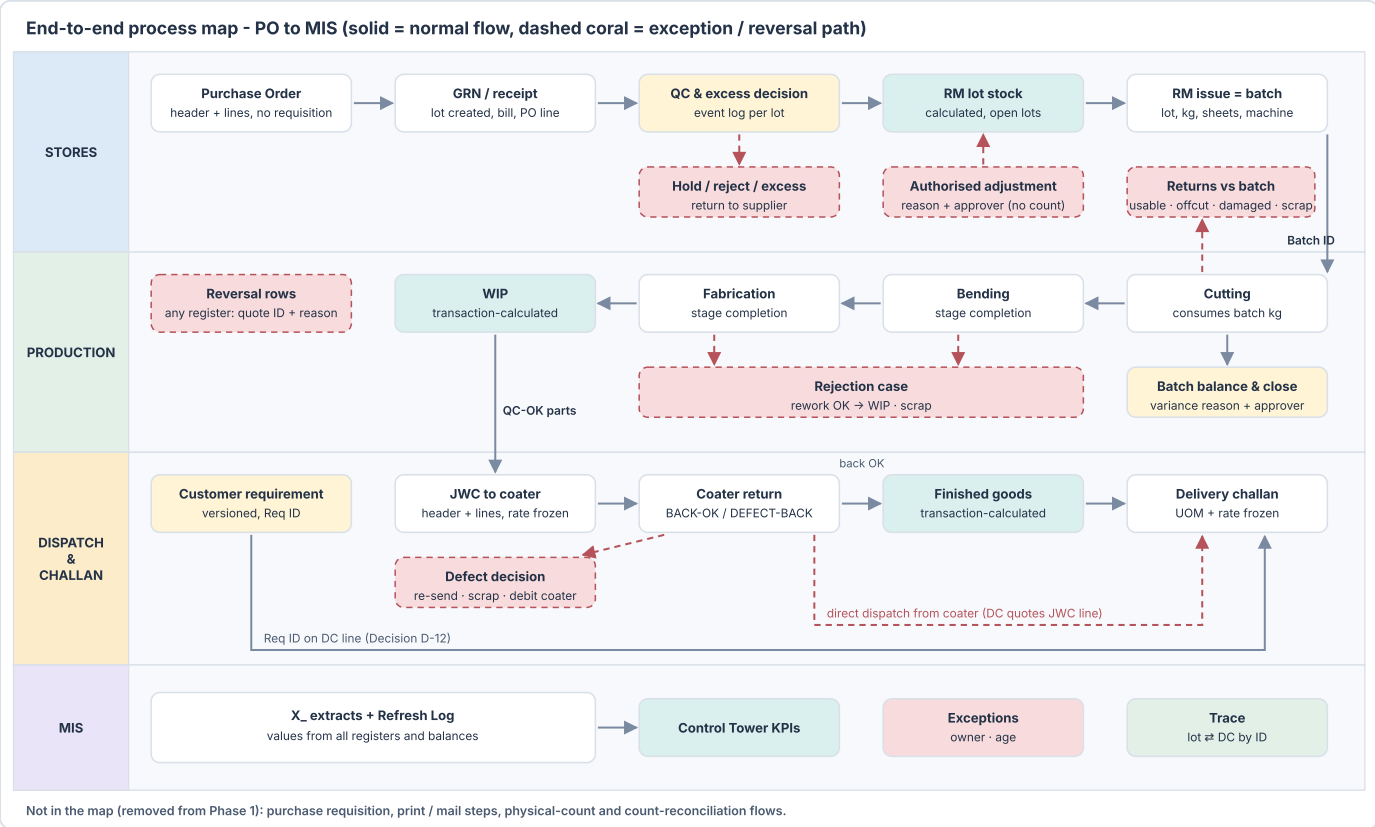


Entry pad, values register and bounded balance

Layer	What it holds	Formulas?	Size
Entry pad	Today's unposted rows, lookups, checks, Ready flag	Yes - all entry logic	~200 rows
Register	Posted transactions, values only, append-only	**None**	up to 100,000 rows per FY file
Balance	Open lots / batches / JWC lines / FG / WIP	Power Query group-by or bounded SUMIFS	open items only

Layer	What it holds	Formulas?	Size
Extract (X_)	Values from another book + lineage columns	None (Power Query output)	as source, needed columns only
Report	KPIs, exceptions, trace	PivotTables / small summaries	hundreds of rows

6. End-to-end process map



End-to-end process map with exception and reversal paths

Exception and reversal pathways

- **GRN:** excess over PO → *Excess decision* (accept / return) in the QC log before the lot can be issued; QC HOLD/REJECTED → lot not issue-eligible; supplier return → *RETURN-TO-SUPPLIER* movement.
- **RM movement:** wrong issue → reversal row quoting the Move ID, then a correct row; usable material back → *RETURN-USABLE* / *RETURN-OFFCUT* against the batch; damaged → *RETURN-DAMAGED* (separate bucket, not issue-eligible); stores scrap → *STORES-SCRAP* against the batch.
- **Production:** over-consumption is blocked; wrong line → reversal; rejection → REJECT case → REWORK-OK (back to WIP) or REWORK-SCRAP; batch closes only through a Batch Close row with reason/approver above tolerance.
- **Coater:** defects back → DEFECT-BACK with decision (re-send, scrap, debit coater); direct dispatch from coater → DC line quoting the JWC line.
- **Dispatch:** cancelled challan → DC status line CANCELLED plus reversal lines; FG cannot go negative.
- **Corrections to stock** outside these paths → authorised stock-adjustment movement only.

Purchase requisition, print/mail steps and physical-count/reconciliation flows are not part of the map.

7. Workbook responsibility matrix

Workbook	Purpose	Owns (system of record)	Imports (values)	Must not do
1 Stores	Purchase, receipt, QC, RM and consumable stock	PO header/line, GRN/lot, Lot QC, RM movements, consumable receipts/issues, adjustments; masters: Material, Grade, Supplier, Consumable, Category, Where-used	X_Product, X_Machine, X_People (from Production)	Hold production, dispatch or KPI history
2 Production	Make parts, consume batches, control yield	Production lines, rejection/rework events, batch close; Batch Balance, WIP Balance; masters: Product, Part, BOM, Stage/Route, Machine, People, Reasons	X_Batch, X_Material (from Stores); X_Customer (from Dispatch)	Re-key stores issues; compute dispatch
3 Dispatch & Challan	Requirements, job work, coaters, FG, delivery	Requirements, JWC header/lines, coater movements, DC header/lines; Open JWC Balance, FG Balance; masters: Customer, Customer price & UOM, Vendor, Coater rate	X_ProdOutput, X_Product, X_People (from Production)	Hold production history
4 MIS	Director reporting and cross-book checks	KPI summaries, exception table, Refresh Log, trace queries	X_ extracts from all three books	Accept transaction entry

7.1 Source-of-truth matrix (one row per data entity)

Entity	System of record	Entered / imported	Primary key	Key foreign keys	Maintained by	Consumed by	Type
PO header	Stores	Entered once (pad)	PO No	Supplier ID	Purchase / Stores	Stores (GRN), MIS	Operational
PO line	Stores	Entered once (pad)	PO Line ID	PO No, Item ID	Purchase / Stores	Stores (GRN), MIS (RM Requirement, PO status)	Operational
GRN / RM lot	Stores	Entered once (pad)	Lot ID	PO Line ID, Supplier ID, Material ID	Stores	Stores (issue), Production (via batch), MIS (trace, stock)	Operational
Lot QC / excess decision	Stores	Entered once (event)	QC Event ID	Lot ID	QC / Purchase	Stores (eligibility), MIS	Operational
RM issue / return movement	Stores	Entered once (pad)	Move ID (issue Move ID = Batch ID)	Lot ID, Batch ID, Machine ID, Person ID	Stores	Production (X_Batch), MIS	Operational
Consumable movement	Stores	Entered once (pad)	Receipt ID / Cons Move ID	Item ID, Supplier ID, Person ID	Stores	MIS	Operational
Authorised stock adjustment	Stores	Entered once (pad, approved)	Adj ID	Lot ID / Item ID	Stores + approver	Stores balances, MIS	Operational
Production record	Production	Entered once (pad)	Prod ID	Batch ID, Lot ID, Product ID, Part ID, Machine ID	Production supervisors	Dispatch (X_ProdOutput), MIS	Operational

Entity	System of record	Entered / imported	Primary key	Key foreign keys	Maintained by	Consumed by	Type
Rejection / rework event	Production	Entered once (pad)	Event ID (Case ID)	Source Prod ID, Batch ID, Lot ID	QC / Production	Production (QC OK), MIS	Operational
Batch close	Production	Entered once (event)	Batch ID	Batch ID	Production + Stores	MIS	Operational
Batch balance	Production	Calculated	Batch ID	Lot ID, Machine ID	System	MIS (X_BatchBal)	Reporting-only (calculated)
WIP balance	Production (in-book stages) / MIS (with JWC & FG)	Calculated	Product ID|Part ID|Stage	Route	System	MIS, Dispatch	Reporting-only (calculated)
Customer requirement	Dispatch	Entered once (pad)	Req ID + Version	Customer ID, Product ID	Dispatch / Sales	MIS (RM Requirement), Dispatch	Operational
JWC header	Dispatch	Entered once (document pad)	JWC No	Vendor ID	Dispatch	MIS	Operational
JWC line	Dispatch	Entered once (document pad)	JWC Line ID	JWC No, Product ID, Part ID, Batch ID	Dispatch	MIS, Dispatch balances	Operational
Coater movement (opening, return, defect)	Dispatch	Entered once (pad)	Coater Move ID	JWC Line ID, Coater ID	Dispatch	Dispatch balances, MIS	Operational
FG balance	Dispatch	Calculated	Product ID (|Part ID)	-	System	Dispatch (DC hard stop), MIS	Reporting-only (calculated)
DC header	Dispatch	Entered once (document pad)	Challan No	Customer ID, Dispatch From	Dispatch	MIS	Operational
DC line	Dispatch	Entered once (document pad)	DC Line ID	Challan No, Product ID, JWC Line ID, Req ID, Batch ID	Dispatch	MIS, Dispatch balances	Operational
BOM (edition + lines)	Production	Master (direct)	BOM Model + Edition + SR No	Product ID, Part ID, Material ID	Production engineering	MIS (RM Requirement)	Master
Customer	Dispatch	Master (direct)	Customer ID	-	Dispatch / Accounts	Production (product), MIS	Master
Customer price & UOM	Dispatch	Master (effective-dated)	Customer ID|Product ID|Effective From	Customer ID, Product ID	Dispatch / Accounts	Dispatch (DC snapshot)	Master
Supplier (RM / consumables)	Stores	Master (direct)	Supplier ID	-	Purchase	MIS	Master
Vendor / coater + rate	Dispatch	Master (effective-dated rate)	Vendor ID; Vendor ID|Effective From	-	Dispatch / Accounts	Dispatch (JWC snapshot), MIS	Master
Product / part	Production	Master (direct)	Product ID; Part ID	Customer ID, Material ID	Production engineering	Stores, Dispatch, MIS (imports)	Master

Entity	System of record	Entered / imported	Primary key	Key foreign keys	Maintained by	Consumed by	Type
Material / grade	Stores	Master (direct)	Material ID; Grade code	-	Stores	Production, MIS (imports)	Master
Machine	Production	Master (direct)	Machine ID	-	Production	Stores, MIS (imports)	Master
People	Production	Master (direct)	Person ID	-	HR / Production	Stores, Dispatch (imports)	Master
KPI / summary	MIS	Calculated on refresh	KPI code + period	-	MIS owner	Directors	Reporting -only
Refresh log / exceptions	Each book (log) / MIS (exceptions)	Written by refresh	Run ID; Exception ID	Source Row ID	MIS owner	All owners	Control

8. Sheet-by-sheet rebuild decision register

Every one of the 120 current sheets is listed with its measured size and the recommended action. *Rows* = non-blank records in the sheet's table (where a table exists); *Formulas* = formula cells; *Used range* = the sheet's dimension. Decisions: **Keep / Simplify / Combine / Removed from Phase 1 / Convert to report**. The full register with inputs/outputs is also in the decision-register workbook.

Decision summary across all 120 sheets - Keep: 16, Simplify: 61, Combine: 19, Convert to report: 4, Removed from Phase 1: 20.

1 Stores (28 sheets)

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
START	Navigation page with hyperlinks to every sheet and a process description.	A1:E55; 26 f	**Simplify**	Keep a plain README; remove links to removed sheets (requisition, print, count) and the print area.	README	n/a	Out: navigation only	None	Stale instructions if not updated with each release; version the README.
1.1 Requisition	Purchase requisition register (requested by, item, kg, stores decision, PO link). 0 data rows; 2,401 formula cells pre-filled.	A1:AB154; 0 rows; 2,401 f	**Removed from Phase 1**	Scope override 1. PO is raised directly from need shown in MIS RM Requirement and the re-order flags. Control impact: purchase authorisation moves to the PO approval field; no PO depends on a requisition reference.	None	0	None	Referenced by 1.2 (Req No/Item/Line, Req kg), Lists AL:AR, names ReqFrom/RqItem/RqLen/RqWid/RqLine/TolReqPO, MIS D_Requisition, 4.10 Checks rows 36 and 38, 4.7 DC Trace chain.	All dependants must be deleted in the same release or #REF! errors will appear.
1.2 Purchase Order	PO lines with header fields repeated on the first line, requisition link, received/pending progress.	A1:AH206; 3 rows; 5,632 f	**Simplify**	Split into PO header and PO line registers; remove requisition columns and the 'Add to PO' mechanic; received/pending become a bounded PO Status view.	S-PO Entry (pad) + tblPOHead + tblPOLine + PO Status (open lines)	100,000 lines design limit (actual ~1-2k/yr expected)	In: supplier, item, qty, rate, dates. Out: open PO lines to GRN; PO status to MIS.	M_Supplier, M_Material, M_Consumable; GRN register for receipts.	TODAY() in 202 rows and 1,597 running-range formulas today; target: no volatile functions, PO Line ID permanent.
1.3 Print PO	PO print layout with Gmail compose link and 'Copy page' link.	A1:K500; 31 f	**Removed from Phase 1**	Scope overrides 2 and 3 (print and mail). Control impact: none on stock; the PO register remains the record. Management decides how a PO document reaches the supplier outside the ERP.	None	0	None	Lists P:AA mail lists, MailTo4, POMailTo, Print_Area name, printerSettings part.	Remove print area, printer settings, Gmail hyperlinks and mail names together.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
1.4 RM Receipt GRN	One row per received lot: size, sheets, kg, theoretical kg, rate, PO link, QC, excess, accepted kg/sheets.	A1:AN352; 187 rows; 5,918 f	**Simplify**	Keep receipt facts; move QC and excess decisions to an append-only QC decision log so decisions after receipt do not overwrite history; Lot ID becomes permanent.	S-GRN Entry (pad) + tblGRN + tblLotQC + Lot Balance	100,000 lots design limit	In: receipt, bill, QC. Out: issue-eligible lots, PO receipt progress, lot rate.	tblPOLine, M_Material, M_Supplier, Settings tolerances.	RM ID built with a running COUNTIF (changes if rows are sorted/inserted); 696 running-range formulas; QC Status blank on all 187 lots.
1.5 Consumable Receipt	Consumable receipts (item, qty, bill rate, supplier, opening lines).	A1:S417; 202 rows; 3,305 f	**Simplify**	Entry pad + values register; opening balances posted once as OPENING movements.	S-Cons Receipt (pad) + tblConsRcpt	100,000	In: item, qty, rate, bill. Out: consumable balance and average rate.	M_Consumable, M_Supplier	413 self-table aggregate formulas today; target none in the register.
1.6 RM Issue & Return	RM issue, return, offcut return, scrap and sample movements with FIFO lot auto-pick, availability and checks.	A1:AD3397; 3,023 rows; 57,683 f	**Simplify**	Largest Stores hotspot. Rebuild as entry pad + values-only movement register + bounded lot balance. Hard stops retained at entry; historic rows never recalculate.	S-RM Move Entry (pad) + tblRMMove + Lot Balance + Batch list	100,000 movements (actual ~6k/yr)	In: material, lot, kg, sheets, machine, person, product. Out: batches to Production; lot balances; issue value.	tblGRN, tblLotQC, tblAdjust, M_Machine (imported), M_Person.	57,683 formula cells; 13,572 running-range and 13,572 self-table SUMIFS formulas (grow with the square of row count); TODAY() in 3,393 rows; FIFO lot pick recalculated for history.
1.6b Print RM Issue Slip	Printable RM issue slip (batch slip) with NOW() and copy-page link.	A1:H30; 31 f	**Removed from Phase 1**	Scope override 2. Control impact: the batch number is no longer carried on paper; Production picks the batch from the imported open-batch list by machine.	None	0	None	SlipIssue name, RMSlipList, Print_Area.	Shop-floor must confirm batch identification without a slip (Decision D-07).
1.7 Consumable Issue	Consumable issues/returns with BOM qty, variance, stock check, value.	A1:V866; 128 rows; 10,346 f	**Simplify**	Entry pad + values register; BOM vs actual moves to MIS; stock check becomes a hard stop at entry against the consumable balance.	S-Cons Issue (pad) + tblConsIssue	100,000	In: item, qty, returned qty, person, where used. Out: consumable balance, cost by where-used.	M_Consumable, M_WhereUsed, M_Person	1,724 TODAY() cells and 862 running-range formulas today; 734 pre-filled blank rows.
1.8 Print Issue Slip	Consumable issue slip print layout with Gmail link.	A1:H413; 79 f	**Removed from Phase 1**	Scope overrides 2 and 3. Control impact: none; the issue register is the record.	None	0	None	IssuePick, MailTo1, Print_Area, printerSettings.	Remove with its names and list columns.
1.9 Stock Adjustment	RM and consumable adjustments; current lines are count-driven corrections (217 lines, net -97,471 kg, no reason code or entered-by on any line).	A1:P251; 217 rows; 1,483 f	**Simplify**	Redesign as an Authorised Stock-Adjustment movement (reason code, reference, requester, approver, value). Count-difference reasons are removed; this is not a count module.	S-Adjust Entry (pad) + tblAdjust	100,000 (expected low)	In: lot/item, +/- qty, reason, approver. Out: lot and consumable balances.	tblGRN, M_Consumable, M_Reason	Today none of the 217 adjustments has a reason code or an approver; target makes both hard stops.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
1.10 Stock Count	Physical stock count entry (273 lines) compared with book stock on the dashboard.	A1:I450; 273 rows; 447 f	**Removed from Phase 1**	Scope override 5. Control impact: stock is controlled through movement validation, QC eligibility and authorised adjustments only.	None	0	None	MIS D_Count/qCount, 4.3 Stock Summary, 4.8 MRM 'Counted (kg)', adjustment text.	Historic count rows are archived read-only with the 2026-27 files, not migrated.
1.11 RM Stock	Lot-by-lot balance, value, ageing and grade summary (spill formulas).	A1:U407; 395 f	**Simplify**	Keep as the operational Lot Balance (needed for availability). Calculated for open lots only, refreshed on posting.	Lot Balance (calculated)	Open lots only (~300-1,000)	In: GRN, QC, movements, adjustments. Out: available kg/sheets by lot; ageing.	tblGRN, tblLotQC, tblRMMove, tblAdjust	ANCHORARRAY/FILTER/SORT spill formulas need Excel 2021/365; replace with Power Query output or bounded SUMIFS.
1.12 Consumable Stock	Consumable stock, value, reorder status; Gmail/Copy page.	A1:K406; 19 f	**Simplify**	Keep as Consumable Balance with reorder and negative flags; remove mail and print area.	Consumable Balance (calculated)	Active items (~250-500)	Out: stock, value, reorder flag.	tblConsRcpt, tblConsIssue, tblAdjust, M_Consumable	Negative stock possible today (2 items); target hard stop at issue.
1.13 Re-order Alert	List of RM and consumables below minimum, with Gmail 'send alert'.	A1:H500; 7 f	**Combine**	Reorder flags live on the two balance tables and on the MIS exception list; separate alert sheet and mail removed.	Flags on Lot/Consumable Balance; MIS exception	n/a	Out: reorder exceptions	Balances	Mail-based alert removed (scope override 3).
1.14 Stores Dashboard	Monthly Stores KPIs (stock, received, issued, turnover, ageing).	A1:L433; 180 f	**Convert to report**	Move KPIs to the MIS Control Tower so there is one stock number; Stores keeps operational balances only.	MIS Control Tower (Stores tiles)	n/a	Out: KPIs	MIS extracts	Duplicate KPI definitions removed.
M_Supplier	Supplier master (Tally ledger mapping, GSTIN, terms, email).	A1:L88; 49 rows; 0 f	**Keep**	Add a stable Supplier ID and Active flag; email kept only as an optional plain reference field (no hyperlink, no mail use).	M_Supplier	<1,000	Out: supplier dropdowns, PO/GRN snapshots	None	Name edits must not change history: PO/GRN store Supplier ID and a name snapshot.
M_Material	Material master with grade, thickness, standard size/weight, average issue, min level, live stock.	A1:M128; 69 rows; 992 f	**Simplify**	Keep identity and standards; remove live stock/avg/reorder formula columns (moved to Lot Balance and MIS).	M_Material	<2,000	Out: material IDs, standard sheet weight, min level	M_Grade	992 formula cells today on 69 materials.
M_Grade	Grade codes used as the prefix of RM IDs.	A1:C34; 10 rows; 0 f	**Keep**	Small stable list.	M_Grade	<100	Out: grade code	None	Code changes must never re-key existing lots.
M_Consumable	Consumable item master with live received/issued/stock formulas.	A1:O248; 97 rows; 2,684 f	**Simplify**	Keep identity, unit, category, min level; live quantities move to Consumable Balance.	M_Consumable	<5,000	Out: item list	M_Cons_Category	2,684 formula cells on 97 items today.
M_Cons_Category	Consumable categories and code prefixes.	A1:B40; 16 rows; 0 f	**Keep**	Small list.	M_Cons_Category	<100	Out: category	None	None
M_Product	Stores copy of products/jobs (249 rows; manual copy from Production on 03-Oct-2026).	A1:C403; 249 rows; 0 f	**Combine**	Product master has three diverging copies (Stores 249, Production 269, Dispatch 278). Stores imports the Production-owned master.	X_Product (import)	<5,000	In: Production product master	Production M_Product	24 products exist in Production but not in Stores today.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
M_WhereUsed	Machines and departments a consumable can be issued to.	A1:B51; 17 rows; 0 f	**Keep**	Keep as the cost-centre list; machine names come from the imported Machine master to avoid a second spelling.	M_WhereUsed	<200	Out: where-used list	X_Machine	Machine spelling drift between books.
M_Person	People who receive issues.	A1:C78; 24 rows; 0 f	**Combine**	One People master (owned by Production, imported into Stores and Dispatch) with Person ID and Active flag.	X_People (import)	<500	In: Production M_Worker	Production	Person names typed differently in each book today.
M_Reason	Reason list for rate, adjustment and other dropdowns.	A1:C64; 14 rows; 0 f	**Simplify**	Keep rate and adjustment reasons; remove count-difference reasons.	M_Reason	<200	Out: reason codes	None	Reason codes must be codes, not free text, for MIS grouping.
M_Contacts	Email contacts and groups for Gmail links.	A1:F44; 10 rows; 0 f	**Removed from Phase 1**	Scope override 3 (mail).	None	0	None	MailTo/MailPick names and Lists columns	Delete together with all mail names.
Settings	Thresholds, dates, prefixes, company print details, requisition settings.	A1:C37; 4 f	**Simplify**	Keep tolerances, FY, prefixes, LockDate and a typed AsOnDate; remove print/mail fields (CoAddr, CoEmail), ReqFrom, TolReqPO and TODAY()-based cells.	Settings	<100	Out: named parameters	None	Settings edits must be logged in the Change Log.
Lists (hidden)	Hidden helper lists built with spill formulas (open lots, POs, mail lists, requisition lists).	A1:AU3026; 35 f	**Simplify**	Replace spill formulas with master-table columns and Power Query outputs; keep the sheet visible and labelled.	Lists (visible, no spill formulas)	<5,000	Out: dropdown sources	Masters, balances	35 ANCHORARRAY-based names fail in Excel 2016/2019.

2 Production (23 sheets)

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
START	Navigation page.	A1:E43; 18 f	**Simplify**	Plain README; remove links to removed sheets.	README	n/a	Navigation	None	Keep in step with releases.
2.1 Production Entry	Every production line: stage, machine, batch, lot, product, part, qty, weight, scrap, batch status, QC.	A1:AK12542; 11,406 rows; 203,113 f	**Simplify**	Largest hotspot in the system (203,113 formula cells). Rebuild as entry pad + values-only register + bounded batch balance. Snapshot Wt/pc and lot at posting.	P-Prod Entry (pad) + tblProd + Batch Balance	100,000 lines (actual ~23k/yr)	In: shift, machine, batch, product, part, qty, scrap. Out: output kg, batch consumption, WIP movements, JWC availability.	X_Batch (Stores), M_Part, M_Product, M_Machine	37,614 self-table SUMIFS + 25,076 running-range formulas; 102,808 formulas read Power Query tables that refresh every 5 minutes; 1,132 pre-filled blank rows.
2.2 Rejection & Rework	In-process rejections with reason, action, rework lot/kg/OK qty.	A1:AA267; 68 rows; 1,842 f	**Simplify**	Event register: REJECT, REWORK-OK, REWORK-SCRAP lines linked to the original Prod ID; original never overwritten.	P-Rej Entry (pad) + tblRejRework	100,000	In: source Prod ID, qty, reason, disposition. Out: QC OK qty, WIP reduction, scrap.	tblProd, M_Reason, M_Stage	Only 13 of 68 rejections reference a Prod ID today; target makes it mandatory for in-process rejections.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
2.3 Machine Log	Breakdown/maintenance log with hours.	A1:K204; 0 rows; 601 f	**Removed from Phase 1**	0 rows entered in six months and capacity figures are marked ASSUMED. Recommend Phase 2 unless Management requires downtime KPIs now (Decision D-09).	None (Phase 2: tblDowntime)	0	None	MIS D_MachineLog, 2.6, 2.7 downtime tiles	Downtime KPIs disappear from dashboards until Phase 2.
2.4 WIP Count	WIP physical count at month end (1,503 lines).	A1:I2100; 1,503 rows; 2,097 f	**Removed from Phase 1**	Scope override 4. WIP becomes transaction-calculated from posted production, rejection and JWC movements.	WIP Balance (calculated) - see Section 13	n/a	None	MIS D_WIPCount, 4.3 Stock Summary	Historic count lines archived, not migrated.
2.5 Daily Production	Output by machine for a picked date plus every line of that day; Gmail/Copy page.	A1:U3000; 10 f	**Simplify**	Keep as a simple daily view (PivotTable on the register filtered by date); remove mail and print area.	Daily Output (PivotTable)	n/a	Out: daily view	tblProd	Spill formulas replaced with a PivotTable refreshed on demand.
2.6 Machine Report	Machine output by month and utilisation against capacity.	A1:N420; 47 f	**Removed from Phase 1**	Utilisation depends on assumed capacities; machine output by month is kept in MIS Production report. Utilisation returns in Phase 2 with real capacity data.	MIS Production & Yield (output by machine)	n/a	None	M_Machine capacity	Decision D-09.
2.7 Production Dashboard	Monthly stage output, rejection, downtime, utilisation tiles.	A1:L431; 77 f	**Convert to report**	Move to MIS so production KPIs have one definition.	MIS Control Tower / Production & Yield	n/a	KPIs	MIS extracts	Duplicate definitions removed.
2.8 Batch Balance	Per batch: issued, returned, output, scrap, variance, status; batch drill-down.	A1:AF2000; 19 f	**Keep**	Keep as the single owner of batch balance; add an append-only Batch Close register with variance reason and approver.	Batch Balance (calculated) + tblBatchClose	Open batches (~500-2,000)	In: X_Batch, tblProd, tblBatchClose. Out: batch status to MIS.	X_Batch (Stores), tblProd	MIS currently recomputes batches separately (qBatch) - two versions of the same number.
M_Machine	Machines, process (stage), capacity, status.	A1:G32; 8 rows; 0 f	**Keep**	Machine ID and process drive Stage; capacity optional until Phase 2.	M_Machine (owner for all books)	<100	Out: machine list to Stores, MIS	None	Single machine spelling across books.
M_Worker	Operators, helpers, QC, maintenance.	A1:D77; 24 rows; 0 f	**Keep**	Becomes the single People master (Person ID, role, active), imported by Stores and Dispatch.	M_People	<1,000	Out: people list	None	Never delete; set Active = No.
M_Product	Products/models with customer, category, BOM model, set weight, made kg.	A1:G400; 269 rows; 792 f	**Keep**	Single Product master for all books (Product ID, Customer ID, category, coating route flag, BOM model, active). Remove 'Made kg' formula (report).	M_Product (owner)	<5,000	Out: product list to Stores, Dispatch, MIS	M_BOM_Model	Three copies today with different row counts.
M_Part	Weight per piece by Product + Part, drawing, material.	A1:I3319; 122 rows; 6,630 f	**Simplify**	Part master with Part ID, drawing no, revision, material, thickness, Wt/pc and effective date; weight snapshot taken at posting.	M_Part	<20,000	Out: Wt/pc, drawing	M_Product, M_Material (import)	6,630 formula cells in the master today; Wt/pc edits currently change historic output kg.
M_BOM	BOM lines: drawing, revision, material, size, blank weight, qty/set, costable, variant.	A1:V577; 246 rows; 2,292 f	**Keep**	Keep with revision and effective date; 'Data Fix' migration column folded into Remarks.	M_BOM	<20,000	Out: set weight, RM requirement	M_Product, M_Material	BOM revisions must be added, not overwritten.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
M_BOM_Model	BOM editions and review dates.	A1:I47; 13 rows; 215 f	**Keep**	Edition header with effective-from; review date typed, status a flag.	M_BOM_Model	<500	Out: edition	M_BOM	10 models overdue for review today (MIS check).
M_Reason	Rejection reasons and groups.	A1:C84; 27 rows; 0 f	**Keep**	Coded list.	M_Reason	<300	Out: reasons	None	None
M_Stage	Places a rejection can be found besides machines.	A1:B44; 7 rows; 0 f	**Combine**	Merge into a Stage & Route master (stage sequence per part family, FG/coating flag) needed for calculated WIP.	M_Stage_Route	<500	Out: stage sequence	M_Part	Route master is new and needs Production sign-off (Decision D-05).
M_Contacts	Mail contacts.	A1:F42; 8 rows; 0 f	**Removed from Phase 1**	Scope override 3.	None	0	None	MailTo1, Lists I:J	Delete with mail names.
Settings	AsOnDate, density, LockDate, CheckFrom, BatchFrom; CELL('filename') path.	A1:C16; 3 f	**Simplify**	Keep density, tolerances, LockDate; replace CELL()/TODAY() with typed values; source path typed once.	Settings	<100	Out: parameters	None	CELL() and TODAY() are volatile.
D_Material (hidden)	Hidden Power Query copy of Stores material master.	A1:F70; 69 rows; 0 f	**Simplify**	Keep as controlled import X_Material (visible, labelled, read-only by convention).	X_Material	<2,000	In: Stores export	Stores tblMaterial	Refresh on open/every 5 minutes today; target manual controlled refresh.
D_Lots (hidden)	Hidden Power Query lot list with issued kg.	A1:K188; 187 rows; 0 f	**Combine**	Replaced by X_Batch (batch carries lot); lot detail not needed separately in Production.	X_Batch	n/a	In: Stores export	Stores	None
D_Issues (hidden)	Hidden Power Query issue/batch list with returns and batch kg.	A1:T3023; 3,022 rows; 0 f	**Simplify**	Keep as X_Batch: open and recently closed batches only (not full history), with issued/returned/scrap kg.	X_Batch	Open + last 90 days (~2,000)	In: Stores export	Stores tblRMMove	Full-history import grows without limit today (3,022 rows).
Lists (hidden)	Hidden dropdown lists (spill formulas, mail lists).	A1:Z3269; 19 f	**Simplify**	Visible list sheet without spill formulas or mail lists.	Lists	<5,000	Dropdown sources	Masters	FILTER/SORTBY/UNIQUE need Excel 2021/365.

3 Dispatch & Challan (23 sheets)

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
START	Navigation page.	A1:E41; 19 f	**Simplify**	Plain README.	README	n/a	Navigation	None	None
3.1 Customer Requirement	Customer schedules with versions, priority, status, dispatched and pending qty.	A1:O111; 7 rows; 643 f	**Keep**	Keep version/revises logic; dispatched qty computed from DC lines carrying Req ID (new optional link).	D-Req Entry (pad) + tblRequirement	100,000	In: customer, product, qty, date, priority. Out: pending requirement to MIS RM Requirement.	M_Customer, X_Product, tblDCLine	Dispatched qty is matched by customer+product today, not by requirement (Decision D-12).

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
3.2 JWC Header	Job-work challan header (vendor, vehicle, e-way bill, purpose, shade, status, coater bill).	A1:Q331; 152 rows; 2,617 f	**Simplify**	Document pad (header + lines) posting to header and line registers; JWC number permanent and sequential per FY.	D-JWC Entry (pad) + tblJWCHead	100,000	In: vendor, date, vehicle, purpose. Out: JWC totals.	M_Vendor	2,617 formula cells for 152 headers today.
3.3 JWC Lines	Parts sent to coaters: product, drawing, batch, lot, qty, kg, sq ft, rate, cost, returns, pending, ageing.	A1:AG4034; 3,508 rows; 88,661 f	**Simplify**	Second-largest hotspot (88,661 formula cells). Lines register stores snapshots (rate, sq ft, lot); returns/pending move to the bounded Open JWC Balance.	tblJWCLine + Open JWC Balance	100,000 lines (actual ~7k/yr)	In: product, part, batch, qty, kg, sq ft/pc. Out: pending at coater, PC cost.	X_ProdOutput (Production), M_Vendor rates, tblCoaterReturn, tblDCLine	Coater rate and basis looked up live from M_Vendor today (history changes when the rate changes).
3.4 Print JWC	Job-work challan print layout with Gmail link.	A1:K1000; 33 f	**Removed from Phase 1**	Scope overrides 2 and 3. Statutory job-work challan document must be produced outside the ERP (Decision D-03).	None	0	None	JWCMailTo, MailTo2, Print_Area	Statutory document risk - see Risk R-14.
3.5 Coater Opening	One-time opening quantities at each coater on go-live (empty).	A1:H104; 0 rows; 100 f	**Simplify**	Keep as a one-time cut-over Opening Balance movement confirmed from coater statements; not a recurring count.	tblCoaterMove (type OPENING)	<1,000	In: coater, product, part, qty, kg. Out: opening at coater.	M_Vendor, X_Product	Must be entered before go-live or at-coater balances start wrong.
3.6 Coater Return	Coated parts back OK / defects, fault, decision, re-sent JWC, debit, status, Mail Coater link.	A1:X104; 0 rows; 1,201 f	**Simplify**	Keep as coater movement register (BACK-OK, DEFECT-BACK, DIRECT-DISPATCH via DC); debit/claim and closure fields kept; Mail Coater removed.	D-Coater Return (pad) + tblCoaterMove	100,000	In: JWC line, qty, kg, type, decision. Out: pending at coater, FG receipts, debit notes.	tblJWCLine	Mail Coater (100 formula cells) removed.
3.7 At Coater	Material at each coater, pending by age (spill formulas).	A1:AE3000; 27 f	**Simplify**	Keep as Open JWC Balance by coater and line with ageing buckets; transaction-calculated.	Open JWC Balance (calculated)	Open lines only	Out: pending qty/kg/sq ft by coater, age	tblJWCLine, tblCoaterMove, tblDCLine	Spill formulas replaced.
3.8 FG Count	Finished goods physical count (450 lines).	A1:G742; 450 rows; 739 f	**Removed from Phase 1**	Scope override 5. Replacement: transaction-calculated FG Balance (Add replacement).	FG Balance (calculated)	Products (~300-1,000)	None	MIS D_FGCount, 4.1 'FG counted' tile	FG receipt rule needs sign-off (Decision D-06).
3.9 DC Header	Delivery challan header (customer, dispatch from, vehicle, driver, LR, invoice, e-way bill, status).	A1:Q455; 184 rows; 3,159 f	**Simplify**	Document pad posting header + lines; DC number permanent and sequential per FY.	D-DC Entry (pad) + tblDCHead	100,000	In: customer, from, transport refs. Out: DC totals.	M_Customer, M_Vendor (dispatch-from coater)	None
3.10 DC Lines	Delivery lines: JWC link, product, part, batch, qty, unit, challan qty, weight, rate, value.	A1:Y1296; 626 rows; 21,329 f	**Simplify**	Lines register with UOM, weight and rate snapshots; FG availability hard stop at entry.	tblDCLine + FG Balance	100,000 lines	In: product, qty, JWC line (if from coater). Out: dispatch qty/kg/value, FG reduction.	M_Customer UOM, Customer Price table, FG Balance, Open JWC Balance	Sale rate and set weight looked up live from M_Product today; 33 lines show Unit '0'.
3.11 Print DC	Delivery challan print layout with Gmail link.	A1:K1000; 33 f	**Removed from Phase 1**	Scope overrides 2 and 3. Statutory delivery challan document must be produced outside the ERP (Decision D-03).	None	0	None	DCMailTo, MailTo1, Print_Area	Statutory document risk - see Risk R-14.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
3.12 Customer Summary	Dispatch kg by customer and month.	A1:N420; 52 f	**Convert to report**	Move to MIS.	MIS Dispatch report	n/a	KPIs	MIS	None
3.13 Dispatch Dashboard	Dispatch and powder-coating KPIs.	A1:L435; 95 f	**Convert to report**	Move to MIS Control Tower.	MIS Control Tower	n/a	KPIs	MIS	None
M_Customer	Customers with GSTIN, Tally ledger, dispatch UOM, email.	A1:L40; 15 rows; 0 f	**Keep**	Add Customer ID and Active; Dispatch UOM mandatory (8 of 15 blank today); email optional plain reference only.	M_Customer (owner)	<1,000	Out: customer list, UOM	None	Blank UOM must be completed before go-live.
M_Vendor	Powder coaters with basis and rates.	A1:K28; 5 rows; 0 f	**Simplify**	Keep vendor identity; move rates to an effective-dated Coater Rate table so a rate change never alters posted JWC cost.	M_Vendor + tblCoaterRate	<200 / <2,000	Out: vendor list, rate at JWC date	None	Live rate lookup today.
M_Product	Dispatch copy of products with HSN, unit, set weight, sale rate, dispatched kg.	A1:I400; 278 rows; 396 f	**Combine**	Use the Production-owned product master (import) plus a Dispatch-owned Customer Price & UOM table (effective-dated).	X_Product + tblCustPrice	<5,000	In: Production master	Production M_Product	9 products exist only here (scrap sale items).
M_Worker	Dispatch people (0 rows).	A1:C44; 0 rows; 0 f	**Combine**	Import the single People master.	X_People	n/a	In: Production	Production	None
M_Contacts	Mail contacts.	A1:F43; 9 rows; 0 f	**Removed from Phase 1**	Scope override 3.	None	0	None	Mail names	Delete with mail names.
Settings	AsOnDate, FY, DC/JWC prefixes and next numbers, GST print fields.	A1:C29; 5 f	**Simplify**	Keep FY, prefixes and counters, tolerances, LockDate; remove print-only GST fields and CELL()/TODAY().	Settings	<100	Parameters	None	Counter edits must be logged.
D_ProdLots (hidden)	Hidden Power Query: cut kg by product/part/lot from Production.	A1:G4321; 4,320 rows; 0 f	**Combine**	Replace with X_ProdOutput (QC-OK quantity available to send by Batch, Product, Part).	X_ProdOutput	Open items	In: Production export	Production	Full history import today (4,320 rows).
D_ProdBatch (hidden)	Hidden Power Query: batch output and QC OK by product/part (1 header row, no data).	A1:M2; 0 rows; 0 f	**Combine**	Merged into X_ProdOutput.	X_ProdOutput	n/a	In: Production export	Production	Currently returns no rows - JWC 'sent > produced' check cannot work.
Lists (hidden)	Hidden dropdown lists incl. mail lists.	A1:AM892; 29 f	**Simplify**	Visible lists without spill formulas or mail columns.	Lists	<5,000	Dropdown sources	Masters	None

4 MIS - report sheets (15)

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
START	Navigation and refresh instructions.	A1:E31; 13 f	**Simplify**	README with the refresh procedure and a link to the Refresh Log.	README	n/a	Navigation	None	None

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
4.1 Control Tower	Five-filter KPI tiles, charts and tables; Gmail/Copy page.	A1:R479; 131 f	**Simplify**	Rebuild as one director dashboard fed by small Power Query summary tables and PivotTables; refresh status panel; 'FG counted' tile replaced by calculated FG.	MIS Control Tower	n/a	Out: KPIs, exceptions	X_ extracts, Refresh Log	51 spill references and FILTER formulas need Excel 2021/365.
4.2 RM Requirement	Open requirements x BOM set weight less stock and on-order = what to buy.	A1:S407; 23 f	**Keep**	Keep as the purchase-need report that replaces the requisition step; computed in Power Query; no mail.	MIS RM Requirement	n/a	Out: buy list by material/thickness	X_Requirement, X_BOM, X_LotBal, X_POStatus	2 requirement lines with no BOM today.
4.3 Stock Summary	RM, consumables, WIP and FG summary incl. 'latest physical count'.	A1:G450; 23 f	**Simplify**	All balances transaction-calculated (RM, consumables, WIP, at coater, FG); physical-count block removed.	MIS Stock & WIP Summary	n/a	Out: stock positions	X_LotBal, X_ConsBal, X_WIP, X_AtCoater, X_FGBal	Physical-count dependency removed.
4.4 Consumables	Consumable use by category/where/person and BOM vs actual.	A1:AA500; 19 f	**Simplify**	PivotTable over X_ConsMove; BOM vs actual kept as optional report.	MIS Consumables	n/a	Out: usage	X_ConsMove	None
4.5 Production & Yield	Output by stage by month, yield, rejection by reason.	A1:L423; 81 f	**Keep**	Power Query monthly summary + PivotTable; adds output by machine (from 2.6).	MIS Production & Yield	n/a	Out: yield KPIs	X_Prod, X_RejRework, X_BatchBal	None
4.6 Lot Trace	Trace one lot from PO to customer.	A1:AN190; 43 f	**Simplify**	Parameter-driven Power Query (pick Lot ID, Refresh) returning the chain; no FILTER formulas; no requisition link.	MIS Lot Trace	n/a	Out: trace chain	X_ extracts	Trace limited to data captured (batch links mandatory from go-live).
4.7 DC Trace	Trace a delivery challan back to JWC, batch, lot, GRN, PO and requisition.	A1:AP187; 44 f	**Simplify**	Parameter-driven Power Query; chain ends at PO (requisition removed).	MIS DC Trace	n/a	Out: trace chain	X_ extracts	None
4.8 MRM Monthly	Monthly review pack; Gmail/Copy page; 'Counted (kg)'.	A1:H503; 103 f	**Simplify**	Keep the pack without mail, print area or count columns.	MIS MRM Monthly	n/a	Out: monthly pack	MIS summaries	None
4.9 MIS Dashboard	Second monthly KPI dashboard duplicating Control Tower tiles.	A1:R33; 234 f	**Combine**	Merged into the Control Tower (one KPI definition each).	MIS Control Tower	n/a	KPIs	MIS	Duplicate tiles removed.
4.10 Checks	Cross-book checks with result, status, where-to-fix and 'Mail the team'.	A1:E49; 113 f	**Keep**	Rebuild as the Exception table (one row per open exception, owner, age); mail column, requisition and count checks removed.	MIS Exceptions (tblException)	Open exceptions (<5,000)	Out: exceptions by owner	X_ extracts, Refresh Log	MIS last refreshed 06-Oct-2026 while Stores holds later PO lines - stale numbers today.
4.11 Scrap Dashboard	Scrap and loss by month, customer, model, part, material, process, batch.	A1:R110; 72 f	**Simplify**	Keep as PivotTable with slicers over a Power Query material-flow table.	MIS Scrap & Yield	n/a	Out: scrap KPIs	Q_MaterialFlow	60 spill references today.
M_Contacts	Mail contacts.	A1:F45; 11 rows; 0 f	**Removed from Phase 1**	Scope override 3.	None	0	None	MailTo names, Checks 'Mail the team'	Delete with mail names.
Settings	DataFolder (absolute path), file names, AsOnDate, months.	A1:E22; 3 f	**Simplify**	Typed DataFolder and file names, period parameters; no CELL()/TODAY().	Settings	<100	Parameters	None	Absolute D: path breaks when the folder moves.

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason and control impact	Proposed final sheet / table	Target rows	Inputs / outputs	Dependencies	Main risks / controls
Lists (hidden)	Hidden lists for Control Tower dropdowns and mail.	A1:AC2191; 582 f	**Simplify**	Visible lists without spill formulas or mail.	Lists	<5,000	Dropdowns	X_ extracts	None

4 MIS - hidden Power Query import sheets (31)

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason	Proposed	Target rows	Source
D_RMReceipt (hidden)	Import of Stores GRN table (all 38 columns). Hidden.	A1:AL188; 187 rows; 0 f	**Simplify**	X_GRN with the columns MIS needs only.	X_GRN	Matches source	Stores workbook table
D_RMIssue (hidden)	Import of Stores RM issue table (all 30 columns). Hidden.	A1:AD3394; 3,023 rows; 0 f	**Simplify**	X_RMMove (values register export).	X_RMMove	Matches source	Stores workbook table
D_ConsReceipt (hidden)	Import of consumable receipts. Hidden.	A1:P203; 202 rows; 0 f	**Combine**	X_ConsMove (receipts and issues in one extract).	X_ConsMove	Matches source	Stores workbook table
D_ConsIssue (hidden)	Import of consumable issues. Hidden.	A1:U129; 128 rows; 0 f	**Combine**	X_ConsMove.	X_ConsMove	Matches source	Stores workbook table
D_Adjust (hidden)	Import of stock adjustments. Hidden.	A1:P218; 217 rows; 0 f	**Simplify**	X_Adjust (authorised adjustments).	X_Adjust	Matches source	Stores workbook table
D_Count (hidden)	Import of physical stock count. Hidden.	A1:I274; 273 rows; 0 f	**Removed from Phase 1**	Scope override 5.	None	0	Stores workbook table
D_Material (hidden)	Import of material master. Hidden.	A1:M125; 69 rows; 0 f	**Simplify**	X_Material.	X_Material	Matches source	Stores workbook table
D_Consumable (hidden)	Import of consumable master. Hidden.	A1:O245; 97 rows; 0 f	**Simplify**	X_ConsItem.	X_ConsItem	Matches source	Stores workbook table
D_POLine (hidden)	Import of PO lines (2 rows while Stores holds 3 - stale). Hidden.	A1:AH203; 2 rows; 0 f	**Simplify**	X_POStatus (open PO lines with received/pending).	X_POStatus	Matches source	Stores workbook table
D_Production (hidden)	Import of all production lines (11,406 rows x 38 columns). Hidden.	A1:AL11407; 11,406 rows; 0 f	**Simplify**	X_Prod with required columns only; monthly summary built in Power Query.	X_Prod	Matches source	Production workbook table
D_MachineLog (hidden)	Import of machine log (empty). Hidden.	A1:K2; 0 rows; 0 f	**Removed from Phase 1**	Machine log deferred to Phase 2.	None	0	Production workbook table
D_Rejection (hidden)	Import of rejections. Hidden.	A1:AB69; 68 rows; 0 f	**Simplify**	X_RejRework.	X_RejRework	Matches source	Production workbook table
D_WIPCount (hidden)	Import of WIP physical count. Hidden.	A1:I1504; 1,503 rows; 0 f	**Removed from Phase 1**	Scope override 4.	None	0	Production workbook table
D_Machine (hidden)	Import of machine master. Hidden.	A1:G9; 8 rows; 0 f	**Simplify**	X_Machine.	X_Machine	Matches source	Production workbook table
D_BOM (hidden)	Import of BOM lines. Hidden.	A1:V574; 246 rows; 0 f	**Combine**	X_BOM (lines with edition).	X_BOM	Matches source	Production workbook table
D_BOMModel (hidden)	Import of BOM editions. Hidden.	A1:I44; 13 rows; 0 f	**Combine**	X_BOM.	X_BOM	Matches source	Production workbook table

Current sheet	Apparent purpose	Used range; rows; formulas	Decision	Reason	Proposed	Target rows	Source
D_ProductP (hidden)	Import of Production product master. Hidden.	A1:G397; 269 rows; 0 f	**Combine**	X_Product (single master).	X_Product	Matches source	Production workbook table
D_DCHead (hidden)	Import of DC headers. Hidden.	A1:Q185; 184 rows; 0 f	**Simplify**	X_DCHead.	X_DCHead	Matches source	Dispatch workbook table
D_DCLine (hidden)	Import of DC lines. Hidden.	A1:Y627; 626 rows; 0 f	**Simplify**	X_DCLine.	X_DCLine	Matches source	Dispatch workbook table
D_JWCHead (hidden)	Import of JWC headers. Hidden.	A1:Q153; 152 rows; 0 f	**Simplify**	X_JWCHead.	X_JWCHead	Matches source	Dispatch workbook table
D_JWCLine (hidden)	Import of JWC lines (3,508 × 34). Hidden.	A1:AH3509; 3,508 rows; 0 f	**Simplify**	X_JWCLine.	X_JWCLine	Matches source	Dispatch workbook table
D_Requirement (hidden)	Import of customer requirements. Hidden.	A1:O8; 7 rows; 0 f	**Simplify**	X_Requirement.	X_Requirement	Matches source	Dispatch workbook table
D_FGCount (hidden)	Import of FG physical count. Hidden.	A1:H451; 450 rows; 0 f	**Removed from Phase 1**	Scope override 5; replaced by X_FGBal (calculated in Dispatch).	X_FGBal	Matches source	Dispatch workbook table
D_CoaterReturn (hidden)	Import of coater returns (empty). Hidden.	A1:Y2; 0 rows; 0 f	**Combine**	X_CoaterMove.	X_CoaterMove	Matches source	Dispatch workbook table
D_CoaterOpening (hidden)	Import of coater opening (empty). Hidden.	A1:I2; 0 rows; 0 f	**Combine**	X_CoaterMove (type OPENING).	X_CoaterMove	Matches source	Dispatch workbook table
D_Customer (hidden)	Import of customer master. Hidden.	A1:L16; 15 rows; 0 f	**Simplify**	X_Customer.	X_Customer	Matches source	Dispatch workbook table
D_Vendor (hidden)	Import of vendor master. Hidden.	A1:K6; 5 rows; 0 f	**Simplify**	X_Vendor.	X_Vendor	Matches source	Dispatch workbook table
D_ProductD (hidden)	Import of Dispatch product master. Hidden.	A1:I397; 278 rows; 0 f	**Combine**	X_Product (single master) + X_CustPrice.	X_Product	Matches source	Dispatch workbook table
D_Requisition (hidden)	Import of requisitions (empty). Hidden.	A1:AB2; 0 rows; 0 f	**Removed from Phase 1**	Scope override 1.	None	0	Stores workbook table
D_Flow (hidden)	Power Query material-flow table built from production, rejection and stores scrap (11,474 rows). Hidden.	A1:Q11475; 11,474 rows; 0 f	**Simplify**	Keep as Q_MaterialFlow (staging query feeding Scrap & Yield); load to the Data Model or a summary only.	Q_MaterialFlow	Matches source	MIS (derived) workbook table
D_Batch (hidden)	Power Query batch balance recomputed inside MIS (413 rows). Hidden.	A1:W414; 413 rows; 0 f	**Combine**	Import Production's Batch Balance (X_BatchBal) instead of recomputing - one owner for batch variance.	X_BatchBal	Matches source	Production workbook table

9. Column-by-column design blueprint

Legend. Class: **IN** user input · **DD** dropdown / master selection · **SN** transaction-time snapshot · **CE** calculated at entry (posted as value) · **LK** lookup / display only (pad) · **CF** exception / control flag · **IM** imported / read-only · **RP** report-only. Mandatory: **M** mandatory · **O** optional · **C** conditional. Validation: **H** hard stop · **W** warning (post with authorised explanation).

Standard audit columns appended to every register (marked "+ audit" below):

#	Column	Type / unit	Class	M/O	Source or rule	Validation	Why needed
A1	Posted At	Date-time (value)	SN	M	Stamped at posting (macro) or typed by the poster (manual fallback); never a formula.	H: required	Audit trail; replaces formula dates.
A2	Posted By	Person ID	DD	M	M_People.	H: required, active person	Accountability.
A3	Entry Source	Code	SN	M	PAD / OPENING / MIGRATION.	H: list	Separates migrated history from live entries.
A4	Reverses ID	ID	DD	O	ID of the posted row being reversed (same register).	H: must exist, not already reversed, same type	Corrections without editing history.
A5	Correction Reason	Code	DD	C	M_Reason (Correction). Mandatory when Reverses ID is filled.	H: required with Reverses ID	Explains every reversal.
A6	Remarks	Text	IN	O	Free text, 250 characters.	-	Context.

9.1 PO header - `tblPOHead`

Book: 1 Stores · *Sheet:* PO Register · *Target volume:* 100,000 design limit + audit columns A1-A6. One row per purchase order. Created independently - there is no requisition reference.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	PO No	ID text PO/26-27/nnn	SN	M	Settings prefix + next number (COUNTA of register + 1, FY-wise file).	H: unique	Commercial reference, permanent.
2	PO Date	Date	IN	M		H: > LockDate and <= As-on date	Ageing, period.
3	Supplier ID	ID	DD	M	M_Supplier (active).	H: active supplier	Stable key.
4	Supplier Name	Text	SN	M	Copied from M_Supplier at posting.	-	History unchanged if master renamed.
5	Delivery By	Date	IN	M		W: earlier than PO Date	Overdue tracking.
6	Payment Terms (days)	Integer	DD	M	Default from supplier; editable.	H: 0-365	Commercial term.
7	Delivery Location	Code	DD	M	List in Settings (e.g. ISKO Factory).	H: list	Receiving point.
8	Approved By	Person ID	DD	M	Purchase approver list.	H: required; W: same as Posted By	Replaces the authorisation the requisition gave.
9	PO Status	Code	DD	M	OPEN at posting; CLOSED/CANCELLED only through a status-change line (never edited in place).	H: list	Open PO list.

9.2 PO line - `tblPOLine`

Book: 1 Stores · *Sheet:* PO Register · *Target volume:* 100,000 + audit columns A1-A6. One row per item on a PO. Received and pending are not stored here; they are calculated in PO Status.

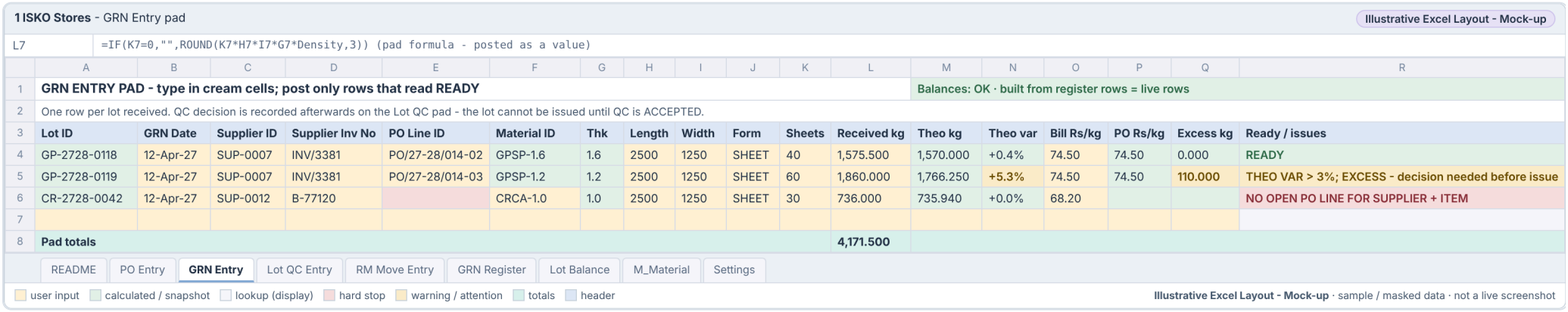
#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	PO Line ID	ID PO No-nn	SN	M	PO No & '-' & line number in the pad.	H: unique	GRN links to this key.
2	PO No	ID	SN	M	From pad header.	H: exists in tblPOHead	Header link.
3	Item Type	Code	DD	M	RM / CONSUMABLE / OTHER.	H: list	Routes the line to GRN or consumable receipt.
4	Item ID	ID	DD	M	M_Material or M_Consumable by Item Type.	H: active item	Stable key.
5	Item Description	Text	SN	M	Copied at posting.	-	History.
6	Grade	Code	SN	C	From material (RM).	-	Lot compatibility.
7	Thk mm	Decimal mm	SN	C	From material (RM).	-	Lot compatibility.
8	Length mm	Integer mm	IN	C	Default standard length.	H: >0 for sheet RM	Theoretical weight.
9	Width mm	Integer mm	IN	C	Default standard width.	H: >0 for sheet RM	Theoretical weight.
10	HSN	Text	SN	O	From item master.	-	Reference for accounts.
11	Ordered Qty	Decimal	IN	M		H: > 0	Pending calculation.
12	Unit	Code KG/NOS/SHEETS/...	SN	M	From item master.	H: list	UOM control.
13	Rate Rs/unit	Decimal Rs	IN	M		H: > 0; W: differs from last GRN rate by > TolRate	Valuation reference.
14	Line Value Rs	Decimal Rs	CE	M	Ordered Qty x Rate.	-	PO value.
15	Due Date	Date	IN	M	Default = header Delivery By.	W: < PO Date	Overdue by line.

9.3 PO Status (calculated view - not a register) - `POStatus`

Book: 1 Stores · *Sheet:* PO Status · *Target volume:* Open lines only. Open PO lines only. Built by Power Query (or bounded SUMIFS) on posting; feeds the GRN dropdown and MIS.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	PO Line ID	ID	IM	M	Open lines from tblPOLine.	-	Key.
2	Supplier ID / Item ID	ID	IM	M		-	Dropdown filters.
3	Ordered Qty	Decimal	IM	M		-	
4	Received Accepted Qty	Decimal	CE	M	Sum of accepted kg/qty on tblGRN by PO Line ID.	-	Progress.
5	Pending Qty	Decimal	CE	M	Ordered - received accepted (min 0).	-	Chasing.
6	Last Receipt Date	Date	CE	O	Latest GRN date.	-	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
7	Overdue Days	Integer	CF	O	As-on date - Due Date when pending > 0.	W: > 0	Exception to MIS.



1 ISKO Stores - GRN Entry pad — Illustrative Excel Layout - Mock-up

L7 `=IF(K7=0,"",ROUND(K7*H7*I7*G7*Density,3)) (pad formula - posted as a value)`

GRN ENTRY PAD - type in cream cells; post only rows that read READY — Balances: OK · built from register rows = live rows

One row per lot received. QC decision is recorded afterwards on the Lot QC pad - the lot cannot be issued until QC is ACCEPTED.

Lot ID	GRN Date	Supplier ID	Supplier Inv No	PO Line ID	Material ID	Thk	Length	Width	Form	Sheets	Received kg	Theo kg	Theo var	Bill Rs/kg	PO Rs/kg	Excess kg	Ready / issues
GP-2728-0118	12-Apr-27	SUP-0007	INV/3381	PO/27-28/014-02	GPSP-1.6	1.6	2500	1250	SHEET	40	1,575.500	1,570.000	+0.4%	74.50	74.50	0.000	READY
GP-2728-0119	12-Apr-27	SUP-0007	INV/3381	PO/27-28/014-03	GPSP-1.2	1.2	2500	1250	SHEET	60	1,860.000	1,766.250	+5.3%	74.50	74.50	110.000	THEO VAR > 3%; EXCESS - decision needed before issue
CR-2728-0042	12-Apr-27	SUP-0012	B-77120		CRCA-1.0	1.0	2500	1250	SHEET	30	736.000	735.940	+0.0%	68.20			NO OPEN PO LINE FOR SUPPLIER + ITEM
Pad totals											**4,171.500**						

README | PO Entry | **GRN Entry** | Lot QC Entry | RM Move Entry | GRN Register | Lot Balance | M_Material | Settings

user input · calculated / snapshot · lookup (display) · hard stop · warning / attention · totals · header

Illustrative Excel Layout - Mock-up · sample / masked data · not a live screenshot

Illustrative Excel Layout - Mock-up: Stores - GRN entry pad (sample data)

9.4 RM receipt / GRN (one row per lot) - `tblGRN`

Book: 1 Stores · *Sheet:* GRN Register · *Target volume:* 100,000 lots + audit columns A1-A6. Receipt facts only. QC and excess decisions are separate events in tblLotQC, so a later decision never overwrites the receipt.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Lot ID	ID e.g. GP-2728-0118	SN	M	Grade code + FY + (COUNTIF of grade code in register + 1). Safe because the register is append-only.	H: unique	Permanent lot identity (replaces sort-sensitive RM ID).
2	GRN Date	Date	IN	M		H: > LockDate, <= As-on date	Period.
3	Supplier ID	ID	DD	M		H: active; must equal PO supplier	Traceability.
4	Supplier Invoice No	Text	IN	M		H: required; W: same supplier + invoice already posted	Bill matching, duplicate guard.
5	Invoice Date	Date	IN	M		W: > GRN Date	
6	PO Line ID	ID	DD	M	Open PO lines of the supplier (PO Status).	H: open line, same supplier and item; opening/migration lots use 'NO-PO' with reason	PO-to-GRN completeness.
7	Material ID	ID	SN	M	From PO line.	H: equals PO line item	Compatibility.
8	Grade	Code	SN	M	From material.	-	
9	Thk mm	Decimal mm	SN	M	From material.	-	
10	Length mm	Integer mm	IN	M	Default from PO line.	W: differs from PO line	Size control.
11	Width mm	Integer mm	IN	M	Default from PO line.	W: differs from PO line	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
12	Form	Code	DD	M	SHEET / COIL / STRIP / OFFCUT.	H: list	
13	Received Sheets	Integer sheets	IN	C	Mandatory when Form = SHEET.	H: integer >= 0	Sheet stock.
14	Received kg	Decimal kg (3 dp)	IN	M	Weighbridge / bill weight.	H: > 0	Stock basis.
15	Theoretical kg	Decimal kg	CE	C	Sheets x L x W x T x density (Settings).	-	Weight check.
16	Theo Variance %	Percent	CE	C	Received / theoretical - 1.	W: |var| > TolTheo	Short-weight detection.
17	Bill Rate Rs/kg	Decimal Rs	IN	M		H: > 0	Lot valuation.
18	PO Rate Rs/kg	Decimal Rs	SN	M	From PO line.	-	Rate check.
19	Rate Variance Reason	Code	DD	C	Required when |bill - PO| > TolRate.	H: required above tolerance	Price control.
20	PO Pending Before (kg)	Decimal kg	SN	M	Pending Qty from PO Status at entry.	-	Evidence for excess.
21	Excess over PO kg	Decimal kg	CE	M	max(0, Received kg - PO Pending Before).	W: > 0 (decision needed in tblLotQC before issue)	Excess control.
22	Lot Value Rs (received)	Decimal Rs	CE	M	Received kg x Bill Rate.	-	Valuation before QC.

9.5 Lot QC and excess decision log - `tblLotQC`

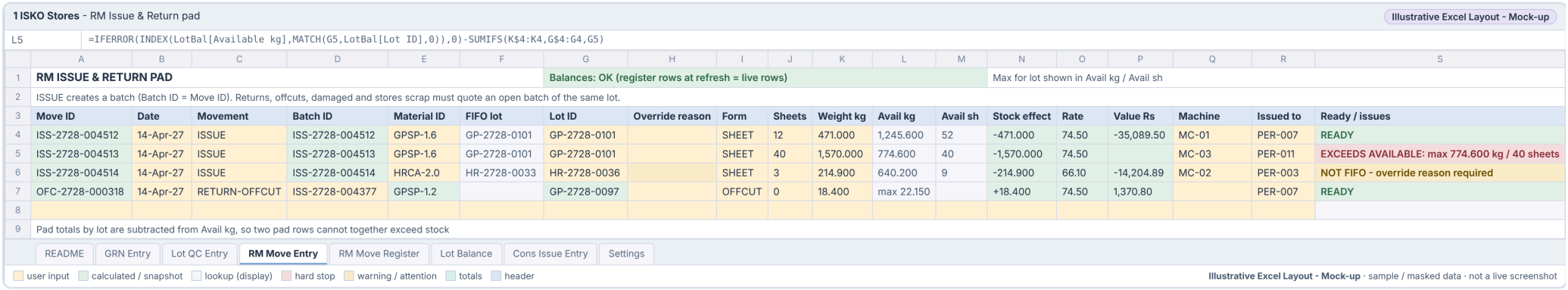
Book: 1 Stores · *Sheet:* Lot QC Register · *Target volume:* 100,000 + audit columns A1-A6. Append-only decisions per lot. The latest row per lot decides issue eligibility.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	QC Event ID	ID QC-2728-nnnnnn	SN	M	Sequence.	H: unique	Audit.
2	Lot ID	ID	DD	M	Lots in QC PENDING / HOLD / excess-pending state.	H: exists	
3	QC Date	Date	IN	M		H: >= GRN Date	
4	QC Status	Code	DD	M	ACCEPTED / ACCEPTED WITH REJECTION / HOLD / REJECTED.	H: list	Issue eligibility.
5	QC Rejected kg	Decimal kg	IN	C	Required for ACCEPTED WITH REJECTION / REJECTED.	H: <= received kg	Accepted kg.
6	QC Rejected Sheets	Integer	IN	C		H: <= received sheets	
7	Excess Decision	Code	DD	C	ACCEPT EXCESS / RETURN EXCESS; required when excess > 0.	H: required before issue	Excess control.
8	Decided By	Person ID	DD	M	QC / Purchase approver list.	H: required	Accountability.
9	Accepted kg (after decision)	Decimal kg	CE	M	Received - QC rejected - (excess if RETURN EXCESS).	-	Lot stock basis.
10	Accepted Sheets	Integer	CE	C	Received sheets - rejected sheets.	-	

9.6 Lot Balance (calculated, open lots only) - `LotBal`

Book: 1 Stores · *Sheet:* Lot Balance · *Target volume:* Open lots (~300-1,000). Recalculated on posting (Power Query group-by, or bounded SUMIFS for open lots). Drives every issue hard stop.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Lot ID	ID	IM	M	Lots with balance <> 0 or activity in last 90 days.	-	
2	Material ID / Grade / Thk / Size	Mixed	IM	M	From tblGRN.	-	Dropdown filters.
3	Issue Eligible	Yes/No	CF	M	Yes only if latest QC = ACCEPTED or ACCEPTED WITH REJECTION and excess decided.	H: issue blocked when No	QC before issue.
4	Accepted kg	Decimal kg	CE	M	Latest tblLotQC accepted kg (0 until QC).	-	
5	Net Movement kg	Decimal kg	CE	M	Sum of Stock Effect kg on tblRMMove.	-	
6	Adjustment kg	Decimal kg	CE	M	Sum of approved tblAdjust kg.	-	
7	Available kg	Decimal kg	CE	M	Accepted + adjustments + net movement.	H: issue > available; CF: < 0 = BREACH	Stock control.
8	Available Sheets	Integer	CE	C	Same logic on sheets.	H: sheets > available	Sheet control.
9	Rate Rs/kg	Decimal Rs	IM	M	Bill rate of the lot.	-	Valuation.
10	Value Rs	Decimal Rs	CE	M	Available kg x rate.	-	
11	Age Days / Age Bucket	Integer / code	CE	M	As-on date - GRN date; 0-30/31-60/61-90/>90.	W: > 90 days	Ageing.
12	Register Rows at Refresh	Integer	CF	M	Row count of tblRMMove when balance was built.	H: <> live count = BALANCES STALE (entry blocked)	Stale-balance guard.



Illustrative Excel Layout - Mock-up: Stores - RM issue and return entry pad (sample data)

9.7 RM issue and return movement register - `tblRMMove`

Book: 1 Stores · *Sheet:* RM Movement Register · *Target volume:* 100,000 movements + audit columns A1-A6. Values only. One ISSUE line = one batch (Batch ID = its Move ID). Returns, offcut returns and stores scrap must quote the batch.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Move ID	ID ISS-/RET-/OFC-/DMG-/SCR-/RTS-2728-nnnnnn	SN	M	Type prefix + FY + (COUNTA of register + pad row).	H: unique	Permanent key; no ROW() dependency.
2	Move Date	Date	IN	M		H: > LockDate, <= As-on date	
3	Movement Type	Code	DD	M	ISSUE / RETURN-USABLE / RETURN-OFFCUT / RETURN-DAMAGED / STORES-SCRAP / RETURN-TO-SUPPLIER.	H: list	Signed stock effect.
4	Batch ID	ID	CE/DD	M	ISSUE: own Move ID. Others: pick an open batch.	H: return/scrap must quote an open batch of the same lot	Batch balance.
5	Material ID	ID	DD	M	Materials with issue-eligible stock (ISSUE) or from batch (others).	H: exists, active	
6	Lot ID	ID	DD	M	Eligible lots of the material, oldest first; FIFO suggestion shown.	H: Issue Eligible = Yes; available > 0; lot material = Material ID	Lot control.
7	FIFO Suggested Lot	ID	LK	-	Oldest eligible lot with stock (pad display only).	-	Guides the pick.
8	Lot Override Reason	Code	DD	C	Required when Lot ID <> FIFO suggestion.	W: required	Explains non-FIFO.
9	Grade / Thk mm / Size	Mixed	SN	M	From lot at posting.	H: thickness/grade must match machine-material rule if defined	Compatibility snapshot.
10	Form	Code	DD	M	SHEET / OFFCUT / COIL / STRIP.	H: list	
11	Sheets	Integer sheets	IN	C	Required when Form = SHEET.	H: integer; issue <= available sheets	Sheet control.
12	Weight kg	Decimal kg (3 dp)	IN	M	Absolute quantity, always positive.	H: > 0; ISSUE <= available kg; return <= batch returnable kg	No negative stock.
13	Stock Effect kg	Decimal kg signed	CE	M	ISSUE, RETURN-TO-SUPPLIER = -kg; RETURN-USABLE, RETURN-OFFCUT = +kg; RETURN-DAMAGED = +kg to DAMAGED bucket only; STORES-SCRAP = 0 to lot.	-	Single signed field for balances.
14	Stock Effect Sheets	Integer signed	CE	C	ISSUE -sheets; RETURN-USABLE +sheets; others 0.	-	
15	Rate Rs/kg	Decimal Rs	SN	M	Lot rate at posting.	-	Issue valuation frozen.
16	Value Rs	Decimal Rs signed	CE	M	Stock Effect kg x rate.	-	
17	Machine ID	ID	DD	C	Required for ISSUE.	H: active machine	Batch destination.
18	Issued To	Person ID	DD	M	M_People.	H: active person	
19	Issued By	Person ID	DD	M	Stores staff list.	H: required	
20	Purpose / Product ID	ID or code	DD	C	Product ID, or SAMPLE / MAINTENANCE / OTHER. Required for ISSUE.	W: blank for ISSUE	Costing link.
21	Available kg Before	Decimal kg	SN	M	Lot available kg shown at entry, frozen at posting.	-	Evidence for audit.
22	Batch Returnable kg Before	Decimal kg	SN	C	For returns: batch kg not yet returned/consumed.	-	Evidence.

9.8 Consumable receipt register - `tblConsRcpt`

Book: 1 Stores · *Sheet:* Consumable Receipt Register · *Target volume:* 100,000 + audit columns A1-A6. Values only.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Receipt ID	ID CR-2728-nnnnnn	SN	M	Sequence.	H: unique	
2	Receipt Date	Date	IN	M		H: > LockDate	
3	Item ID	ID	DD	M	M_Consumable (active).	H: active	
4	Item Name / Category / Unit	Text / code	SN	M	From master.	-	History.
5	Qty	Decimal (unit)	IN	M		H: > 0	
6	Bill Rate Rs	Decimal Rs	IN	M		H: >= 0; W: 0	
7	Value Rs	Decimal Rs	CE	M	Qty x rate.	-	
8	Receipt Type	Code	DD	M	PURCHASE / OPENING (floor returns are RETURN rows in the issue register).	H: list	
9	Supplier ID	ID	DD	C	Required for PURCHASE.	H: required for PURCHASE	
10	PO Line ID	ID	DD	O	Consumable PO lines if used.	W: blank for PURCHASE	Optional PO link.
11	Supplier Bill No	Text	IN	C		W: duplicate supplier + bill	

9.9 Consumable issue / return register - `tblConsIssue`

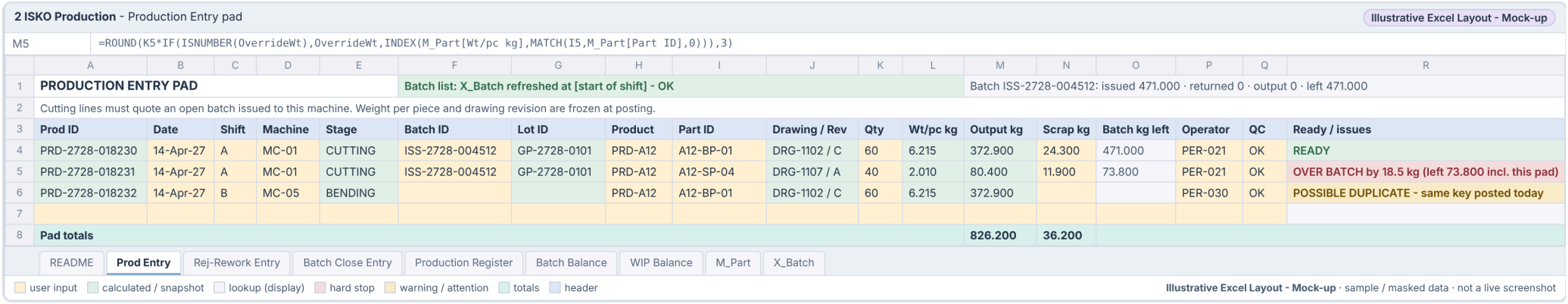
Book: 1 Stores · *Sheet:* Consumable Issue Register · *Target volume:* 100,000 + audit columns A1-A6. Values only. Returns are separate rows with Movement = RETURN.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Cons Move ID	ID CI-2728-nnnnnn	SN	M	Sequence.	H: unique	
2	Move Date	Date	IN	M		H: > LockDate	
3	Movement	Code	DD	M	ISSUE / RETURN.	H: list	
4	Item ID	ID	DD	M		H: active	
5	Item Name / Unit	Text	SN	M		-	
6	Qty	Decimal	IN	M	Positive.	H: > 0; ISSUE <= consumable balance	No negative stock.
7	Stock Effect Qty	Decimal signed	CE	M	-qty issue, +qty return.	-	
8	Rate Rs (avg at posting)	Decimal Rs	SN	M	Moving average from Consumable Balance at posting.	-	Valuation frozen.
9	Value Rs	Decimal Rs signed	CE	M		-	
10	Issued To	Person ID	DD	M		H: active	
11	Where Used	Code	DD	M	M_WhereUsed.	H: list	Cost centre.
12	Product ID / Job	ID	DD	O		-	Costing (optional).
13	Balance Before	Decimal	SN	M	Shown at entry.	-	Evidence.

9.10 Authorised stock-adjustment movement (not a count module) - `tblAdjust`

Book: 1 Stores · *Sheet:* Adjustment Register · *Target volume:* 100,000 (expected low) + audit columns A1-A6. Only for operationally necessary corrections. No count entry, no count-versus-book comparison.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Adj ID	ID ADJ-2728-nnnnn	SN	M	Sequence.	H: unique	
2	Adj Date	Date	IN	M		H: > LockDate	
3	Stock Type	Code	DD	M	RM-LOT / CONSUMABLE.	H: list	
4	Lot ID / Item ID	ID	DD	M		H: exists	
5	Direction	Code	DD	M	INCREASE / DECREASE.	H: list	
6	Qty / kg	Decimal	IN	M	Positive.	H: > 0; DECREASE <= available	No negative stock.
7	Sheets	Integer	IN	C	RM sheet lots.	H: integer	
8	Rate Rs	Decimal Rs	SN	M	Lot rate / average rate.	-	
9	Value Rs	Decimal Rs signed	CE	M		W: above Settings AdjValueLimit needs MD approval	Materiality.
10	Reason Code	Code	DD	M	WEIGHING ERROR / ENTRY ERROR (with reference) / DAMAGE WRITE-OFF / SUPPLIER CREDIT NOTE / OPENING MIGRATION.	H: required; count-difference codes do not exist	Every correction explained.
11	Reference ID	Text/ID	IN	M	Source document or posted ID being corrected.	H: required	Traceability.
12	Requested By	Person ID	DD	M		H: required	
13	Approved By	Person ID	DD	M	Approver list.	H: required and <> Requested By	Segregation of duties.
14	Approval Date	Date	IN	M		H: >= Adj Date	



Illustrative Excel Layout - Mock-up: Production - production entry pad (sample data)

9.11 Production entry register - `tblProd`

Book: 2 Production · *Sheet:* Production Register · *Target volume:* 100,000 lines + audit columns A1-A6. Values only. Cutting lines consume a Stores batch; later stages move WIP between stages.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Prod ID	ID PRD-2728-nnnnnn	SN	M	Sequence (COUNTA of register + pad row).	H: unique	Permanent; replaces ROW()-based PRD-nnnnn.
2	Prod Date	Date	IN	M		H: > LockDate, <= As-on date	
3	Shift	Code	DD	M	A / B / C / GENERAL.	H: list	Blank on all 11,406 history rows today.
4	Machine ID	ID	DD	M	M_Machine (active).	H: active	
5	Stage	Code	SN	M	Machine process (CUTTING/BENDING/FABRICATION...).	-	WIP stage.
6	Batch ID	ID	DD	C	CUTTING: open batches issued to this machine (X_Batch). Later stages: per Decision D-04.	H: CUTTING requires an open batch of this machine	Lot traceability.
7	Lot ID	ID	SN	C	From batch.	-	Trace.
8	Material ID / Grade / Thk / Size	Mixed	SN	C	From batch lot.	H: part material/thickness must equal lot (where part master defines it)	Compatibility.
9	Product ID	ID	DD	M	M_Product (active).	H: active	
10	Part ID	ID	DD	M	Parts of the product (M_Part).	H: belongs to product	
11	Drawing No / Rev	Text	SN	M	From part master.	-	Revision at time of make.
12	Qty (nos)	Integer nos	IN	M		H: integer > 0	
13	Wt/pc kg	Decimal kg	SN	M	Part master weight at posting.	-	Output kg frozen.
14	Wt/pc Override kg	Decimal kg	IN	O	Only with reason.	W: differs > 10% from master	Real weight when known.
15	Output kg	Decimal kg	CE	M	Qty x (override or master Wt/pc).	H (CUTTING): output + scrap <= batch kg left + tolerance	Batch consumption.
16	RM Sheets Used	Integer	IN	O	Cutting from sheets.	H: <= batch sheets left	Sheet reconciliation.
17	Offcut Size Used	Text LxW mm	IN	O	When cutting from offcut.	-	
18	Scrap kg	Decimal kg	IN	C	Required for CUTTING (0 allowed).	H: >= 0	Yield.
19	Batch kg Left Before	Decimal kg	SN	C	From Batch Balance at entry.	-	Evidence.
20	Operator	Person ID	DD	M	M_People.	H: active	
21	QC Status	Code	DD	M	OK / HOLD (rejections recorded in tblRejRework).	H: list	HOLD blocks next stage and JWC.
22	QC By	Person ID	DD	C	Required when HOLD.	-	
23	Dup Key	Text	CE	M	Date|Shift|Machine|Batch|Part|Qty.	W: key already posted (split/rework rule)	Duplicate posting guard.

9.12 Rejection and rework event register - `tblRejRework`

Book: 2 Production · *Sheet:* Rejection & Rework Register · *Target volume:* 100,000 + audit columns A1-A6. Event lines: REJECT creates the case; REWORK-OK and REWORK-SCRAP close it. The original production row is never edited.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Event ID	ID REJ-/RWK-2728-nnnnn	SN	M		H: unique	
2	Event Date	Date	IN	M		H: > LockDate	
3	Event Type	Code	DD	M	REJECT / REWORK-OK / REWORK-SCRAP.	H: list	
4	Case ID	ID	CE/DD	M	REJECT: own Event ID. Rework events: open cases.	H: rework must quote an open case	Closure.
5	Source Prod ID	ID	DD	M	Production line where the parts were made.	H: required for in-process rejections; W: QC-inspection finds without source need reason	Links to batch and lot.
6	Detected at Stage / Place	Code	DD	M	M_Stage_Route + machines.	H: list	
7	Product ID / Part ID / Drawing	IDs	SN	M	From Source Prod ID.	-	
8	Batch ID / Lot ID / Material	IDs	SN	C	From Source Prod ID.	-	Traceability.
9	Qty	Integer nos	IN	M		H: REJECT qty <= source qty - already rejected; rework qty <= case open qty	No over-rejection.
10	kg	Decimal kg	CE	M	Qty x Wt/pc snapshot of source.	-	Scrap kg.
11	Reason Group / Reason	Code	DD	C	Required for REJECT.	H: list	Pareto.
12	Responsibility	Code	DD	C	IN-HOUSE / SUPPLIER / COATER / CUSTOMER.	H: list	
13	Disposition	Code	DD	C	REWORK / SCRAP / RETURN-TO-STORES / USE-AS-IS (concession).	H: list; USE-AS-IS needs approver	
14	Approved By	Person ID	DD	C	Required for USE-AS-IS and SCRAP above limit.	H	
15	Operator	Person ID	DD	O		-	

9.13 Batch Balance (calculated) and batch close logic - `BatchBal`

Book: 2 Production · *Sheet:* Batch Balance · *Target volume:* Open batches (~500-2,000). One row per open batch, rebuilt on posting or refresh. A batch closes only through a tblBatchClose row.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Batch ID	ID	IM	M	From X_Batch (Stores ISSUE lines).	-	
2	Issue Date / Machine / Lot / Material / Size	Mixed	IM	M	From X_Batch.	-	
3	Issued kg / Sheets	Decimal / integer	IM	M	Stores issue.	-	
4	Returned kg (usable + offcut)	Decimal kg	IM	M	Stores returns quoting the batch.	-	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
5	Stores Scrap kg	Decimal kg	IM	M	STORES-SCRAP lines quoting the batch.	-	
6	Net Batch kg	Decimal kg	CE	M	Issued - returned.	-	
7	Cut Output kg	Decimal kg	CE	M	Sum Output kg of CUTTING lines on the batch.	-	
8	Cut Scrap kg	Decimal kg	CE	M	Sum Scrap kg of CUTTING lines.	-	
9	kg Left	Decimal kg	CE	M	Net - output - cut scrap - stores scrap.	H (entry): new cutting line cannot exceed kg Left + tolerance	Over-consumption block.
10	Variance kg / Var %	Decimal / %	CE	M	kg Left at close; % of net.	W: |var| > Settings tolerance (2%, min 1 kg)	Yield control.
11	Variance Value Rs	Decimal Rs	CE	M	Variance kg x lot rate.	-	Loss value.
12	Lines / Last Prod Date / Days Open	Integer / date	CE	M		W: open > 7 days	Ageing.
13	Status	Code	CF	M	NOT STARTED / IN PROGRESS / READY TO CLOSE / OVER-CONSUMED / CLOSED.	CF: OVER-CONSUMED	Batch control.
14	tblBatchClose: Batch ID, Close Date, Closed By, Variance kg (SN), Var % (SN), Variance Reason (DD), Approved By (DD)	Register	SN/DD	M	Posted when production and Stores agree the batch is finished; values frozen.	H: reason + approver when |var| > tolerance	Batch closure audit.

9.14 WIP Balance (transaction-calculated - no count) - `WIPBal`

Book: 2 Production · *Sheet:* WIP Balance · *Target volume:* Product x part x stage (~5-20k). WIP by Product, Part and Stage from posted movements only (see Section 13 rule). Built by Power Query on refresh.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Product ID / Part ID	IDs	IM	M		-	
2	Stage	Code	IM	M	Route stage.	-	
3	Qty In	Integer	CE	M	Qty completed at this stage (tblProd).	-	
4	Qty Moved On	Integer	CE	M	Qty completed at the next route stage, sent on JWC from this stage, or transferred to FG.	-	
5	Qty Rejected	Integer	CE	M	REJECT events detected at this stage (net of REWORK-OK returned).	-	
6	WIP Qty	Integer	CE	M	In - moved on - rejected.	CF: < 0 = route/posting error	Never 'corrected' by a count.
7	WIP kg / Value Rs	Decimal	CE	M	WIP qty x Wt/pc; x average lot rate.	-	Valuation.
8	Oldest Open Date	Date	CE	O		W: > Settings WIPAge days	Ageing.

9.15 Product / model master (single owner: Production) - `M_Product`

Book: 2 Production · *Sheet:* M_Product · *Target volume:* <5,000. Imported read-only by Stores, Dispatch, MIS.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Product ID	ID	IN	M	Short code, never reused.	H: unique	Stable key.
2	Product Name	Text	IN	M	Display name.	H: unique	
3	Customer ID	ID	DD	M	M_Customer (imported from Dispatch).	H: exists	
4	Category / Class	Code	DD	M	ODU / IDU / PLENUM / ...	H: list	
5	BOM Model	ID	DD	O	M_BOM_Model.	-	
6	Coating Route	Code	DD	M	POWDER COAT / NO COATING / MIXED (by part).	H: list	FG receipt rule.
7	Active	Yes/No	DD	M		-	Never delete.
8	Effective From	Date	IN	M		-	

9.16 Part master - `M_Part`

Book: 2 Production · *Sheet:* M_Part · *Target volume:* <20,000. One row per Part ID and revision.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Part ID	ID	IN	M	Product code + part code.	H: unique	Replaces 'Product|Part' text key.
2	Product ID	ID	DD	M		H: exists	
3	Part Name	Text	IN	M		-	
4	Drawing No / Rev / Rev Date	Text / date	IN	M		W: new revision must be a new row	Revision control.
5	Material ID / Thk mm	ID / decimal	DD	M	M_Material (import).	H: exists	Lot compatibility.
6	Blank L x W mm	Integer	IN	O		-	
7	Wt/pc kg	Decimal kg	IN	M	Blank or finished weight per piece.	H: > 0	Output kg.
8	Sq ft / pc	Decimal	IN	O	Coating area.	-	JWC cost.
9	Route	Code list	DD	M	Stage sequence (M_Stage_Route).	H: exists	Calculated WIP.
10	Active / Effective From	Yes/No / date	DD/IN	M		-	

9.17 BOM lines and editions - `M_BOM`

Book: 2 Production · *Sheet:* M_BOM / M_BOM_Model · *Target volume:* <20,000. Edition header (M_BOM_Model) + lines (M_BOM).

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	BOM Model / Edition	ID	DD	M		H: exists	
2	Product ID	ID	SN	M		-	
3	SR No / Level / Parent SR	Integer	IN	M		H: parent exists	Structure.
4	Part ID / Drawing No / Rev	IDs	DD	M		H: exists	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
5	Material ID / Thk / Width / Length	Mixed	SN	M	From part.	-	
6	Blank Wt kg / Qty per Set / Wt per Set kg	Decimal	IN/CE	M	Wt per set = blank wt x qty.	-	RM requirement.
7	Costable / Variant	Code	DD	M		-	
8	Effective From / Review Due	Date	IN	M	Typed dates (no TODAY()).	W: review overdue	BOM governance.

9.18 Customer requirement (versioned) - `tblRequirement`

Book: 3 Dispatch & Challan · *Sheet:* Requirement Register · *Target volume:* 100,000 + audit columns A1-A6. A revision is a new row that quotes the row it revises.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Req ID	ID REQ-2728-nnnnn	SN	M		H: unique	
2	Version / Revises Req ID	Integer / ID	CE/DD	M	Version = revised version + 1.	H: revised row must be latest	Version control.
3	Entry Date	Date	IN	M		-	
4	Customer ID / Product ID	IDs	DD	M		H: product belongs to customer	
5	Required Qty / Unit	Decimal / code	IN/SN	M	Unit from customer UOM.	H: > 0	
6	Required Date / Priority	Date / integer	IN	M		-	
7	Status	Code	DD	M	OPEN / ON HOLD / SUPERSEDED / CLOSED.	H: list	
8	Dispatched Qty / Pending Qty	Decimal	RP	-	Calculated in Requirement Status from DC lines quoting Req ID.	-	Not stored.

9.19 Job Work Challan header - `tblJWCHead`

Book: 3 Dispatch & Challan · *Sheet:* JWC Register · *Target volume:* 100,000 + audit columns A1-A6. Posted from the JWC document pad together with its lines.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	JWC No	ID JWC/26-27/nnnn	SN	M	Prefix + next number, sequential per FY.	H: unique, no gaps	Statutory serial.
2	JWC Date	Date	IN	M		H: > LockDate	
3	Vendor ID	ID	DD	M	M_Vendor (active coater/sub-contractor).	H: active	
4	Vendor Name	Text	SN	M		-	
5	Vehicle No / e-Way Bill No	Text	IN	C	e-Way bill when value above threshold.	W: blank above threshold	Transport reference.
6	Purpose / Shade	Code / text	DD/IN	M		H: list	
7	JWC Status	Code	DD	M	OPEN at posting; CLOSED/CANCELLED by status line.	H: list	
8	Coater Bill No	Text	IN	O	Added via status line when billed.	-	Cost matching.

9.20 Job Work Challan lines - `tblJWCLine`

Book: 3 Dispatch & Challan · *Sheet:* JWC Register · *Target volume:* 100,000 lines + audit columns A1-A6. Values only; returns and pending are calculated in Open JWC Balance.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	JWC Line ID	ID JWC No-nn	SN	M		H: unique	Return/dispatch key.
2	JWC No / JWC Date / Vendor ID	IDs / date	SN	M	From header (snapshot on every line).	-	Flat line register.
3	Product ID / Part ID / Drawing No	IDs	DD/SN	M		H: active	
4	Batch ID	ID	DD	C	Batches with QC-OK quantity of this part available to send (X_ProdOutput).	H: required from go-live; qty <= available to send	Trace to lot.
5	Lot ID / Lot Material / Lot Size	IDs	SN	C	From batch.	-	
6	Coating	Code	DD	M	POWDER COAT / NO COATING / OTHER.	H: list	
7	Qty NOS	Integer	IN	M		H: > 0	
8	Wt kg	Decimal kg	CE	M	Qty x Wt/pc (or typed weighbridge kg).	W: differs > 5% from Qty x Wt/pc	
9	Sq ft / pc	Decimal	SN	C	From part master.	W: 0 for SQFT basis	
10	Sq ft	Decimal	CE	C	Qty x sq ft/pc, or kg / conversion when not defined.	-	
11	Rate Basis	Code	SN	M	From tblCoaterRate effective on JWC date.	-	Frozen.
12	PC Rate	Decimal Rs	SN	M	Effective rate; override needs reason.	W: override	Frozen.
13	PC Cost Rs	Decimal Rs	CE	M	Basis qty x rate.	-	Cost frozen.
14	Goods Value Rs	Decimal Rs	CE	O	Qty x goods rate (for challan value / e-way bill).	-	

9.21 Coater movement register (opening, returns, defects, re-send) - `tblCoaterMove`

Book: 3 Dispatch & Challan · *Sheet:* Coater Movement Register · *Target volume:* 100,000 + audit columns A1-A6. Every change to material at a coater other than the JWC itself and direct dispatch (which is a DC line).

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Coater Move ID	ID CRT-2728-nnnnn	SN	M		H: unique	
2	Move Date	Date	IN	M		H: > LockDate	
3	Move Type	Code	DD	M	OPENING (cut-over only) / BACK-OK / DEFECT-BACK / DEFECT-AT-COATER-SCRAP.	H: list; OPENING only before go-live date	
4	Coater ID	ID	DD	M		H: active	
5	JWC Line ID	ID	DD	C	Open JWC lines of the coater. Not used for OPENING.	H: qty <= line pending	No over-return.
6	Coater Challan No	Text	IN	C	Coater's return document.	W: blank	
7	Product ID / Part ID / Batch ID	IDs	SN	M	From JWC line.	-	
8	Qty NOS / Wt kg	Integer / decimal	IN/CE	M	kg = qty x line kg/qty.	H: > 0	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
9	Defect Description / Whose Fault	Text / code	IN/DD	C	Required for DEFECT types.	H	
10	Decision	Code	DD	C	REWORK AT COATER / RE-SEND / SCRAP / DEBIT COATER.	H: required for defects	
11	Re-sent on JWC Line ID	ID	DD	O	When re-sent.	-	Closure.
12	Debit to Coater Rs	Decimal Rs	IN	C	Required when DEBIT COATER.	H: > 0	Claim.
13	Case Status	Code	DD	M	OPEN / CLOSED (closed by a later line quoting this ID).	-	

9.22 Open JWC / at-coater balance (calculated) - `JWCBal`

Book: 3 Dispatch & Challan · *Sheet:* Open JWC Balance · *Target volume:* Open lines. Open JWC lines only, rebuilt on posting.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	JWC Line ID / Coater / Product / Part / Batch	IDs	IM	M		-	
2	Sent Qty / kg / Sq ft	Decimal	IM	M		-	
3	Back OK Qty	Integer	CE	M	BACK-OK moves.	-	
4	Defect Back Qty	Integer	CE	M	DEFECT-BACK moves.	-	
5	Direct Dispatch Qty	Integer	CE	M	DC lines quoting the JWC line.	-	
6	Scrapped at Coater Qty	Integer	CE	M		-	
7	Pending Qty / kg / Sq ft	Decimal	CE	M	Sent - back OK - defect back - direct dispatch - scrapped.	CF: < 0 = BREACH	Material at coater.
8	Days Pending / Age Bucket	Integer	CE	M	As-on date - JWC date.	W: > Settings CoaterAge days	Chasing.
9	Coater opening (cut-over) is a separate block by coater/product		IM		From OPENING moves.		Opening balance.

9.23 Finished-goods balance (transaction-calculated - no count) - `FGBal`

Book: 3 Dispatch & Challan · *Sheet:* FG Balance · *Target volume:* Products (~300-1,000). By Product (and Part where sold as parts).

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Product ID / Part ID	IDs	IM	M		-	
2	FG Opening (cut-over)	Decimal	IM	M	One-time migration opening, approved (not a count module).	-	Start point.
3	FG In - coater BACK-OK	Decimal	CE	M	tblCoaterMove BACK-OK.	-	
4	FG In - uncoated completion	Decimal	CE	M	Final-route-stage production for NO COATING items (Decision D-06).	-	
5	FG Out - dispatched from factory	Decimal	CE	M	DC lines with Dispatch From = factory.	-	
6	FG Out - rejected at final QC	Decimal	CE	M	tblRejRework at stage FINAL.	-	
7	FG Adjustments (authorised)	Decimal	CE	M	Approved FG adjustment lines only.	-	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
8	FG Available	Decimal	CE	M	Opening + in - out + adjustments.	H (DC entry): dispatch qty <= FG Available; CF: < 0	No dispatch above FG.

3 ISKO Dispatch & Challan - DC document pad — Illustrative Excel Layout - Mock-up

K5 `=IF(H5="KG",J5,IF(H5="TON",ROUND(J5/1000,3),G5)) UOM from Customer-Product table, frozen at posting`

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	**DELIVERY CHALLAN PAD (one challan at a time: header, then lines)**									FG Balance: OK · JWC Balance: OK					
2	**Challan No**	DC/27-28/0191	**Date**	15-Apr-27	**Customer**	CUST-01		**UOM**	QTY	**Dispatch from**	FACTORY	**Vehicle**	GJ06-XX-0000	**LR / Invoice**	[ref] / [ref]
3	**e-Way Bill**	[number]	**Driver**	[name]	**Status**	DISPATCHED		**Lines**	**3**	Header copied onto every line at posting (flat line register)					
4	**DC Line ID**	**Product**	**Part ID**	**JWC Line ID**	**Batch ID**	**Type**	**Qty**	**UOM**	**Wt/unit**	**Wt kg**	**Challan qty**	**Rate**	**Value Rs**	**FG avail**	**Ready / issues**
5	DC/27-28/0191-01	PRD-A12	(set)			SALE	40	QTY	41.950	1,678.000	40	173.24	2,90,696.72	64	READY
6	DC/27-28/0191-02	PRD-B07	(set)			SALE	30	QTY	28.400	852.000	30	165.10		22	ABOVE FG AVAILABLE: max 22
7	DC/27-28/0191-03	PRD-A12	A12-SP-04	JWC/27-28/0419-01	ISS-2728-004401	SALE	20	QTY	2.010	40.200	20	173.24	6,964.25	50 at coater	READY (from coater: JWC pending 50)
8															
9	**Challan totals**						**90**			**2,570.200**			**2,97,660.97**		

README | Req Entry | JWC Entry | Coater Move Entry | **DC Entry** | DC Register | FG Balance | Open JWC Balance | M_Customer

user input · calculated / snapshot · lookup (display) · hard stop · warning / attention · totals · header

Illustrative Excel Layout - Mock-up · sample / masked data · not a live screenshot

Illustrative Excel Layout - Mock-up: Dispatch - delivery challan document pad (sample data)

9.24 Delivery Challan header - `tblDCHead`

Book: 3 Dispatch & Challan · *Sheet:* DC Register · *Target volume:* 100,000 + audit columns A1-A6. Posted from the DC document pad with its lines.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Challan No	ID DC/26-27/nnnn	SN	M	Prefix + next number (Settings DCNext), sequential per FY.	H: unique, no gaps	Statutory serial.
2	DC Date	Date	IN	M		H: > LockDate, <= As-on date	
3	Customer ID / Customer Name	ID / text	DD/SN	M		H: active	
4	Dispatch From	Code	DD	M	FACTORY or a Coater ID.	H: list	FG vs at-coater.
5	Vehicle No / Driver / LR No	Text	IN	C		W: blank	
6	Invoice No / e-Way Bill No	Text	IN	C		W: blank when value above threshold	Accounts link.
7	DC Status	Code	DD	M	DISPATCHED / DELIVERED / CANCELLED (status line).	H: list	

9.25 Delivery Challan lines - `tblDCLine`

Book: 3 Dispatch & Challan · *Sheet:* DC Register · *Target volume:* 100,000 lines + audit columns A1-A6. Values only. UOM and rate frozen at posting.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	DC Line ID	ID Challan No-nn	SN	M		H: unique	

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
2	Challan No / DC Date / Customer ID / Dispatch From	Mixed	SN	M	From header.	-	
3	Req ID	ID	DD	O	Open requirement of customer + product.	W: blank (Decision D-12)	Requirement closure.
4	JWC Line ID	ID	DD	C	Required when Dispatch From is a coater.	H: required from coater; qty <= JWC pending	At-coater reduction.
5	Product ID / Part ID / Description	IDs	DD/SN	M		H: active; product belongs to customer	
6	Batch ID	ID	SN/DD	C	From JWC line, or picked for factory stock.	W: blank	Trace.
7	Dispatch Type	Code	DD	M	SALE / REPLACEMENT / SAMPLE / RETURN-TO-CUSTOMER-OWNED.	H: list	
8	Qty	Decimal	IN	M	Pieces/sets.	H: > 0; <= FG available (factory) or JWC pending (coater)	No over-dispatch.
9	Dispatch UOM	Code QTY/NOS/KG/TON/SET	SN	M	Customer-Product UOM table, else M_Customer Dispatch UOM.	H: blank UOM blocks posting	Customer-specific unit.
10	Wt per Unit kg	Decimal kg	SN	M	Set weight / part weight at posting.	-	Frozen.
11	Wt kg	Decimal kg	CE	M	Qty x Wt per unit (or weighbridge override).	W: override	
12	Challan Qty	Decimal	CE	M	UOM = KG → Wt kg; TON → Wt kg/1000; QTY/NOS/SET → Qty.	-	What the challan states.
13	Sq ft	Decimal	CE	O	From JWC line proportion.	-	
14	Rate Rs / Rate Basis	Decimal / code	SN	M	Customer Price table effective on DC date.	W: override with reason	Frozen value.
15	Value Rs	Decimal Rs	CE	M	Challan Qty x rate (per basis).	-	
16	FG Available Before	Decimal	SN	M	Shown at entry.	-	Evidence.

9.26 Controlled import / export (staging) table - standard columns - `X_Extract`

Book: All · *Sheet:* X_* tables · *Target volume:* Matches source. Every X_ table is a values-only Power Query output (or pasted values in the fallback). These columns are added to every extract.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Source Book	Text	IM	M	File name read.	-	Lineage.
2	Source Table	Text	IM	M	Register name.	-	
3	Source File Modified	Date-time	IM	M	From Folder.Files [Date modified].	CF: older than Settings MaxAgeHours = STALE	Staleness.
4	Extract Run ID	ID RUN-yymmdd-hhmm	IM	M	Same for all rows of one refresh.	-	Duplicate import guard.
5	Source Row ID	ID	IM	M	The register primary key.	H: duplicates in extract = refresh FAILED	Uniqueness.
6	Business columns	Per table	IM	M	Only columns consumers need (listed in Section 12).	-	

9.27 Refresh Log (one row per extract per refresh) - `tblRefreshLog`

Book: All · *Sheet:* Refresh Log · *Target volume:* Grows ~50 rows/day. Written by Power Query (or by hand in the fallback).

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Run ID	ID	IM	M		-	
2	Refreshed At	Date-time value	IM	M	DateTime.LocalNow() inside Power Query (static value, not a worksheet formula).	-	Visible refresh time.
3	Source File / Version	Text	IM	M	File name + Settings version cell of the source book.	CF: version mismatch	
4	Source Modified	Date-time	IM	M		CF: stale	
5	Rows Read / Distinct IDs	Integer	IM	M		CF: not equal = DUPLICATES	
6	Control Total kg / Rs	Decimal	IM	M	Sum of key quantity/value columns.	CF: <> source control total	Reconciliation after refresh.
7	Period Covered	Date range	IM	M	Min/max business date.	-	
8	Status	Code	CF	M	OK / STALE / MISSING FILE / MISMATCH / DUPLICATES.	CF	Shown on every dashboard.

9.28 MIS exception / check table - `tblException`

Book: 4 MIS · *Sheet:* Exceptions · *Target volume:* Open exceptions (<5,000). One row per open exception (not one row per check). Built by Power Query from extracts.

#	Final column (in order)	Type / unit	Class	M/O	Source or calculation rule	Validation (H / W)	Why needed in Phase 1
1	Exception ID	Text	IM	M	Check code + source row ID (stable across refreshes).	-	Tracking.
2	Check Code / Check Name	Code / text	IM	M	From the check catalogue (Section 15).	-	
3	Severity	Code	IM	M	BLOCKER / HIGH / MEDIUM / INFO.	-	
4	Book / Register / Source Row ID	Text / ID	IM	M		-	Drill-through.
5	Detail	Text	IM	M	Values that triggered it.	-	
6	Owner (role)	Code	IM	M	Stores / Production / Dispatch / Finance / MIS.	-	Action owner.
7	First Seen / Days Open	Date / integer	IM	M	Kept by matching Exception ID with the previous run.	W: > 3 days	Ageing.
8	Action Note	Text	IN	O	Typed by owner in a side table keyed by Exception ID.	-	Follow-up.

10. Master data and unique-ID standards

10.1 Master ownership

Master	Owner book	Key	Rules
Material, Grade	Stores	Material ID (e.g. GPSP-1.6), Grade code (GP)	Never re-key; Active flag; standard size and sheet weight
Supplier	Stores	Supplier ID (SUP-0001)	Tally ledger name kept as reference; email optional text
Consumable item, Category	Stores	Item ID (ABR-001)	Category prefix + sequence
Product / model	Production	Product ID (short code)	One master for all books (replaces 3 copies); Customer ID; coating route
Part	Production	Part ID (product + part code)	Drawing no, revision, material, thickness, Wt/pc, route
BOM model / edition / lines	Production	Model + Edition + SR No	New edition = new rows; effective-from dates
Stage & route	Production	Stage code; Route ID	Sequence per part family; FG point (D-05)
Machine	Production	Machine ID (MC-01)	Process = stage; capacity Phase 2
People	Production	Person ID (PER-001)	Role; Active; used by all books
Customer, Customer price & UOM	Dispatch	Customer ID (CUS-001)	Dispatch UOM mandatory; price effective-dated
Vendor / coater, Coater rate	Dispatch	Vendor ID (VEN-001)	Rate basis and rate effective-dated
Reasons	Owning book	Reason code	Codes, not free text; count reasons deleted

10.2 ID standard

Record	Format	Example	Generated by
PO / PO line	PO/yy-yy/nnn ; PO No-nn	PO/27-28/014-02	Settings prefix + next number
Lot	Grade code-FY-nnnn	GP-2728-0118	Grade code + COUNTIF of grade in register + 1
QC event	QC-FY-nnnnnn	QC-2728-000231	Sequence
RM movement / batch	ISS/RET/OFC/DMG/SCR/RTS-FY-nnnnnn	ISS-2728-004512 (= Batch ID)	Sequence shared by the register
Consumable receipt / issue	CR-/CI-FY-nnnnnn	CI-2728-000877	Sequence
Adjustment	ADJ-FY-nnnnn	ADJ-2728-00012	Sequence
Production line	PRD-FY-nnnnnn	PRD-2728-018230	Sequence
Rejection / rework	REJ-/RWK-FY-nnnnn	REJ-2728-00341	Sequence
Requirement	REQ-FY-nnnnn + version	REQ-2728-00045 v2	Sequence
JWC / JWC line	JWC/yy-yy/nnnn ; JWC No-nn	JWC/27-28/0412-03	Settings prefix + next number
Coater movement	CRT-FY-nnnnn	CRT-2728-00090	Sequence
DC / DC line	DC/yy-yy/nnnn ; DC No-nn	DC/27-28/0191-01	Settings prefix + next number (current DCNext = 191)
Extract run	RUN-yymmdd-hhmm	RUN-270415-0915	Power Query

Rules: IDs are written as values at posting and never recalculated; the FY element prevents collisions after archival; sequences are `COUNTA(register key) + pad row number`, which is safe only because registers are append-only and one person posts at a time; statutory document numbers (DC, JWC) must have no gaps - a cancelled number stays in the register with status CANCELLED.

10.3 Naming and revision rules

Names are labels, IDs are keys. A renamed product keeps its ID. A new drawing revision is a new Part row with a new effective date; the old row is set inactive, never overwritten. Master deletions are not allowed; set *Active = No*.

11. Formula and performance architecture

11.1 What stays a formula

Location	Formulas	Rule
Entry pads (≈200 rows)	Lookups (one MATCH per key + INDEX), calculated values, checks, Ready flag	Recalculate on every edit - cheap because small
Balance tables	One aggregate per open item (or Power Query)	Rebuilt on posting / refresh
Registers	**None**	Values only
Masters	Only simple row-local helpers (e.g. sheet weight)	No aggregates over transactions
MIS	Small summaries over Power Query outputs, PivotTables	Refresh only

Formula budget: ≤ 40,000 formula cells per book at 100,000 register rows (today: up to 217,775 at 12,538 rows).

11.2 Calculated once at posting, not on every edit

IDs; snapshot rates, weights, UOM and names; stock effect; value; available-before evidence; duplicate key. These are pad formulas whose *results* are posted.

11.3 Posting mechanism - VBA versus no macros

	A. Signed "Post" macro (preferred if D-02 allows)	B. Manual paste-values (fallback / no macros)
How	Button validates Ready rows, writes values to the register, stamps Posted At/By, clears the pad, refreshes balances	User selects Ready rows, *Copy → Paste Values* into the next register rows, types Posted At/By, clears pad, *Refresh All*
Control strength	Hard stops cannot be bypassed; consistent	Depends on procedure; pad control total must equal register delta
Security	`.xlsm`, digitally signed, macros enabled only for that signature; one small module (~150 lines)	No macros
Backup / audit	Macro writes a posting log row	Poster records the control total in the posting log
Compatibility	Excel 2016+ Windows	Any Excel
Maintenance	Needs someone able to read VBA	None

Both use identical pads and registers, so a book can move from B to A later.

11.4 Why this scales better

Aspect	Current design (measured)	Target design
Formula cells in largest book	217,775 at 12,538 rows	<= 40,000 at 100,000 rows (pads + balances)
Work per new row	~18-25 formulas, several scanning the whole table	0 in the register; pad row lookups only
Aggregates per edit	Per-row SUMIFS / expanding ranges over the table (grows with rows²)	Balances for open items, rebuilt on posting
Volatile triggers	TODAY() in 5,571 cells; CELL(); 5-minute background refresh	None
IDs	ROW() / running counts	Values written at posting

Aspect	Current design (measured)	Target design
Imports	Whole tables, all columns, blank rows included	Needed columns, non-blank keys, lineage, logged
History volume per file	Unlimited growth	One FY per file (~23k production rows/yr)

12. Cross-workbook integration and MIS refresh design

12.1 Method comparison

Method	Pros	Cons	Use
Live external-link formulas	Instant	Fragile paths, slow, break on rename; recalc chains	**Not used**
Copy / paste-as-values export	Works in any Excel	Manual; duplicate risk	Fallback when Power Query unavailable
Power Query from the saved source book	Already used; replace-not-append; no formulas; lineage columns	Reads last *saved* version; needs Excel with Get & Transform	**Recommended**
VBA-assisted refresh	Can add logging and checks	Needs macros	Optional with D-02

12.2 Extract catalogue

Extract	From (owner)	To	Columns that travel	Refresh point
X_Batch	Stores tblRMMove (ISSUE lines + returns/scrap per batch)	Production	Batch ID, Move Date, Machine ID, Lot ID, Material ID, Grade, Thk, Size, Issued kg, Issued sheets, Returned kg, Stores scrap kg, Rate	Start of shift; on demand
X_Material	Stores M_Material	Production, MIS	Material ID, Grade, Thk, Std L/W, Std sheet kg, Active	Weekly / on change
X_Product / X_Part / X_Machine / X_People	Production masters	Stores, Dispatch, MIS	IDs, names, customer, route, Wt/pc, sq ft/pc, active	Weekly / on change
X_Customer	Dispatch M_Customer	Production, MIS	Customer ID, name, group, Dispatch UOM, active	Weekly / on change
X_ProdOutput	Production register + rejection events	Dispatch	Batch ID, Product ID, Part ID, Stage, QC-OK qty available to send, kg	Start of shift; on demand
X_GRN, X_LotQC, X_RMMove, X_Adjust, X_ConsMove, X_POStatus, X_LotBal, X_ConsBal	Stores registers and balances	MIS	Register columns needed for KPIs and trace + lineage	Before MIS review
X_Prod, X_RejRework, X_BatchBal, X_WIP, X_BOM	Production registers and balances	MIS	As above	Before MIS review
X_Requirement, X_JWCHead, X_JWCLine, X_CoaterMove, X_JWCBal, X_FGBal, X_DCHead, X_DCLine, X_CustPrice	Dispatch registers and balances	MIS	As above	Before MIS review
Control Totals	Each book (rows, kg, Rs per register and month)	MIS	Book, register, month, rows, kg, Rs	Every save (formula-free: written at posting)

Each extract also carries the standard lineage columns (Section 9, *X_ extract*).

12.3 Refresh points

When	Who	What
Start of shift	Production / Dispatch	Refresh X_Batch, X_ProdOutput (after Stores/Production saved)
After each posting	Book owner	Refresh own balances (automatic with macro)
End of shift	Each owner	Save, back up, write day control totals
Before MIS review (daily 10:00 suggested)	MIS owner	Refresh All in MIS, read Refresh Log, review exceptions
Month close	Finance	Lock date, close totals, archive copy

12.4 How a user knows a refresh succeeded

The MIS header and every operational book show a **Refresh status panel**: last run time, each source with *OK / STALE / MISSING FILE / MISMATCH / DUPLICATES*, rows read and control totals. Anything except *OK* turns the panel coral and the Control Tower shows a banner; KPIs from that source are labelled *stale*.

12.5 Duplicates, corrections and file problems

- **Duplicates:** extracts *replace* their table on each refresh (never append) and filter on a non-blank primary key; the Refresh Log compares rows with distinct IDs.
- **Corrections:** happen only in the owning book as reversal rows, so the next refresh carries them automatically.
- **File closed / moved / renamed / missing:** Power Query reads the saved file (closed or open); a missing file gives *MISSING FILE* and the previous extract is kept, clearly marked stale.
- **Updated by another person:** only the owner edits a book; a consumer reading during an unsaved edit gets the last saved state and the *Source Modified* time shows it.
- **Reconciliation after refresh:** control totals (rows, kg, Rs) of each extract equal the source register totals written in the source book's *Control Totals* table.
- **Delay tolerance:** KPIs, trends and trace may be up to one shift old. Stock availability, batch kg left, JWC pending and FG available must be validated in the owning book at the moment of entry - never from MIS.

13. Inventory and production control design

13.1 RM lot balance and availability

Lot available kg = accepted kg (latest QC decision) + approved adjustments + signed stock effect of all movements. Sheets follow the same rule. Balances are rebuilt for open lots only after each posting; the pad subtracts quantities already typed in the pad for the same lot, so two pad rows cannot together exceed stock. A **stale guard** compares the register row count stored in the balance with the live count; a difference blocks entry until refresh.

13.2 Issue and return rules

Movement	Lot effect	Must quote	Hard stops
ISSUE	− kg, − sheets	Machine, person, purpose	Lot eligible; kg ≤ available; sheets ≤ available
RETURN-USABLE	+ kg, + sheets	Batch	≤ batch returnable; same lot
RETURN-OFFCUT	+ kg (sheets 0)	Batch	≤ batch returnable
RETURN-DAMAGED	+ kg to DAMAGED bucket (not issue-eligible)	Batch	≤ batch returnable
STORES-SCRAP	0 to lot; batch consumption	Batch	≤ batch returnable
RETURN-TO-SUPPLIER	− kg	Lot, supplier document	≤ available

13.3 Batch balance, closure, yield and scrap

Batch kg left = issued − returned − cut output − cut scrap − stores scrap. A cutting line cannot exceed kg left plus tolerance (Settings, today 2% with a 1 kg minimum). Closing a batch posts a *Batch Close* row that freezes variance kg, variance % and variance value; above tolerance it needs a reason and an approver. Yield = cut output / net batch kg on closed batches.

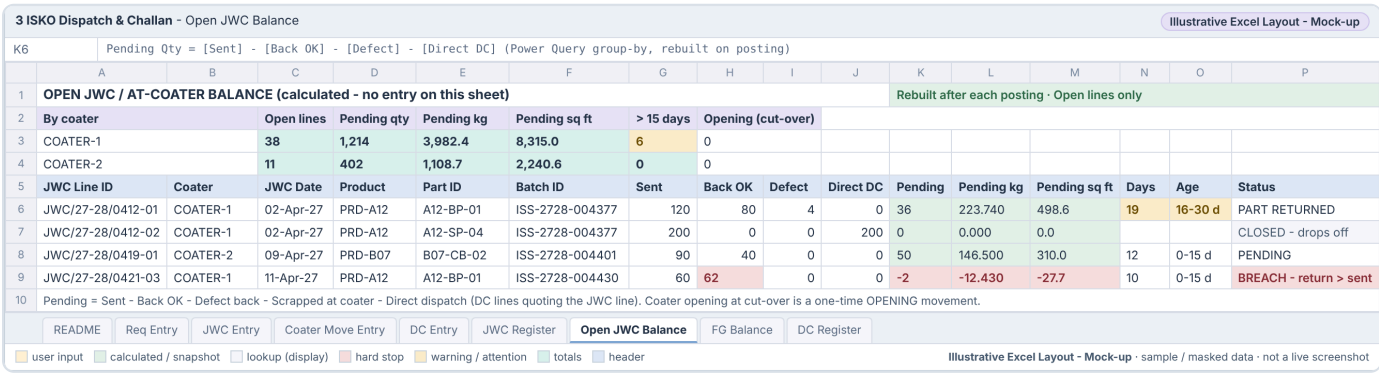
13.4 Rework

Rejections are cases: REJECT (from a source Prod ID) → REWORK-OK (quantity returns to WIP at the detection stage) or REWORK-SCRAP. The original production line is never edited; QC-OK quantity = produced − rejected + reworked OK.

13.5 Transaction-calculated WIP (no counts)

For each Product + Part + Stage on the part's route: **WIP = completed at this stage − completed at the next route stage − sent on JWC from this stage − transferred to FG − rejected at this stage + rework OK returned to this stage.** Opening WIP at cut-over is a one-time approved migration entry derived from closing book figures (D-08), not a count. Negative WIP is an exception (route or posting error), never a reason to post a count. The route master is new and needs Production sign-off (D-05).

14. JWC, subcontractor and dispatch controls



3 ISKO Dispatch & Challan - Open JWC Balance — Illustrative Excel Layout - Mock-up

K6: `Pending Qty = [Sent] - [Back OK] - [Defect] - [Direct DC] (Power Query group-by, rebuilt on posting)`

OPEN JWC / AT-COATER BALANCE (calculated - no entry on this sheet) — Rebuilt after each posting · Open lines only

By coater	Open lines	Pending qty	Pending kg	Pending sq ft	> 15 days	Opening (cut-over)
COATER-1	38	1,214	3,982.4	8,315.0	6	0
COATER-2	11	402	1,108.7	2,240.6	0	0

JWC Line ID	Coater	JWC Date	Product	Part ID	Batch ID	Sent	Back OK	Defect	Direct DC	Pending	Pending kg	Pending sq ft	Days	Age	Status
JWC/27-28/0412-01	COATER-1	02-Apr-27	PRD-A12	A12-BP-01	ISS-2728-004377	120	80	4	0	36	223.740	498.6	19	16-30 d	PART RETURNED
JWC/27-28/0412-02	COATER-1	02-Apr-27	PRD-A12	A12-SP-04	ISS-2728-004377	200	0	0	200	0	0.000	0.0			CLOSED - drops off
JWC/27-28/0419-01	COATER-2	09-Apr-27	PRD-B07	B07-CB-02	ISS-2728-004401	90	40	0	0	50	146.500	310.0	12	0-15 d	PENDING
JWC/27-28/0421-03	COATER-1	11-Apr-27	PRD-A12	A12-BP-01	ISS-2728-004430	60	62	0	0	-2	-12.430	-27.7	10	0-15 d	BREACH - return > sent

Pending = Sent - Back OK - Defect back - Scrapped at coater - Direct dispatch (DC lines quoting the JWC line). Coater opening at cut-over is a one-time OPENING movement.

README | Req Entry | JWC Entry | Coater Move Entry | DC Entry | JWC Register | **Open JWC Balance** | FG Balance | DC Register

user input · calculated / snapshot · lookup (display) · hard stop · warning / attention · totals · header

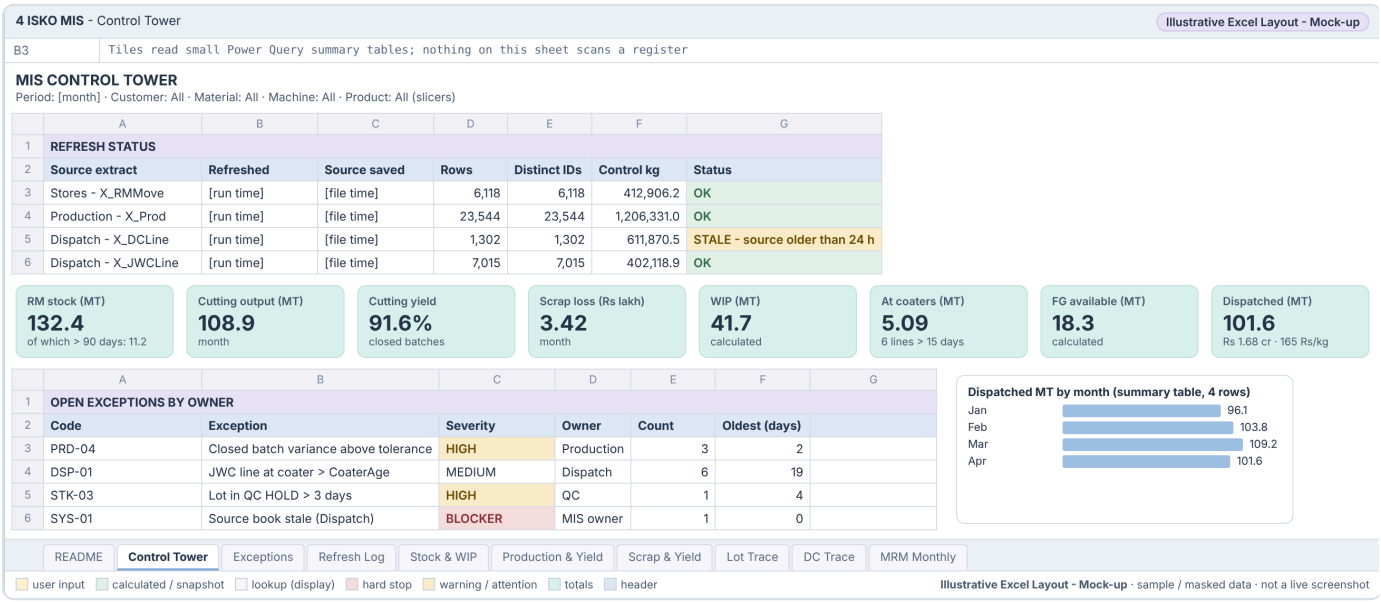
Illustrative Excel Layout - Mock-up · sample / masked data · not a live screenshot

Illustrative Excel Layout - Mock-up: Open JWC / coater balance (sample data)

- **Pending at coater** per JWC line = sent − BACK-OK − DEFECT-BACK − scrapped at coater − direct dispatch (DC lines quoting the JWC line). Coater opening at cut-over is a one-time OPENING movement confirmed from the coater's statement (not a count). Returns and direct dispatch above pending are blocked.
- **Return / debit logic:** every defect carries *whose fault* and a *decision* (rework at coater, re-send, scrap, debit coater); a debit needs an amount; a case closes when a later movement quotes it.
- **Cost:** coater rate and basis come from an effective-dated rate table at the JWC date and are frozen on the line (today they re-price history).
- **Customer UOM:** the dispatch UOM comes from a Customer-Product UOM table, else the customer default; QTY / NOS / SET use quantity, KG uses weight, TON uses weight ÷ 1,000. Blank UOM blocks posting (8 of 15 customers are blank today, D-18).
- **FG availability:** FG = opening + coater BACK-OK + uncoated final-stage completion − dispatch from factory − final-QC rejection ± authorised FG adjustment. A DC line from the factory cannot exceed FG available; from a coater it cannot exceed JWC pending.
- **Traceability:** DC line → JWC line → batch → lot → GRN → PO, using IDs on every row.

Risk R-14 - statutory documents. Removing print removes the only place the ERP produces the delivery challan and job-work challan documents. Goods movements still need those documents. Management must decide how they are produced (Decision D-03) before go-live; the ERP will keep the challan number, lines and status as the record.

15. Management dashboards and exception reporting



Illustrative Excel Layout - Mock-up: MIS Control Tower (sample data)

PivotTables and small summary tables built by Power Query are preferred over formula dashboards: they work in Excel 2016, refresh in one step and do not scan registers on every edit. Charts read only the summaries (hundreds of rows). Filters: period, customer, material, machine, product (slicers).

15.1 KPI definitions

KPI	Definition	Source table	Filters	Action owner
RM stock (kg, Rs) and RM > 90 days	Sum of Available kg / Value on open lots; part older than 90 days	X_LotBal (Stores Lot Balance)	Material, grade, thickness	Stores head
RM issued (kg)	Sum of -Stock Effect kg on ISSUE minus returns in period	X_RMMove	Month, machine, material	Stores head
Cutting output (kg)	Sum Output kg of CUTTING lines	X_Prod	Month, machine, product	Production head
Cutting yield %	Cut output kg / net batch kg of closed batches	X_BatchBal	Month, machine, material	Production head
Batch variance over tolerance (count, kg, Rs)	Closed batches with |var| > tolerance	X_BatchBal	Month, machine	Production head
Scrap kg and loss Rs	Cut scrap + stores scrap + rejection scrap; x lot rate	Q_MaterialFlow	Month, process, material	Production head
In-process rejection %	REJECT kg / output kg	X_RejRework, X_Prod	Month, stage, reason	QC
WIP (kg, Rs) and oldest WIP	Transaction-calculated WIP balance	X_WIP	Product, stage	Production head
At coater (kg) and lines > age limit	Pending kg on open JWC lines; count over CoaterAge days	X_JWCBal	Coater	Dispatch head
Powder-coating cost (Rs, Rs/kg)	Sum PC Cost Rs / sent kg	X_JWCLine	Month, coater	Dispatch head / Finance
FG available (qty, kg)	Transaction-calculated FG balance	X_FGBal	Product, customer	Dispatch head
Dispatched (kg, Rs) and realisation Rs/kg	Sum Wt kg / Value Rs of DC lines (SALE)	X_DCLine	Month, customer, product	MD

KPI	Definition	Source table	Filters	Action owner
Requirement pending and on-time %	Required - dispatched against Req ID; lines dispatched by required date	X_Requirement, X_DCLine	Customer, product	Dispatch head
Overdue PO lines	Pending qty > 0 and due date passed	X_POStatus	Supplier	Purchase
Consumable spend (Rs)	Value of consumable issues net of returns	X_ConsMove	Month, where used	Stores head
Open exceptions (BLOCKER / HIGH) and age	Count and days open	tblException	Owner, severity	Each owner; MD reviews
Data freshness	Refresh Log status per source	tblRefreshLog	-	MIS owner

15.2 Exception catalogue

Code	Exception	Severity	Owner
STK-01	Lot or consumable with negative available balance	BLOCKER	Stores
STK-02	Issue posted from a lot not issue-eligible (QC / excess)	BLOCKER	Stores
STK-03	Lots in QC PENDING / HOLD more than 3 days	HIGH	QC
STK-04	Excess over PO without decision	HIGH	Purchase
STK-05	Adjustment without reason, reference or independent approver	BLOCKER	Stores / Finance
PUR-01	PO lines past due date with pending quantity	MEDIUM	Purchase
PUR-02	GRN rate differs from PO rate beyond tolerance	MEDIUM	Purchase
PRD-01	Cutting line without batch (from go-live)	BLOCKER	Production
PRD-02	Batch over-consumed (output + scrap > net batch kg + tolerance)	BLOCKER	Production
PRD-03	Batch open more than 7 days	MEDIUM	Production
PRD-04	Closed batch variance above tolerance	HIGH	Production
PRD-05	Part material/thickness differs from batch lot	HIGH	Production
PRD-06	Negative WIP for product/part/stage	HIGH	Production
PRD-07	Rejection rework case open more than 7 days	MEDIUM	QC
DSP-01	JWC line pending at coater beyond CoaterAge days	MEDIUM	Dispatch
DSP-02	Return or direct dispatch above JWC pending (negative pending)	BLOCKER	Dispatch
DSP-03	DC line from factory above FG available (negative FG)	BLOCKER	Dispatch
DSP-04	DC line with blank UOM or customer without dispatch UOM	BLOCKER	Dispatch
DSP-05	JWC line without batch (from go-live)	HIGH	Dispatch
DSP-06	Coater defect case without decision	MEDIUM	Dispatch
MST-01	Posted ID not found in master (product, part, machine, person)	HIGH	Master owner

Code	Exception	Severity	Owner
MST-02	BOM edition overdue for review / requirement product with no BOM	MEDIUM	Production engineering
SYS-01	Source book missing, stale or wrong version	BLOCKER	MIS owner
SYS-02	Duplicate IDs in an extract	BLOCKER	MIS owner
SYS-03	Closed-period control totals changed	BLOCKER	Finance
SYS-04	Reversal quoting a missing or already-reversed ID	HIGH	Book owner

Exceptions are one row per open problem with an owner and age, rebuilt on each MIS refresh. Hard stops in the books should keep BLOCKER exceptions at zero; any BLOCKER appearing means a control was bypassed and is reviewed the same day.

16. Data migration, archival and go-live

16.1 History and archival policy

Option	For	Against
Keep 100,000+ rows in one live register	Simple search	Slower open/save, larger backups, more risk per file
FY-wise files (recommended)	Each register holds one year (Production ≈ 23k rows/yr) - well inside the 100,000 target; clean statutory numbering	Needs carry-forward of open balances
Half-year files	Fallback if a PC fails the test	More files

Policy: at FY close, archive the four books read-only (file attribute and a separate folder; a checksum is stored in the Close Log). Carry forward as OPENING rows: open lots (kg, sheets, rate), open batches, WIP by product/part/stage, at-coater balances by JWC line, FG by product, open POs, open requirements. IDs include the FY so they never collide. MIS reads current and archived years through a folder query (*Folder.Files*), so history stays searchable and trace queries cross years. Nothing is deleted because it is old.

16.2 Migration and go-live steps

1. **Freeze and back up** the current books (read-only archive with checksums).
2. **Clean masters:** one product master (merge 249/269/278 rows), Person IDs, Customer UOM (8 blanks), coater rates with effective dates, route master.
3. **Map** current columns to target columns (Appendix B); history either migrated as values with *Entry Source = MIGRATION* or left in the archived books (D-17).
4. **Opening balances at cut-over** (D-08): RM lots, consumables, WIP, at coater, FG, open POs, open requirements - taken from the closing figures of the current books, reviewed by Finance and posted as OPENING movements. This is a one-time migration, not a count module.
5. **Validate:** totals by lot, batch, coater and product match the approved closing figures; zero BLOCKER exceptions.
6. **Parallel run** one month (March 2027 suggested) with daily comparison of key totals.
7. **Cut-over** on 1 April 2027 (start of FY 2027-28) so numbering, archival and opening balances align.
8. **Hypercare** for two months with daily exception review.

17. Stress-test and acceptance plan

Planned Validation - Not Yet Tested. No Excel timing has been measured. The only measurement made is file size: a synthetic register of 100,000 rows × 32 columns written as values by Python (openpyxl) is **14.8 MB** (10k: 1.5 MB; 25k: 3.7 MB; 50k: 7.4 MB). Today's Production book is 7.2 MB at 12,538 rows; growing the current design linearly would give roughly 58 MB at 100,000 rows (estimate).

Test environment to record: PC model, CPU, RAM, disk type, Windows version, Excel version/build and bitness, calculation mode (automatic), add-ins, file location (local / network), file size, row counts per register, elapsed time (median of three), memory/CPU from Task Manager, errors, pass/fail. Run on the **slowest** PC that will be used.

Volumes: each major register at 10,000 / 25,000 / 50,000 / 100,000 rows of synthetic or masked data, and a combined system of 400,000+ register rows across the four books with masters and extracts.

17.1 Provisional acceptance thresholds

Operation	Target at 10k-25k rows	Target at 100k rows / 400k system	How measured
Open workbook (largest book)	<= 10 s	<= 20 s	Stopwatch from double-click to ready (status bar 'Ready').
Save workbook	<= 5 s	<= 15 s	Ctrl+S to 'Ready'.
Recalculate after typing one pad cell	<= 0.5 s	<= 1 s	No visible 'Calculating' delay; Application.CalculateFull time recorded separately.
Full recalculation (Ctrl+Alt+F9)	<= 5 s	<= 15 s	VBA timer or stopwatch (diagnostic only).
Filter / search register by ID, batch, material, part, customer, date	<= 2 s	<= 3 s	AutoFilter / Ctrl+F on the Table.
Post 20 pad lines (macro)	<= 5 s	<= 10 s	Button click to confirmation.
Post 20 pad lines (manual fallback)	<= 2 min user time	<= 2 min user time	Observed user time incl. control-total check.
Refresh balances after posting	<= 10 s	<= 20 s	Data > Refresh All on the book.
Run stock / batch / dispatch checks	<= 10 s	<= 20 s	Refresh of check queries.
Refresh MIS (all extracts, 400k+ source rows)	<= 90 s	<= 180 s	Refresh All to Refresh Log complete.
Open / update dashboard after refresh	<= 3 s	<= 5 s	Slicer or period change to redraw.
Close and reopen after save - integrity	Totals identical	Totals identical	Control totals before close = after reopen.
Detect missing / mismatched source and duplicate import	100% of seeded faults	100% of seeded faults	Seeded fault tests in the checklist.
File size per book	<= 10 MB	<= 25 MB	File properties. Indicative: a 100,000 × 32 values register written by openpyxl measured 14.8 MB.

These are **targets requiring validation on the company's computers.** After the pilot, the MIS owner records actual medians; Management keeps or tightens thresholds that pass with margin, and for any that fail chooses: shorter archive period, fewer extract columns, Data Model instead of sheet tables, or a Phase 2 database.

17.2 Test matrix

ID	Operation	Volumes	Steps	Pass condition
T-01	Open workbook	10k / 25k / 50k / 100k per register; 400k+ system	Cold start, close other apps, open largest book three times, record median	Within threshold
T-02	Save workbook	Same	Change one pad cell, save three times, record median	Within threshold
T-03	Recalculate after a normal entry	Same	Type 20 pad rows; record any delay; run full recalc diagnostic	Within threshold; no 'Not responding'
T-04	Filter / search	Same	Filter by ID, batch, material, part, customer, date range	Within threshold
T-05	Post a transaction	Same	Post 1, 20 and 200 pad rows (macro and manual)	IDs unique, values frozen, control totals match
T-06	Stock / batch / dispatch checks	Same	Seed 10 over-issue, 10 over-consumption, 10 over-dispatch attempts	100% blocked at entry; flags shown
T-07	Refresh MIS extracts	400k+ combined	Refresh All; read Refresh Log	Within threshold; all rows OK; control totals equal
T-08	Open / update dashboard	400k+ combined	Change period and filters	Within threshold
T-09	Close and reopen after save	Same	Record totals, close, reopen, compare	Identical
T-10	Missing / renamed / moved source file	Any	Rename Stores book; refresh MIS	Refresh Log MISSING FILE; dashboard banner; no silent zeros
T-11	Stale source	Any	Edit Stores without saving; refresh MIS	Refresh Log STALE (modified time older than MaxAgeHours) where applicable
T-12	Duplicate import	Any	Duplicate 5 rows in a source register copy; refresh	SYS-02 raised; extract status DUPLICATES
T-13	Stale balance guard	Any	Paste a posting without refreshing; try to enter an issue	Entry blocked with STALE flag
T-14	Reversal	Any	Reverse an issue, a production line and a DC line	Balances restored; original unchanged; reversal of reversal blocked
T-15	Period lock	Any	Enter a date <= LockDate	Rejected
T-16	Closed-period change detection	Any	Edit a value in a closed month (copy)	SYS-03 raised
T-17	Excel version compatibility	Any	Open pilot books on every target PC version	No #NAME?, no compatibility warnings

Sign-off: MIS owner (execution), department head of each book (functional), Finance (totals), MD (go-live).

18. Risk register

ID	Risk	Likelihood	Impact	Preventive control	Detection	Owner	Fallback
R-01	Excel version on some PCs lacks functions or Power Query	Medium	High	Excel 2016 function set; version audit before build (D-01)	Go-live checklist; formula errors in pilot	IT / MIS owner	Install a common Excel version; or manual paste-values transfer
R-02	Macros not permitted	Medium	Medium	Macro-free core design; VBA only for convenience posting	Decision D-02	MD	Manual paste-values posting procedure with control totals
R-03	Two people edit the same book at once (shared folder)	High	High	One owner per book; open read-only for others; book-in-use note in README	Excel 'locked for editing' message; Refresh Log	Book owners	Restore last backup; re-post from pad copies
R-04	User pastes over validation or formulas (no sheet protection)	High	Medium	Registers hold values; pads small; formula-integrity check; training	Pad FORMULA CHANGED flag; MIS exceptions	Book owners	Rebuild pad from template file
R-05	Balances stale after posting (refresh forgotten)	Medium	High	Row-count stale guard blocks entry until refresh	STALE flag on pad header	Stores / Production owners	Refresh; re-validate pad rows
R-06	100,000-row target not met on company PCs	Medium	High	Values-only registers; FY archive; stress test before go-live	Benchmark plan Section 17	MIS owner	Shorter archive period (half-year files) or move registers to Access/SQL Express in Phase 2
R-07	Migrated history inconsistent (blank QC, lots without PO, balancing lines)	High	Medium	Migrate history as values with Entry Source = MIGRATION; open balances from approved closing figures	Opening balance reconciliation sign-off	Finance	Keep old books read-only as archive
R-08	WIP/FG rules mis-specified (routes, FG point)	Medium	High	Route master and FG rule signed off (D-05, D-06); pilot	Negative WIP/FG flags	Production head	Adjust route; authorised FG adjustment
R-09	Imported file missing, moved or renamed	Medium	Medium	Typed folder parameter; fixed file names; README	Refresh Log MISSING FILE	MIS owner	Re-point parameter; previous extract kept
R-10	Lost or corrupted file	Low	High	Daily dated backups; weekly off-machine copy; registers append-only	Restore test monthly	IT	Restore and re-post day's pad copies
R-11	Masters edited carelessly (renamed or deleted rows)	Medium	Medium	IDs not names; never delete; Active flag; Change Log	Exception: posted ID not in master	Master owners	Re-add master row
R-12	Users resist entry pad + posting	Medium	Medium	Same columns and colours as today; short training; pilot champions	Pilot feedback	Department heads	Adjust pad layout
R-13	Closed-period data changed	Low	High	LockDate; Close Log totals	BLOCKER exception CLOSED PERIOD CHANGED	Finance	Restore period from close copy
R-14	Statutory delivery / job-work challan document no longer produced after print removal	High	High	Decision D-03 before go-live (e.g. from accounting software or pre-numbered books)	Go-live checklist	MD / Accounts	Keep current challan practice outside the ERP
R-15	Exceptions ignored	Medium	Medium	Exception owners and ageing on Control Tower	Days-open on tblException	MD	Weekly exception review in MRM

19. Implementation roadmap

Gate / phase	When (proposed)	Content	Sign-off
Gate 0 - Design approval	Nov 2026	Approve this plan; take decisions D-01 to D-18; freeze scope.	MD, department heads
Build 1 - Templates and masters	Dec 2026	Masters with IDs; Settings; pad + register templates; synthetic data generator.	MIS owner / implementer
Stress test 1 (synthetic data)	Dec 2026 - Jan 2027	Run Section 17 at 10k-100k on the company's slowest target PC.	MIS owner
Pilot 1 - Stores	Jan 2027	PO, GRN, QC, RM movements, consumables, adjustments with real users on a copy.	Stores head
Pilot 2 - Production	Feb 2027	Production entry, rejection/rework, batch balance and close, WIP calculation.	Production head
Pilot 3 - Dispatch	Feb 2027	Requirement, JWC, coater movements, FG balance, DC.	Dispatch head
MIS reconciliation	Mar 2027	Extracts, Refresh Log, exceptions, Control Tower; numbers reconciled to pilots.	MIS owner / Finance
Stress test 2 + parallel run	Mar 2027	Full-volume test; one month parallel with old books; closing balances agreed.	All heads, Finance
Go-live gate	1 Apr 2027 (FY 2027-28)	Opening balances posted; old books archived read-only.	MD
Hypercare	Apr - May 2027	Daily exception review; weekly performance log.	MIS owner

Each gate passes only when its tests pass and the department head signs. A failed stress test stops the next pilot, not the whole programme.

20. Management decisions required

ID	Decision required	Why it matters	Area
D-01	Excel version and bitness on every PC that will use the books; whether Excel 2021/365-only functions may be used.	Determines function set and Power Query availability.	Platform
D-02	Are macros (a signed 'Post' button) allowed?	Macro posting is faster and enforces hard stops; fallback is manual paste-values.	Platform
D-03	How will the statutory delivery challan and job-work challan documents be produced once print is removed?	Goods movement documents are still required outside the ERP.	Compliance
D-04	Must bending/fabrication lines carry the Batch ID (full lot traceability) or only cutting?	Affects entry effort and trace depth.	Process
D-05	Approve the Stage & Route master (stage sequence per part; where WIP ends).	Required for transaction-calculated WIP.	Process
D-06	Which event puts goods into FG: coater BACK-OK; final-stage completion for uncoated items; assemblies?	Required for FG balance and the DC hard stop.	Process
D-07	How does the shop floor identify the batch without a printed issue slip?	Print removed; dropdown by machine proposed.	Process
D-08	Cut-over date and approved opening balances (RM lots, WIP, at coater, FG) from closing book figures.	Opening balances are a one-time migration entry, not a count module.	Finance
D-09	Machine downtime and utilisation in Phase 1 or Phase 2?	Machine log unused for six months; capacities assumed.	Scope
D-10	Adjustment approval matrix (value limits, approvers).	Authorised adjustment movement.	Controls
D-11	Tolerances: theoretical weight, rate variance, batch variance, coater age, WIP age.	Hard-stop and warning limits.	Controls
D-12	Must DC lines quote the customer Req ID?	Enables exact requirement closure.	Process
D-13	Archive policy: FY-wise files (recommended) and retention period.	Keeps registers within the tested volume.	Data
D-14	Folder location, single-editor rule per book, backup owner.	Multi-user risk.	IT
D-15	Company legal name to show on books (files say 'ISKO ENGINEERING LIMITED').	Consistency.	Admin
D-16	Approve provisional performance thresholds (Section 17).	Acceptance criteria.	Platform
D-17	Migrate Apr 2026 - cut-over history into new registers (values, MIGRATION tag) or keep only in archived old books?	Trace and MIS history.	Data
D-18	Dispatch UOM for the 8 customers with a blank UOM.	Hard stop on DC lines.	Master data

Top three: D-01/D-02 (platform: Excel version and macro policy), D-03 (statutory challan documents after print removal), D-05/D-06/D-08 (WIP/FG rules and cut-over opening balances).

20.1 Business requirements - decisions

Area	Requirement	Decision	How
A Stores & Purchase	PO created independently (supplier, dates, terms, location, qty/unit/rate/value, receipt progress, pending) - no requisition reference	Simplify	Header + line registers; progress in PO Status view.
A Stores & Purchase	No PO print / layout / macro / send-mail	Remove from Phase 1	1.3 Print PO, mail lists and names removed.
A Stores & Purchase	GRN: sheets, kg, size, bill, PO link, theoretical kg, QC accepted/hold/rejected, rejected kg/sheets, excess, decision tracking	Keep	Receipt + QC decision log.

Area	Requirement	Decision	How
A Stores & Purchase	Issue only after QC accepted; block held/rejected/invalid lots	Keep	Hard stop via Issue Eligible.
A Stores & Purchase	RM lots, lot-level valuation, traceability receipt → issue → production → dispatch	Keep	Lot ID permanent; batch carries lot.
A Stores & Purchase	RM issue and return by batch/issue ID with lot selection/override, material/grade match, kg and sheets, rate/value, issued-to, machine, purpose, audit notes	Simplify	tblRMMove.
A Stores & Purchase	Hard-stop insufficient stock and invalid returns; show max available kg and sheets; no negative stock	Keep	Pad hard stop + stale guard.
A Stores & Purchase	Usable returns, offcuts, damaged, scrap, adjustments tracked separately	Simplify	Movement types; DAMAGED bucket; authorised adjustment.
A Stores & Purchase	Consumables: receipt, issue, returns, authorised adjustment, reorder alert, value, min level; no count	Simplify	Two registers + Consumable Balance.
A Stores & Purchase	Distinguish movements from summaries; reports calculated from transactions	Add	Register / balance / report layers.
B Production	Production entry tied to batch and lot with date, shift, machine, stage, product, part, qty, weight, output, scrap, operator, QC	Keep	tblProd.
B Production	Stages: cutting, bending, fabrication, rework/rejection (+ confirmed others)	Keep	Stages in files: Cutting, Bending, Fabrication; rejections also at Powder Coating and Final.
B Production	Batch balance: issued, returns, output, scrap, variance kg/%/value, status, lines	Keep	Batch Balance + Batch Close.
B Production	Prevent duplicated or invalid postings	Add	Dup Key, batch hard stop.
B Production	Rejection/rework linked to original record, batch and lot; original auditable	Simplify	Event register.
B Production	WIP by item/part/stage from posted transactions only	Add	WIP Balance (replaces 2.4 WIP Count).
B Production	Machine downtime/logging	Decide with Management	Default: Phase 2 (D-09).
B Production	BOM/product/part masters with drawing, revision, material, size, blank weight, qty/set, variant, effective date	Combine	Single product master; Part ID.
C Dispatch & Job Work	Customer requirement with versions, qty, dates, priority, status, dispatched, pending	Keep	Pending via Req ID (D-12).
C Dispatch & Job Work	JWC header and lines with full traceability, cost, returns, pending, ageing	Simplify	Lines register + Open JWC Balance.
C Dispatch & Job Work	Coater opening, returns/rejections, re-send, responsibility, debit, closure, outstanding	Simplify	tblCoaterMove.
C Dispatch & Job Work	FG balance from transactions; no FG count	Add	FG Balance (replaces 3.8).
C Dispatch & Job Work	DC header and lines with transport refs, status, product/batch, qty, UOM, weight, rate/value, links to requirement and JWC	Simplify	DC document pad.
C Dispatch & Job Work	Customer-specific dispatch UOM (QTY / NOS / KG)	Keep	Customer-Product UOM table; snapshot.
C Dispatch & Job Work	No JWC/DC print or mail	Remove from Phase 1	3.4, 3.11, Mail Coater, mail lists removed.
C Dispatch & Job Work	Prevent duplicate dispatch, dispatch above FG, invalid JWC links	Keep	Hard stops.
D MIS	MIS is a reporting layer, no re-entry	Simplify	Only filter/parameter cells are typed in MIS.
D MIS	Review Control Tower ... Scrap Dashboard	Simplify	See Section 8 decisions.
D MIS	Single source of truth per KPI and refresh mechanism	Add	KPI table Section 15; Refresh Log.

Area	Requirement	Decision	How
D MIS	Visible refresh date/time, source file/version, period, stale/missing warning	Add	Refresh status panel.
D MIS	Director KPIs and exceptions only where they support a decision	Simplify	17 KPIs, exception catalogue.
D MIS	Drill-through from summary to source record ID	Simplify	Source Row ID on every extract; trace queries.

21. Appendices

Appendix A - Full sheet inventory (measured)

Workbook	Sheet	State	Used range	Cells	Formula cells	Table	Data rows	Validations	CF rules	Protected	Freeze	Decision
1 Stores	START	visible	A1:E55	132	26	-	-	0	0	Yes	No	Simplify
1 Stores	1.1 Requisition	visible	A1:AB154	4,263	2,401	tblRequisition	0	9	2	Yes	Yes	Removed from Phase 1
1 Stores	1.2 Purchase Order	visible	A1:AH206	6,953	5,632	tblPOLine	3	12	9	Yes	Yes	Simplify
1 Stores	1.3 Print PO	visible	A1:K500	2,156	31	-	-	2	0	Yes	No	Removed from Phase 1
1 Stores	1.4 RM Receipt GRN	visible	A1:AN352	13,325	5,918	tblRMReceipt	187	15	2	Yes	Yes	Simplify
1 Stores	1.5 Consumable Receipt	visible	A1:S417	6,650	3,305	tblConsReceipt	202	5	2	Yes	Yes	Simplify
1 Stores	1.6 RM Issue & Return	visible	A1:AD3397	101,864	57,683	tblRMIssue	3,023	12	2	Yes	Yes	Simplify
1 Stores	1.6b Print RM Issue Slip	visible	A1:H30	82	31	-	-	1	0	Yes	No	Removed from Phase 1
1 Stores	1.7 Consumable Issue	visible	A1:V866	18,152	10,346	tblConsIssue	128	7	2	Yes	Yes	Simplify
1 Stores	1.8 Print Issue Slip	visible	A1:H413	1,217	79	-	-	2	0	Yes	No	Removed from Phase 1
1 Stores	1.9 Stock Adjustment	visible	A1:P251	3,987	1,483	tblAdjust	217	7	1	Yes	Yes	Simplify
1 Stores	1.10 Stock Count	visible	A1:I450	4,035	447	tblCount	273	4	0	Yes	Yes	Removed from Phase 1
1 Stores	1.11 RM Stock	visible	A1:U407	6,202	395	-	-	1	2	Yes	No	Simplify
1 Stores	1.12 Consumable Stock	visible	A1:K406	3,220	19	-	-	1	2	Yes	No	Simplify
1 Stores	1.13 Re-order Alert	visible	A1:H500	3,087	7	-	-	1	2	Yes	Yes	Combine
1 Stores	1.14 Stores Dashboard	visible	A1:L433	2,294	180	-	-	1	2	Yes	No	Convert to report
1 Stores	M_Supplier	visible	A1:L88	1,034	0	tblSupplier	49	2	1	Yes	Yes	Keep
1 Stores	M_Material	visible	A1:M128	1,640	992	tblMaterial	69	1	0	Yes	Yes	Simplify
1 Stores	M_Grade	visible	A1:C34	98	0	tblGrade	10	0	0	Yes	Yes	Keep
1 Stores	M_Consumable	visible	A1:O248	3,692	2,684	tblConsumable	97	3	0	Yes	Yes	Simplify
1 Stores	M_Cons_Category	visible	A1:B40	78	0	tblConsCat	16	0	0	Yes	Yes	Keep
1 Stores	M_Product	visible	A1:C403	1,205	0	tblJob	249	1	0	Yes	Yes	Combine
1 Stores	M_WhereUsed	visible	A1:B51	100	0	tblWhere	17	1	0	Yes	Yes	Keep
1 Stores	M_Person	visible	A1:C78	229	0	tblPerson	24	0	0	Yes	Yes	Combine

Workbook	Sheet	State	Used range	Cells	Formula cells	Table	Data rows	Validations	CF rules	Protected	Freeze	Decision
1 Stores	M_Reason	visible	A1:C64	188	0	tblReason	14	2	0	Yes	No	Simplify
1 Stores	M_Contacts	visible	A1:F44	254	0	tblContact	10	1	1	Yes	Yes	Removed from Phase 1
1 Stores	Settings	visible	A1:C37	99	4	-	-	0	0	Yes	No	Simplify
1 Stores	Lists	hidden	A1:AU3026	10,469	35	-	-	0	0	Yes	No	Simplify
2 Production	START	visible	A1:E43	91	18	-	-	0	0	Yes	No	Simplify
2 Production	2.1 Production Entry	visible	A1:AK12542	464,012	203,113	tblProduction	11,406	20	1	Yes	Yes	Simplify
2 Production	2.2 Rejection & Rework	visible	A1:AA267	7,160	1,842	tblRejection	68	18	1	Yes	Yes	Simplify
2 Production	2.3 Machine Log	visible	A1:K204	2,225	601	tblMachineLog	0	4	0	Yes	Yes	Removed from Phase 1
2 Production	2.4 WIP Count	visible	A1:I2100	18,885	2,097	tblWIPCount	1,503	7	0	Yes	Yes	Removed from Phase 1
2 Production	2.5 Daily Production	visible	A1:U3000	14,234	10	-	-	1	0	Yes	No	Simplify
2 Production	2.6 Machine Report	visible	A1:N420	3,056	47	-	-	0	0	Yes	No	Removed from Phase 1
2 Production	2.7 Production Dashboard	visible	A1:L431	1,386	77	-	-	1	2	Yes	No	Convert to report
2 Production	2.8 Batch Balance	visible	A1:AF2000	19,737	19	-	-	3	3	Yes	Yes	Keep
2 Production	M_Machine	visible	A1:G32	212	0	tblMachine	8	2	0	Yes	Yes	Keep
2 Production	M_Worker	visible	A1:D77	302	0	tblOperator	24	3	0	Yes	Yes	Keep
2 Production	M_Product	visible	A1:G400	2,787	792	tblProduct	269	2	0	Yes	Yes	Keep
2 Production	M_Part	visible	A1:I3319	29,855	6,630	tblPart	122	1	0	Yes	Yes	Simplify
2 Production	M_BOM	visible	A1:V577	12,652	2,292	tblBOM	246	2	0	Yes	Yes	Keep
2 Production	M_BOM_Model	visible	A1:I47	407	215	tblBOMModel	13	0	0	Yes	Yes	Keep
2 Production	M_Reason	visible	A1:C84	248	0	tblReason	27	1	0	Yes	No	Keep
2 Production	M_Stage	visible	A1:B44	86	0	tblStage	7	0	0	Yes	No	Combine
2 Production	M_Contacts	visible	A1:F42	242	0	tblContact	8	1	1	Yes	Yes	Removed from Phase 1
2 Production	Settings	visible	A1:C16	40	3	-	-	0	0	Yes	No	Simplify
2 Production	D_Material	hidden	A1:F70	416	0	qMaterial	69	0	0	No	No	Simplify
2 Production	D_Lots	hidden	A1:K188	1,881	0	qLots	187	0	0	No	No	Combine
2 Production	D_Issues	hidden	A1:T3023	52,091	0	qIssues	3,022	0	0	No	No	Simplify
2 Production	Lists	hidden	A1:Z3269	10,948	19	-	-	0	0	Yes	No	Simplify
3 Dispatch & Challan	START	visible	A1:E41	99	19	-	-	0	0	Yes	No	Simplify
3 Dispatch & Challan	3.1 Customer Requirement	visible	A1:O111	1,643	643	tblRequirement	7	4	0	Yes	Yes	Keep

Workbook	Sheet	State	Used range	Cells	Formula cells	Table	Data rows	Validations	CF rules	Protected	Freeze	Decision
3 Dispatch & Challan	3.2 JWC Header	visible	A1:Q331	5,596	2,617	tblJWCHead	152	5	1	Yes	Yes	Simplify
3 Dispatch & Challan	3.3 JWC Lines	visible	A1:AG4034	133,059	88,661	tblJWCLine	3,508	7	2	Yes	Yes	Simplify
3 Dispatch & Challan	3.4 Print JWC	visible	A1:K1000	6,485	33	-	-	2	0	Yes	No	Removed from Phase 1
3 Dispatch & Challan	3.5 Coater Opening	visible	A1:H104	818	100	tblCoaterOpening	0	3	0	Yes	Yes	Simplify
3 Dispatch & Challan	3.6 Coater Return	visible	A1:X104	2,451	1,201	tblCoaterReturn	0	10	1	Yes	Yes	Simplify
3 Dispatch & Challan	3.7 At Coater	visible	A1:AE3000	50,705	27	-	-	2	4	Yes	No	Simplify
3 Dispatch & Challan	3.8 FG Count	visible	A1:G742	5,182	739	tblFGCount	450	5	0	Yes	Yes	Removed from Phase 1
3 Dispatch & Challan	3.9 DC Header	visible	A1:Q455	7,710	3,159	tblDCHead	184	4	1	Yes	Yes	Simplify
3 Dispatch & Challan	3.10 DC Lines	visible	A1:Y1296	32,353	21,329	tblDCLine	626	9	2	Yes	Yes	Simplify
3 Dispatch & Challan	3.11 Print DC	visible	A1:K1000	5,459	33	-	-	2	0	Yes	No	Removed from Phase 1
3 Dispatch & Challan	3.12 Customer Summary	visible	A1:N420	1,989	52	-	-	0	0	Yes	No	Convert to report
3 Dispatch & Challan	3.13 Dispatch Dashboard	visible	A1:L435	1,415	95	-	-	1	2	Yes	No	Convert to report
3 Dispatch & Challan	M_Customer	visible	A1:L40	458	0	tblCustomer	15	2	1	Yes	Yes	Keep
3 Dispatch & Challan	M_Vendor	visible	A1:K28	288	0	tblVendor	5	3	1	Yes	Yes	Simplify
3 Dispatch & Challan	M_Product	visible	A1:I400	3,584	396	tblProduct	278	3	1	Yes	Yes	Combine
3 Dispatch & Challan	M_Worker	visible	A1:C44	128	0	tblWorker	0	2	0	Yes	No	Combine
3 Dispatch & Challan	M_Contacts	visible	A1:F43	248	0	tblContact	9	1	1	Yes	Yes	Removed from Phase 1
3 Dispatch & Challan	Settings	visible	A1:C29	71	5	-	-	0	0	Yes	No	Simplify
3 Dispatch & Challan	D_ProdLots	hidden	A1:G4321	30,247	0	qProdLots	4,320	0	0	No	No	Combine
3 Dispatch & Challan	D_ProdBatch	hidden	A1:M2	19	0	qProdBatch	0	0	0	No	No	Combine
3 Dispatch & Challan	Lists	hidden	A1:AM892	6,053	29	-	-	0	0	Yes	No	Simplify
4 MIS	START	visible	A1:E31	77	13	-	-	0	0	Yes	No	Simplify
4 MIS	4.1 Control Tower	visible	A1:R479	3,991	131	-	-	6	5	Yes	Yes	Simplify
4 MIS	4.2 RM Requirement	visible	A1:S407	4,929	23	-	-	1	1	Yes	No	Keep
4 MIS	4.3 Stock Summary	visible	A1:G450	2,600	23	-	-	0	1	Yes	No	Simplify
4 MIS	4.4 Consumables	visible	A1:AA500	9,995	19	-	-	1	0	Yes	No	Simplify
4 MIS	4.5 Production & Yield	visible	A1:L423	2,925	81	-	-	1	2	Yes	No	Keep
4 MIS	4.6 Lot Trace	visible	A1:AN190	1,310	43	-	-	1	0	Yes	No	Simplify
4 MIS	4.7 DC Trace	visible	A1:AP187	1,123	44	-	-	1	0	Yes	No	Simplify
4 MIS	4.8 MRM Monthly	visible	A1:H503	2,405	103	-	-	2	0	Yes	No	Simplify

Workbook	Sheet	State	Used range	Cells	Formula cells	Table	Data rows	Validations	CF rules	Protected	Freeze	Decision
4 MIS	4.9 MIS Dashboard	visible	A1:R33	377	234	-	-	1	2	Yes	No	Combine
4 MIS	4.10 Checks	visible	A1:E49	210	113	-	-	0	2	Yes	No	Keep
4 MIS	4.11 Scrap Dashboard	visible	A1:R110	965	72	-	-	9	3	Yes	No	Simplify
4 MIS	M_Contacts	visible	A1:F45	260	0	tblContact	11	1	1	Yes	Yes	Removed from Phase 1
4 MIS	Settings	visible	A1:E22	50	3	-	-	0	0	Yes	No	Simplify
4 MIS	Lists	hidden	A1:AC2191	3,287	582	-	-	0	0	Yes	No	Simplify
4 MIS	D_RMReceipt	hidden	A1:AL188	5,225	0	qRMReceipt	187	0	0	No	No	Simplify
4 MIS	D_RMIssue	hidden	A1:AD3394	79,943	0	qRMIssue	3,023	0	0	No	No	Simplify
4 MIS	D_ConsReceipt	hidden	A1:P203	3,046	0	qConsReceipt	202	0	0	No	No	Combine
4 MIS	D_ConsIssue	hidden	A1:U129	2,326	0	qConsIssue	128	0	0	No	No	Combine
4 MIS	D_Adjust	hidden	A1:P218	2,620	0	qAdjust	217	0	0	No	No	Simplify
4 MIS	D_Count	hidden	A1:I274	2,187	0	qCount	273	0	0	No	No	Removed from Phase 1
4 MIS	D_Material	hidden	A1:M125	1,346	0	qMaterial	69	0	0	No	No	Simplify
4 MIS	D_Consumable	hidden	A1:O245	3,087	0	qConsumable	97	0	0	No	No	Simplify
4 MIS	D_POLine	hidden	A1:AH203	5,694	0	qPOLine	2	0	0	No	No	Simplify
4 MIS	D_Production	hidden	A1:AL11407	296,551	0	qProduction	11,406	0	0	No	No	Simplify
4 MIS	D_MachineLog	hidden	A1:K2	12	0	qMachineLog	0	0	0	No	No	Removed from Phase 1
4 MIS	D_Rejection	hidden	A1:AB69	1,524	0	qRejection	68	0	0	No	No	Simplify
4 MIS	D_WIPCount	hidden	A1:I1504	10,374	0	qWIPCount	1,503	0	0	No	No	Removed from Phase 1
4 MIS	D_Machine	hidden	A1:G9	61	0	qMachine	8	0	0	No	No	Simplify
4 MIS	D_BOM	hidden	A1:V574	6,536	0	qBOM	246	0	0	No	No	Combine
4 MIS	D_BOMModel	hidden	A1:I44	306	0	qBOMModel	13	0	0	No	No	Combine
4 MIS	D_ProductP	hidden	A1:G397	1,888	0	qProductP	269	0	0	No	No	Combine
4 MIS	D_DCHead	hidden	A1:Q185	2,225	0	qDCHead	184	0	0	No	No	Simplify
4 MIS	D_DCLine	hidden	A1:Y627	11,829	0	qDCLine	626	0	0	No	No	Simplify
4 MIS	D_JWCHead	hidden	A1:Q153	1,993	0	qJWCHead	152	0	0	No	No	Simplify
4 MIS	D_JWCLine	hidden	A1:AH3509	102,388	0	qJWCLine	3,508	0	0	No	No	Simplify
4 MIS	D_Requirement	hidden	A1:O8	110	0	qRequirement	7	0	0	No	No	Simplify
4 MIS	D_FGCount	hidden	A1:H451	3,137	0	qFGCount	450	0	0	No	No	Removed from Phase 1

Workbook	Sheet	State	Used range	Cells	Formula cells	Table	Data rows	Validations	CF rules	Protected	Freeze	Decision
4 MIS	D_CoaterReturn	hidden	A1:Y2	26	0	qCoaterReturn	0	0	0	No	No	Combine
4 MIS	D_CoaterOpening	hidden	A1:I2	10	0	qCoaterOpening	0	0	0	No	No	Combine
4 MIS	D_Customer	hidden	A1:L16	79	0	qCustomer	15	0	0	No	No	Simplify
4 MIS	D_Vendor	hidden	A1:K6	51	0	qVendor	5	0	0	No	No	Simplify
4 MIS	D_ProductD	hidden	A1:I397	2,116	0	qProductD	278	0	0	No	No	Combine
4 MIS	D_Requisition	hidden	A1:AB2	30	0	qRequisition	0	0	0	No	No	Removed from Phase 1
4 MIS	D_Flow	hidden	A1:Q11475	170,761	0	qFlow	11,474	0	0	No	No	Simplify
4 MIS	D_Batch	hidden	A1:W414	9,522	0	qBatch	413	0	0	No	No	Combine

Appendix B - Current column inventory and actions

Every column of every Excel Table in the Stores, Production and Dispatch books, with the recommended action. MIS import tables (D_*) mirror these source tables column for column and follow the same actions; they are listed in full in the decision-register workbook.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.1 Requisition / tblRequisition	1	Req No (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	2	Add to Req	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	3	Req Line (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	4	Date	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	5	Requested By	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	6	Department (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	7	Item	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	8	Length mm	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	9	Width mm	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	10	Size mm (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	11	Qty kg	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	12	Required By	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	13	For Product / Job	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	14	Remarks	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	15	Stock now kg (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	16	On order kg (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	17	Stores Check (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	18	Stores Decision	Remove	-	Purchase requisition removed (scope override 1).

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.1 Requisition / tblRequisition	19	Decided By	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	20	PO No (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	21	PO Qty kg (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	22	Balance kg (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	23	Received kg (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	24	Lots (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	25	Status (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	26	Days Open (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	27	Check (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.1 Requisition / tblRequisition	28	Month (calc)	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.2 Purchase Order / tblPOLine	1	Line ID (calc)	Rename	PO Line ID	Becomes PO No-nn; ROW()-based ID removed.
1 Stores	1.2 Purchase Order / tblPOLine	2	Add to PO	Remove	-	Pad header replaces the ‘add to existing PO’ mechanic.
1 Stores	1.2 Purchase Order / tblPOLine	3	PO No	Keep	PO No (tblPOHead + line)	
1 Stores	1.2 Purchase Order / tblPOLine	4	Date (calc)	Rename	PO Date (header)	
1 Stores	1.2 Purchase Order / tblPOLine	5	Supplier (calc)	Rename	Supplier ID + Supplier Name (header)	Stable ID + name snapshot.
1 Stores	1.2 Purchase Order / tblPOLine	6	Delivery By	Keep	Delivery By (header)	
1 Stores	1.2 Purchase Order / tblPOLine	7	Payment Terms	Keep	Payment Terms (days) (header)	
1 Stores	1.2 Purchase Order / tblPOLine	8	Delivery To	Rename	Delivery Location (header)	
1 Stores	1.2 Purchase Order / tblPOLine	9	Status (calc)	Keep	PO Status (header)	Status lines only.
1 Stores	1.2 Purchase Order / tblPOLine	10	Req No	Remove	-	Purchase requisition removed (scope override 1).
1 Stores	1.2 Purchase Order / tblPOLine	11	Req Item	Remove	-	Requisition removed.
1 Stores	1.2 Purchase Order / tblPOLine	12	Req Line (calc)	Remove	-	Requisition removed.
1 Stores	1.2 Purchase Order / tblPOLine	13	Item	Rename	Item ID + Item Description	
1 Stores	1.2 Purchase Order / tblPOLine	14	Type (calc)	Rename	Item Type	
1 Stores	1.2 Purchase Order / tblPOLine	15	Grade (calc)	Keep	Grade (snapshot)	
1 Stores	1.2 Purchase Order / tblPOLine	16	Thk mm (calc)	Keep	Thk mm (snapshot)	
1 Stores	1.2 Purchase Order / tblPOLine	17	Length mm	Keep	Length mm	
1 Stores	1.2 Purchase Order / tblPOLine	18	Width mm	Keep	Width mm	
1 Stores	1.2 Purchase Order / tblPOLine	19	Size mm (calc)	Remove	-	Display text derivable from L x W.
1 Stores	1.2 Purchase Order / tblPOLine	20	HSN	Keep	HSN (snapshot)	
1 Stores	1.2 Purchase Order / tblPOLine	21	Qty	Rename	Ordered Qty	
1 Stores	1.2 Purchase Order / tblPOLine	22	Unit (calc)	Keep	Unit	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.2 Purchase Order / tblPOLine	23	Rate	Rename	Rate Rs/unit	
1 Stores	1.2 Purchase Order / tblPOLine	24	Amount Rs (calc)	Rename	Line Value Rs	
1 Stores	1.2 Purchase Order / tblPOLine	25	Lead Days	Remove	-	Derivable (Due - PO Date).
1 Stores	1.2 Purchase Order / tblPOLine	26	Due (calc)	Rename	Due Date	
1 Stores	1.2 Purchase Order / tblPOLine	27	Received (calc)	Move to report	PO Status: Received Accepted Qty	
1 Stores	1.2 Purchase Order / tblPOLine	28	Last Receipt (calc)	Move to report	PO Status: Last Receipt Date	
1 Stores	1.2 Purchase Order / tblPOLine	29	Pending (calc)	Move to report	PO Status: Pending Qty	
1 Stores	1.2 Purchase Order / tblPOLine	30	Req kg (calc)	Remove	-	Requisition removed.
1 Stores	1.2 Purchase Order / tblPOLine	31	Req left kg (calc)	Remove	-	Requisition removed.
1 Stores	1.2 Purchase Order / tblPOLine	32	Progress (calc)	Move to report	PO Status: Overdue Days	TODAY()-based text removed.
1 Stores	1.2 Purchase Order / tblPOLine	33	Check (calc)	Move to report	Pad control column (not posted)	
1 Stores	1.2 Purchase Order / tblPOLine	34	Month (calc)	Remove	-	Derived in reports.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	1	RM ID (calc)	Rename	Lot ID	Running-COUNTIF ID replaced by a permanent ID.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	2	Date	Rename	GRN Date	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	3	Material	Rename	Material ID	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	4	Grade (calc)	Keep	Grade (snapshot)	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	5	Code (calc)	Remove	-	Grade code used inside Lot ID only.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	6	Thk mm (calc)	Keep	Thk mm (snapshot)	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	7	Length mm	Keep	Length mm	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	8	Width mm	Keep	Width mm	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	9	Size mm (calc)	Remove	-	Display only.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	10	Qty Sheets	Rename	Received Sheets	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	11	Wt kg	Rename	Received kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	12	Theo kg (calc)	Rename	Theoretical kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	13	Var % (calc)	Rename	Theo Variance %	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	14	Bill Rate Rs/kg	Keep	Bill Rate Rs/kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	15	Rate Rs/kg (calc)	Combine	Bill Rate Rs/kg / PO Rate Rs/kg	Fallback to PO rate removed: rate is mandatory.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	16	Rate Reason	Rename	Rate Variance Reason	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	17	Value Rs (calc)	Rename	Lot Value Rs (received)	Accepted value in Lot Balance.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	18	Supplier	Rename	Supplier ID	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	19	GRN / Bill No	Combine	Supplier Invoice No (+ Invoice Date)	Split GRN number from bill number.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	20	PO No	Combine	PO Line ID	PO No derivable from PO Line ID.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	21	PO Line pick	Remove	-	Search helper replaced by filtered dropdown.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	22	PO Line (calc)	Rename	PO Line ID	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	23	Status	Remove	-	Cancellation through reversal line.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	24	QC Status	Combine	tblLotQC: QC Status	Decision becomes an event.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	25	QC Rejected kg	Combine	tblLotQC: QC Rejected kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	26	QC By	Combine	tblLotQC: Decided By	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	27	QC Date	Combine	tblLotQC: QC Date	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	28	Can issue (calc)	Move to report	Lot Balance: Issue Eligible	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	29	PO balance kg (calc)	Rename	PO Pending Before (kg)	Snapshot instead of running SUMIFS.
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	30	Excess kg (calc)	Rename	Excess over PO kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	31	Excess decision	Combine	tblLotQC: Excess Decision	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	32	Excess by	Combine	tblLotQC: Decided By	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	33	Rejected Sheets	Combine	tblLotQC: QC Rejected Sheets	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	34	Accepted kg (calc)	Combine	tblLotQC: Accepted kg	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	35	Accepted Sheets (calc)	Combine	tblLotQC: Accepted Sheets	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	36	Remarks	Keep	Remarks (audit)	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	37	Check (calc)	Move to report	Pad control column	
1 Stores	1.4 RM Receipt GRN / tblRMReceipt	38	Month (calc)	Remove	-	Derived in reports.
1 Stores	1.5 Consumable Receipt / tblConsReceipt	1	Rcpt ID (calc)	Rename	Receipt ID	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	2	Date	Rename	Receipt Date	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	3	Item Name	Combine	Item ID + Item Name	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	4	Item Code (calc)	Rename	Item ID	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	5	Category (calc)	Keep	Category (snapshot)	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	6	Unit (calc)	Keep	Unit (snapshot)	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	7	Qty	Keep	Qty	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	8	Bill Rate Rs	Keep	Bill Rate Rs	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	9	Rate Rs (calc)	Remove	-	Duplicate of bill rate.
1 Stores	1.5 Consumable Receipt / tblConsReceipt	10	Value Rs (calc)	Keep	Value Rs	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	11	Type	Rename	Receipt Type	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	12	Supplier	Rename	Supplier ID	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	13	Bill No	Rename	Supplier Bill No	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	14	Remarks	Keep	Remarks	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	15	Check (calc)	Move to report	Pad control	
1 Stores	1.5 Consumable Receipt / tblConsReceipt	16	Month (calc)	Remove	-	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.6 RM Issue & Return / tblRMIssue	1	Issue ID (calc)	Rename	Move ID	ROW()-based ISS-nnnnn replaced.
1 Stores	1.6 RM Issue & Return / tblRMIssue	2	Batch ID (calc)	Keep	Batch ID	= Move ID of the ISSUE line; COUNTIFS-based ISKO-yymmdd-nnn removed.
1 Stores	1.6 RM Issue & Return / tblRMIssue	3	Date	Rename	Move Date	
1 Stores	1.6 RM Issue & Return / tblRMIssue	4	Movement	Rename	Movement Type	SAMPLE folds into ISSUE + Purpose; REJECTED → RETURN-TO-SUPPLIER; DAMAGED added.
1 Stores	1.6 RM Issue & Return / tblRMIssue	5	Pick Material	Rename	Material ID	
1 Stores	1.6 RM Issue & Return / tblRMIssue	6	Lot override	Combine	Lot ID + Lot Override Reason	Lot is always picked; FIFO is a suggestion.
1 Stores	1.6 RM Issue & Return / tblRMIssue	7	Against Issue	Combine	Batch ID	Returns quote the batch.
1 Stores	1.6 RM Issue & Return / tblRMIssue	8	RM ID (calc)	Rename	Lot ID	Formula FIFO auto-pick removed.
1 Stores	1.6 RM Issue & Return / tblRMIssue	9	Material (calc)	Combine	Material ID	
1 Stores	1.6 RM Issue & Return / tblRMIssue	10	Grade (calc)	Keep	Grade (snapshot)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	11	Thk mm (calc)	Keep	Thk mm (snapshot)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	12	Size mm (calc)	Keep	Size (snapshot)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	13	Machine	Rename	Machine ID	
1 Stores	1.6 RM Issue & Return / tblRMIssue	14	Form	Keep	Form	Strict list (typos SHET/SHEE found today).
1 Stores	1.6 RM Issue & Return / tblRMIssue	15	Qty Sheets	Rename	Sheets	
1 Stores	1.6 RM Issue & Return / tblRMIssue	16	Wt kg	Rename	Weight kg	
1 Stores	1.6 RM Issue & Return / tblRMIssue	17	Net kg (calc)	Rename	Stock Effect kg	
1 Stores	1.6 RM Issue & Return / tblRMIssue	18	Net sheets (calc)	Rename	Stock Effect Sheets	
1 Stores	1.6 RM Issue & Return / tblRMIssue	19	Rate Rs/kg (calc)	Keep	Rate Rs/kg (snapshot)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	20	Value Rs (calc)	Keep	Value Rs	
1 Stores	1.6 RM Issue & Return / tblRMIssue	21	Issued To	Keep	Issued To (Person ID)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	22	Issued By	Keep	Issued By (Person ID)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	23	Product / Job	Rename	Purpose / Product ID	
1 Stores	1.6 RM Issue & Return / tblRMIssue	24	Remarks	Keep	Remarks (audit)	
1 Stores	1.6 RM Issue & Return / tblRMIssue	25	Available kg (calc)	Rename	Available kg Before (snapshot)	Per-row SUMIFS replaced by Lot Balance lookup.
1 Stores	1.6 RM Issue & Return / tblRMIssue	26	Available sheets (calc)	Combine	Pad display + Lot Balance	
1 Stores	1.6 RM Issue & Return / tblRMIssue	27	Can return kg (calc)	Rename	Batch Returnable kg Before	
1 Stores	1.6 RM Issue & Return / tblRMIssue	28	Lot Check (calc)	Move to report	Pad control column	
1 Stores	1.6 RM Issue & Return / tblRMIssue	29	Check (calc)	Move to report	Pad control column	
1 Stores	1.6 RM Issue & Return / tblRMIssue	30	Month (calc)	Remove	-	Derived in reports.
1 Stores	1.10 Stock Count / tblCount	1	Count Date	Remove	-	Physical count removed (scope override 5).

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.10 Stock Count / tblCount	2	Type	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	3	Level	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	4	Item	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	5	Qty	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	6	Wt kg	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	7	Counted By	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	8	Remarks	Remove	-	Physical count removed (scope override 5).
1 Stores	1.10 Stock Count / tblCount	9	Month (calc)	Remove	-	Physical count removed (scope override 5).
1 Stores	1.9 Stock Adjustment / tblAdjust	1	Adj ID (calc)	Keep	Adj ID	Permanent ID.
1 Stores	1.9 Stock Adjustment / tblAdjust	2	Date	Rename	Adj Date	
1 Stores	1.9 Stock Adjustment / tblAdjust	3	Type	Rename	Stock Type	
1 Stores	1.9 Stock Adjustment / tblAdjust	4	RM ID	Combine	Lot ID / Item ID	
1 Stores	1.9 Stock Adjustment / tblAdjust	5	Consumable Item	Combine	Lot ID / Item ID	
1 Stores	1.9 Stock Adjustment / tblAdjust	6	Material (calc)	Remove	-	From lot.
1 Stores	1.9 Stock Adjustment / tblAdjust	7	Qty	Combine	Direction + Qty / kg	Signed values replaced by Direction.
1 Stores	1.9 Stock Adjustment / tblAdjust	8	Wt kg	Combine	Direction + Qty / kg	
1 Stores	1.9 Stock Adjustment / tblAdjust	9	Rate (calc)	Keep	Rate Rs (snapshot)	
1 Stores	1.9 Stock Adjustment / tblAdjust	10	Value (calc)	Keep	Value Rs	
1 Stores	1.9 Stock Adjustment / tblAdjust	11	Reason Code	Keep	Reason Code (mandatory)	Blank on all 217 lines today.
1 Stores	1.9 Stock Adjustment / tblAdjust	12	Reason	Combine	Reference ID + Remarks	
1 Stores	1.9 Stock Adjustment / tblAdjust	13	Entered By	Rename	Requested By (+ Approved By new)	Blank on all 217 lines today.
1 Stores	1.9 Stock Adjustment / tblAdjust	14	Status	Remove	-	Cancellation by reversal.
1 Stores	1.9 Stock Adjustment / tblAdjust	15	Check (calc)	Move to report	Pad control	
1 Stores	1.9 Stock Adjustment / tblAdjust	16	Month (calc)	Remove	-	
1 Stores	M_Supplier / tblSupplier	1	Supplier	Keep	Supplier	Master / reference field.
1 Stores	M_Supplier / tblSupplier	2	Supplies	Keep	Supplies	Master / reference field.
1 Stores	M_Supplier / tblSupplier	3	Contact	Keep	Contact	Master / reference field.
1 Stores	M_Supplier / tblSupplier	4	Phone	Keep	Phone	Master / reference field.
1 Stores	M_Supplier / tblSupplier	5	Email	Keep	Email (optional plain reference)	No hyperlink; no mail use.
1 Stores	M_Supplier / tblSupplier	6	Address	Keep	Address	Master / reference field.
1 Stores	M_Supplier / tblSupplier	7	City	Keep	City	Master / reference field.
1 Stores	M_Supplier / tblSupplier	8	State Code	Keep	State Code	Master / reference field.
1 Stores	M_Supplier / tblSupplier	9	GSTIN	Keep	GSTIN	Master / reference field.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	M_Supplier / tblSupplier	10	Payment Terms	Keep	Payment Terms	Master / reference field.
1 Stores	M_Supplier / tblSupplier	11	Tally Ledger	Keep	Tally Ledger	Master / reference field.
1 Stores	M_Supplier / tblSupplier	12	Tally Group	Keep	Tally Group	Master / reference field.
1 Stores	M_Material / tblMaterial	1	Material	Keep	Material	Master / reference field.
1 Stores	M_Material / tblMaterial	2	Grade (calc)	Keep	Grade	Master / reference field.
1 Stores	M_Material / tblMaterial	3	Code (calc)	Keep	Code	Master / reference field.
1 Stores	M_Material / tblMaterial	4	Thk mm (calc)	Keep	Thk mm	Master / reference field.
1 Stores	M_Material / tblMaterial	5	Std L mm	Keep	Std L mm	Master / reference field.
1 Stores	M_Material / tblMaterial	6	Std W mm	Keep	Std W mm	Master / reference field.
1 Stores	M_Material / tblMaterial	7	Std Wt/Sheet kg (calc)	Keep	Std Wt/Sheet kg	Master / reference field.
1 Stores	M_Material / tblMaterial	8	Avg Issue kg/month (calc)	Move to report	MIS	
1 Stores	M_Material / tblMaterial	9	Min Level kg (calc)	Keep	Min Level kg	Master / reference field.
1 Stores	M_Material / tblMaterial	10	Stock kg (calc)	Move to report	Lot Balance	
1 Stores	M_Material / tblMaterial	11	Reorder (calc)	Move to report	Lot Balance flag	
1 Stores	M_Material / tblMaterial	12	Active	Keep	Active	Master / reference field.
1 Stores	M_Material / tblMaterial	13	Customer Tag	Keep	Customer Tag	
1 Stores	M_Grade / tblGrade	1	Grade	Keep	Grade	Master / reference field.
1 Stores	M_Grade / tblGrade	2	Code	Keep	Code	Master / reference field.
1 Stores	M_Grade / tblGrade	3	Description	Keep	Description	Master / reference field.
1 Stores	M_Consumable / tblConsumable	1	Item Code (calc)	Keep	Item Code	Master / reference field.
1 Stores	M_Consumable / tblConsumable	2	Item Name	Keep	Item Name	Master / reference field.
1 Stores	M_Consumable / tblConsumable	3	Category	Keep	Category	Master / reference field.
1 Stores	M_Consumable / tblConsumable	4	Unit	Keep	Unit	Master / reference field.
1 Stores	M_Consumable / tblConsumable	5	Avg Rate Rs (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	6	Received (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	7	Issued (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	8	Adjusted (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	9	Stock (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	10	Stock Value Rs (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	11	Avg Issue/month (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	12	Min Level (calc)	Keep	Min Level	Master / reference field.
1 Stores	M_Consumable / tblConsumable	13	Status (calc)	Move to report	Consumable Balance	Live formula column removed from master.
1 Stores	M_Consumable / tblConsumable	14	Last Supplier (calc)	Keep	Last Supplier	Master / reference field.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	M_Consumable / tblConsumable	15	Active	Keep	Active	Master / reference field.
1 Stores	M_Cons_Category / tblConsCat	1	Category	Keep	Category	Master / reference field.
1 Stores	M_Cons_Category / tblConsCat	2	Prefix	Keep	Prefix	Master / reference field.
1 Stores	M_Product / tblJob	1	Product / Job	Remove	X_Product (import)	Stores copy of products replaced by import.
1 Stores	M_Product / tblJob	2	Customer	Remove	X_Product (import)	Stores copy of products replaced by import.
1 Stores	M_Product / tblJob	3	Active	Remove	X_Product (import)	Stores copy of products replaced by import.
1 Stores	M_WhereUsed / tblWhere	1	Where Used	Keep	Where Used	Master / reference field.
1 Stores	M_WhereUsed / tblWhere	2	Type	Keep	Type	Master / reference field.
1 Stores	M_Person / tblPerson	1	Name	Combine	X_People (import)	Single people master.
1 Stores	M_Person / tblPerson	2	Department	Combine	X_People (import)	Single people master.
1 Stores	M_Person / tblPerson	3	Usual Machine	Combine	X_People (import)	Single people master.
1 Stores	M_Reason / tblReason	1	Reason	Keep	Reason	Coded reasons; count reasons removed.
1 Stores	M_Reason / tblReason	2	Used For	Keep	Used For	Coded reasons; count reasons removed.
1 Stores	M_Reason / tblReason	3	Active	Keep	Active	Coded reasons; count reasons removed.
1 Stores	M_Contacts / tblContact	1	Name	Remove	-	Mail contacts removed (scope override 3).
1 Stores	M_Contacts / tblContact	2	Team	Remove	-	Mail contacts removed (scope override 3).
1 Stores	M_Contacts / tblContact	3	Groups	Remove	-	Mail contacts removed (scope override 3).
1 Stores	M_Contacts / tblContact	4	Email	Remove	-	Mail contacts removed (scope override 3).
1 Stores	M_Contacts / tblContact	5	Phone	Remove	-	Mail contacts removed (scope override 3).
1 Stores	M_Contacts / tblContact	6	Note	Remove	-	Mail contacts removed (scope override 3).
1 Stores	1.7 Consumable Issue / tblConsIssue	1	Issue ID (calc)	Rename	Cons Move ID	
1 Stores	1.7 Consumable Issue / tblConsIssue	2	Date	Rename	Move Date	
1 Stores	1.7 Consumable Issue / tblConsIssue	3	Item Name	Combine	Item ID + Item Name	
1 Stores	1.7 Consumable Issue / tblConsIssue	4	Item Code (calc)	Rename	Item ID	
1 Stores	1.7 Consumable Issue / tblConsIssue	5	Category (calc)	Remove	-	From master in reports.
1 Stores	1.7 Consumable Issue / tblConsIssue	6	Unit (calc)	Keep	Unit (snapshot)	
1 Stores	1.7 Consumable Issue / tblConsIssue	7	Qty	Keep	Qty (ISSUE rows)	
1 Stores	1.7 Consumable Issue / tblConsIssue	8	Returned Qty	Combine	Separate RETURN row	One movement per row.
1 Stores	1.7 Consumable Issue / tblConsIssue	9	Net Qty (calc)	Rename	Stock Effect Qty	
1 Stores	1.7 Consumable Issue / tblConsIssue	10	BOM Qty	Move to report	MIS Consumables (BOM vs actual)	
1 Stores	1.7 Consumable Issue / tblConsIssue	11	Variance (calc)	Move to report	MIS Consumables	
1 Stores	1.7 Consumable Issue / tblConsIssue	12	Var % (calc)	Move to report	MIS Consumables	
1 Stores	1.7 Consumable Issue / tblConsIssue	13	Stock Check (calc)	Combine	Hard stop at entry + Balance Before	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
1 Stores	1.7 Consumable Issue / tblConsIssue	14	Rate Rs (calc)	Rename	Rate Rs (avg at posting)	
1 Stores	1.7 Consumable Issue / tblConsIssue	15	Value Rs (calc)	Keep	Value Rs	
1 Stores	1.7 Consumable Issue / tblConsIssue	16	Issued To	Keep	Issued To	
1 Stores	1.7 Consumable Issue / tblConsIssue	17	Where Used	Keep	Where Used	
1 Stores	1.7 Consumable Issue / tblConsIssue	18	Purpose / Job	Combine	Product ID / Job	
1 Stores	1.7 Consumable Issue / tblConsIssue	19	Product	Combine	Product ID / Job	
1 Stores	1.7 Consumable Issue / tblConsIssue	20	Check (calc)	Move to report	Pad control	
1 Stores	1.7 Consumable Issue / tblConsIssue	21	Month (calc)	Remove	-	
2 Production	2.1 Production Entry / tblProduction	1	Prod ID (calc)	Keep	Prod ID	Permanent sequence replaces ROW()-based ID.
2 Production	2.1 Production Entry / tblProduction	2	Date	Rename	Prod Date	
2 Production	2.1 Production Entry / tblProduction	3	Shift	Keep	Shift (mandatory)	
2 Production	2.1 Production Entry / tblProduction	4	Machine	Rename	Machine ID	
2 Production	2.1 Production Entry / tblProduction	5	Stage (calc)	Keep	Stage (snapshot)	
2 Production	2.1 Production Entry / tblProduction	6	Batch	Rename	Batch ID	
2 Production	2.1 Production Entry / tblProduction	7	Batch used (calc)	Combine	Batch ID	Formula inheritance (running XLOOKUP) removed.
2 Production	2.1 Production Entry / tblProduction	8	Issue ID (calc)	Remove	-	Batch ID = Issue Move ID.
2 Production	2.1 Production Entry / tblProduction	9	Material	Keep	Material ID (snapshot)	
2 Production	2.1 Production Entry / tblProduction	10	Lot override	Combine	Batch ID	Lot comes from batch.
2 Production	2.1 Production Entry / tblProduction	11	RM ID (calc)	Rename	Lot ID (snapshot)	
2 Production	2.1 Production Entry / tblProduction	12	Grade (calc)	Keep	Grade (snapshot)	
2 Production	2.1 Production Entry / tblProduction	13	Thk mm (calc)	Keep	Thk mm (snapshot)	
2 Production	2.1 Production Entry / tblProduction	14	Size mm (calc)	Keep	Size (snapshot)	
2 Production	2.1 Production Entry / tblProduction	15	Kg left (calc)	Rename	Batch kg Left Before (snapshot)	Per-row SUMIFS removed.
2 Production	2.1 Production Entry / tblProduction	16	RM Form	Remove	-	From batch.
2 Production	2.1 Production Entry / tblProduction	17	RM Length mm	Combine	Offcut Size Used	
2 Production	2.1 Production Entry / tblProduction	18	RM Width mm	Combine	Offcut Size Used	
2 Production	2.1 Production Entry / tblProduction	19	RM Sheets	Rename	RM Sheets Used	
2 Production	2.1 Production Entry / tblProduction	20	RM kg (calc)	Remove	-	Theoretical kg shown in pad only.
2 Production	2.1 Production Entry / tblProduction	21	Product	Rename	Product ID	
2 Production	2.1 Production Entry / tblProduction	22	Part	Rename	Part ID	
2 Production	2.1 Production Entry / tblProduction	23	Qty	Rename	Qty (nos)	
2 Production	2.1 Production Entry / tblProduction	24	Wt/pc kg	Rename	Wt/pc Override kg	
2 Production	2.1 Production Entry / tblProduction	25	Wt/pc used (calc)	Rename	Wt/pc kg (snapshot)	Live M_Part lookup removed.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
2 Production	2.1 Production Entry / tblProduction	26	Total Wt kg	Combine	Wt/pc Override kg	
2 Production	2.1 Production Entry / tblProduction	27	Wt kg (calc)	Rename	Output kg	
2 Production	2.1 Production Entry / tblProduction	28	Scrap kg	Keep	Scrap kg	
2 Production	2.1 Production Entry / tblProduction	29	Batch done	Combine	tblBatchClose	Closure becomes an event.
2 Production	2.1 Production Entry / tblProduction	30	Batch variance kg (calc)	Move to report	Batch Balance	
2 Production	2.1 Production Entry / tblProduction	31	Operator	Keep	Operator (Person ID)	
2 Production	2.1 Production Entry / tblProduction	32	QC Status	Keep	QC Status	
2 Production	2.1 Production Entry / tblProduction	33	QC By	Keep	QC By	
2 Production	2.1 Production Entry / tblProduction	34	QC OK Qty (calc)	Move to report	Calculated from tblRejRework	SUMIFS per row removed.
2 Production	2.1 Production Entry / tblProduction	35	Source	Rename	Entry Source (audit)	
2 Production	2.1 Production Entry / tblProduction	36	Check (calc)	Move to report	Pad control	
2 Production	2.1 Production Entry / tblProduction	37	Month (calc)	Remove	-	
2 Production	2.2 Rejection & Rework / tblRejection	1	Rej ID (calc)	Rename	Event ID	
2 Production	2.2 Rejection & Rework / tblRejection	2	Date	Rename	Event Date	
2 Production	2.2 Rejection & Rework / tblRejection	3	Stage	Combine	Detected at Stage / Place	
2 Production	2.2 Rejection & Rework / tblRejection	4	Machine	Combine	Detected at Stage / Place	
2 Production	2.2 Rejection & Rework / tblRejection	5	Product	Keep	Product ID (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	6	Part	Keep	Part ID (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	7	Drawing No	Keep	Drawing (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	8	Qty	Keep	Qty	
2 Production	2.2 Rejection & Rework / tblRejection	9	Wt kg	Rename	kg	
2 Production	2.2 Rejection & Rework / tblRejection	10	Reason Group	Keep	Reason Group	
2 Production	2.2 Rejection & Rework / tblRejection	11	Reason	Keep	Reason	
2 Production	2.2 Rejection & Rework / tblRejection	12	Type	Rename	Responsibility	
2 Production	2.2 Rejection & Rework / tblRejection	13	Action	Rename	Disposition	
2 Production	2.2 Rejection & Rework / tblRejection	14	Operator	Keep	Operator	
2 Production	2.2 Rejection & Rework / tblRejection	15	Rejected At	Combine	Detected at Stage / Place	
2 Production	2.2 Rejection & Rework / tblRejection	16	Batch (Prod ID) (calc)	Rename	Source Prod ID	Mandatory for in-process rejections.
2 Production	2.2 Rejection & Rework / tblRejection	17	Batch (Issue) (calc)	Keep	Batch ID (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	18	Lot (RM ID) (calc)	Keep	Lot ID (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	19	Material (calc)	Keep	Material (snapshot)	
2 Production	2.2 Rejection & Rework / tblRejection	20	Rework Status	Combine	Event Type REWORK-OK / REWORK-SCRAP	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
2 Production	2.2 Rejection & Rework / tblRejection	21	Rework Lot	Remove	-	Rework links to Case ID.
2 Production	2.2 Rejection & Rework / tblRejection	22	Rework kg	Combine	Rework events kg	
2 Production	2.2 Rejection & Rework / tblRejection	23	OK Qty	Combine	REWORK-OK Qty	
2 Production	2.2 Rejection & Rework / tblRejection	24	Closed On	Combine	Rework event date	
2 Production	2.2 Rejection & Rework / tblRejection	25	Source	Rename	Entry Source (audit)	
2 Production	2.2 Rejection & Rework / tblRejection	26	Check (calc)	Move to report	Pad control	
2 Production	2.2 Rejection & Rework / tblRejection	27	Month (calc)	Remove	-	
2 Production	2.3 Machine Log / tblMachineLog	1	Log ID (calc)	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	2	Date	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	3	Machine	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	4	Type	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	5	From	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	6	To	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	7	Hours (calc)	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	8	Reason	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	9	Action Taken	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	10	Attended By	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	2.3 Machine Log / tblMachineLog	11	Month (calc)	Remove	-	Machine log deferred to Phase 2 (0 rows used).
2 Production	M_Worker / tblOperator	1	Operator	Keep	Operator	Master / reference field.
2 Production	M_Worker / tblOperator	2	Usual Machine	Keep	Usual Machine	Master / reference field.
2 Production	M_Worker / tblOperator	3	Shift	Keep	Shift	Master / reference field.
2 Production	M_Worker / tblOperator	4	Role	Keep	Role	Master / reference field.
2 Production	M_Product / tblProduct	1	Product	Keep	Product	Master / reference field.
2 Production	M_Product / tblProduct	2	Customer	Keep	Customer	Master / reference field.
2 Production	M_Product / tblProduct	3	Category	Keep	Category	Master / reference field.
2 Production	M_Product / tblProduct	4	Class	Keep	Class	Master / reference field.
2 Production	M_Product / tblProduct	5	BOM Model	Keep	BOM Model	Master / reference field.
2 Production	M_Product / tblProduct	6	Set Wt kg (BOM) (calc)	Keep	Set Wt kg (from BOM)	
2 Production	M_Product / tblProduct	7	Made kg (calc)	Move to report	MIS	
2 Production	2.4 WIP Count / tblWIPCount	1	Count Date	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	2	Item	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	3	Part	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	4	Stage	Remove	-	WIP physical count removed (scope override 4).

Workbook	Sheet / table	#	Current column	Action	Final column	Note
2 Production	2.4 WIP Count / tblWIPCount	5	Qty	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	6	Wt kg	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	7	Counted By	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	8	Source	Remove	-	WIP physical count removed (scope override 4).
2 Production	2.4 WIP Count / tblWIPCount	9	Month (calc)	Remove	-	WIP physical count removed (scope override 4).
2 Production	M_Part / tblPart	1	Key (calc)	Rename	Part ID	Permanent key replaces 'Product|Part' text.
2 Production	M_Part / tblPart	2	Product	Keep	Product	Master / reference field.
2 Production	M_Part / tblPart	3	Part	Keep	Part	Master / reference field.
2 Production	M_Part / tblPart	4	Drawing No	Keep	Drawing No	Master / reference field.
2 Production	M_Part / tblPart	5	Material	Keep	Material	Master / reference field.
2 Production	M_Part / tblPart	6	Thk mm	Keep	Thk mm	Master / reference field.
2 Production	M_Part / tblPart	7	Wt/pc kg	Keep	Wt/pc kg	Master / reference field.
2 Production	M_Part / tblPart	8	Source	Remove	-	Migration note.
2 Production	M_Part / tblPart	9	Lines made (calc)	Move to report	MIS	
2 Production	M_BOM / tblBOM	1	Model	Keep	Model	Master / reference field.
2 Production	M_BOM / tblBOM	2	Product	Keep	Product	Master / reference field.
2 Production	M_BOM / tblBOM	3	SR No	Keep	SR No	Master / reference field.
2 Production	M_BOM / tblBOM	4	Level	Keep	Level	Master / reference field.
2 Production	M_BOM / tblBOM	5	Parent SR	Keep	Parent SR	Master / reference field.
2 Production	M_BOM / tblBOM	6	Drawing No	Keep	Drawing No	Master / reference field.
2 Production	M_BOM / tblBOM	7	Part Name	Keep	Part Name	Master / reference field.
2 Production	M_BOM / tblBOM	8	Rev	Keep	Rev	Master / reference field.
2 Production	M_BOM / tblBOM	9	Rev Date	Keep	Rev Date	Master / reference field.
2 Production	M_BOM / tblBOM	10	Material	Keep	Material	Master / reference field.
2 Production	M_BOM / tblBOM	11	Thk mm	Keep	Thk mm	Master / reference field.
2 Production	M_BOM / tblBOM	12	Width mm	Keep	Width mm	Master / reference field.
2 Production	M_BOM / tblBOM	13	Length mm	Keep	Length mm	Master / reference field.
2 Production	M_BOM / tblBOM	14	Blank Wt kg (calc)	Keep	Blank Wt kg	Master / reference field.
2 Production	M_BOM / tblBOM	15	Qty/Set	Keep	Qty/Set	Master / reference field.
2 Production	M_BOM / tblBOM	16	Costable (calc)	Keep	Costable	Master / reference field.
2 Production	M_BOM / tblBOM	17	Material Code (calc)	Keep	Material Code	Master / reference field.
2 Production	M_BOM / tblBOM	18	Wt per Set kg (calc)	Keep	Wt per Set kg	Master / reference field.
2 Production	M_BOM / tblBOM	19	PC Reqd	Keep	PC Reqd	Master / reference field.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
2 Production	M_BOM / tblBOM	20	Variant	Keep	Variant	Master / reference field.
2 Production	M_BOM / tblBOM	21	Remarks	Keep	Remarks	Master / reference field.
2 Production	M_BOM / tblBOM	22	Data Fix	Combine	Remarks	Migration note.
2 Production	M_BOM_Model / tblBOMModel	1	BOM Model	Keep	BOM Model	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	2	Product	Keep	Product	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	3	BOM Lines (calc)	Keep	BOM Lines	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	4	Costable Lines (calc)	Keep	Costable Lines	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	5	Set Wt kg (calc)	Keep	Set Wt kg	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	6	BOM Edition	Keep	BOM Edition	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	7	Effective From	Keep	Effective From	Master / reference field.
2 Production	M_BOM_Model / tblBOMModel	8	Next Review (calc)	Rename	Review Due (typed date)	No TODAY().
2 Production	M_BOM_Model / tblBOMModel	9	Review Status (calc)	Move to report	MIS exception MST-02	
2 Production	M_Reason / tblReason	1	Reason	Keep	Reason	Coded reasons; count reasons removed.
2 Production	M_Reason / tblReason	2	Reason Group	Keep	Reason Group	Coded reasons; count reasons removed.
2 Production	M_Reason / tblReason	3	Active	Keep	Active	Coded reasons; count reasons removed.
2 Production	M_Stage / tblStage	1	Rejected At	Combine	M_Stage_Route	Stage & route master.
2 Production	M_Stage / tblStage	2	Process	Combine	M_Stage_Route	Stage & route master.
2 Production	M_Contacts / tblContact	1	Name	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Contacts / tblContact	2	Team	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Contacts / tblContact	3	Groups	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Contacts / tblContact	4	Email	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Contacts / tblContact	5	Phone	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Contacts / tblContact	6	Note	Remove	-	Mail contacts removed (scope override 3).
2 Production	M_Machine / tblMachine	1	Machine	Keep	Machine	Master / reference field.
2 Production	M_Machine / tblMachine	2	Process	Keep	Process	Master / reference field.
2 Production	M_Machine / tblMachine	3	Capacity kg/day	Keep	Capacity kg/day (optional until Phase 2)	
2 Production	M_Machine / tblMachine	4	Shifts/day	Keep	Shifts/day (optional)	
2 Production	M_Machine / tblMachine	5	Hours/shift	Keep	Hours/shift (optional)	
2 Production	M_Machine / tblMachine	6	Status	Keep	Status	Master / reference field.
2 Production	M_Machine / tblMachine	7	Note	Keep	Note	Master / reference field.
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	1	Req ID (calc)	Keep	Req ID	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	2	Version (calc)	Keep	Version	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	3	Revises	Rename	Revises Req ID	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	4	Entry Date	Keep	Entry Date	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	5	Customer	Rename	Customer ID	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	6	Product	Rename	Product ID	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	7	Req Qty	Rename	Required Qty	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	8	Req Date	Rename	Required Date	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	9	Priority	Keep	Priority	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	10	Status	Keep	Status	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	11	Remark	Keep	Remarks	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	12	Req kg (calc)	Move to report	MIS RM Requirement	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	13	Dispatched Qty (calc)	Move to report	Requirement Status	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	14	Pending Qty (calc)	Move to report	Requirement Status	
3 Dispatch & Challan	3.1 Customer Requirement / tblRequirement	15	Month (calc)	Remove	-	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	1	JWC No (calc)	Keep	JWC No	Permanent sequential.
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	2	Date	Rename	JWC Date	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	3	Vendor	Rename	Vendor ID + Vendor Name	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	4	Vehicle No	Keep	Vehicle No	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	5	e-Way Bill	Keep	e-Way Bill No	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	6	Purpose	Keep	Purpose	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	7	Shade	Keep	Shade	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	8	Status	Rename	JWC Status	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	9	Coater Bill No	Keep	Coater Bill No	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	10	Remarks	Keep	Remarks	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	11	Check (calc)	Move to report	Pad control	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	12	Lines (calc)	Move to report	JWC totals view	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	13	Qty NOS (calc)	Move to report	JWC totals view	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	14	Wt kg (calc)	Move to report	JWC totals view	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	15	SqFt (calc)	Move to report	JWC totals view	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	16	PC Cost Rs (calc)	Move to report	JWC totals view	
3 Dispatch & Challan	3.2 JWC Header / tblJWCHead	17	Month (calc)	Remove	-	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	1	Line ID (calc)	Rename	JWC Line ID	JWC No-nn instead of ROW()-based JL-nnnnn.
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	2	JWC No	Keep	JWC No (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	3	Date (calc)	Keep	JWC Date (snapshot)	Header value copied once at posting.
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	4	Vendor (calc)	Keep	Vendor ID (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	5	Product	Rename	Product ID	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	6	Drawing No	Keep	Drawing No (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	7	Part / Description	Rename	Part ID	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	8	Batch	Rename	Batch ID	Mandatory from go-live.
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	9	Lot (RM ID) (calc)	Rename	Lot ID (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	10	Lot Material (calc)	Keep	Lot Material (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	11	Lot Size mm (calc)	Keep	Lot Size (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	12	Coating	Keep	Coating	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	13	Qty NOS	Keep	Qty NOS	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	14	Wt kg	Keep	Wt kg	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	15	SqFt/pc	Keep	Sq ft / pc (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	16	SqFt (calc)	Keep	Sq ft	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	17	Basis (calc)	Rename	Rate Basis (snapshot)	Live M_Vendor lookup removed.
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	18	Rate Override	Combine	PC Rate (override with reason)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	19	PC Rate (calc)	Keep	PC Rate (snapshot)	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	20	PC Cost Rs (calc)	Keep	PC Cost Rs	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	21	Eff Rs/kg (calc)	Move to report	MIS Dispatch & JWC	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	22	Goods Rs/NOS	Combine	Goods Value Rs	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	23	Goods Value Rs (calc)	Keep	Goods Value Rs	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	24	Back OK Qty (calc)	Move to report	Open JWC Balance	Per-row SUMIFS removed.
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	25	Direct Disp Qty (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	26	Defect Qty (calc)	Move to report	Open JWC Balance	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	27	Pending Qty (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	28	Pending kg (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	29	Pending SqFt (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	30	Days Pending (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	31	Line Status (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	32	Check (calc)	Move to report	Pad control	
3 Dispatch & Challan	3.3 JWC Lines / tblJWCLine	33	Month (calc)	Remove	-	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	1	Coater	Rename	Coater ID (tblCoaterMove OPENING)	One-time cut-over balance.
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	2	Product	Rename	Product ID	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	3	Part / Description	Rename	Part ID	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	4	Qty NOS	Keep	Qty NOS	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	5	Wt kg	Keep	Wt kg	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	6	Confirmed By	Rename	Posted By + Remarks (coater statement ref.)	Confirmation from coater statement, not a count.
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	7	Remarks	Keep	Remarks	
3 Dispatch & Challan	3.5 Coater Opening / tblCoaterOpening	8	As On (calc)	Rename	Move Date	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	1	Return ID (calc)	Rename	Coater Move ID	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	2	Date	Rename	Move Date	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	3	Coater (calc)	Rename	Coater ID	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	4	JWC No	Combine	JWC Line ID	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	5	JWC Line	Rename	JWC Line ID	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	6	Return Type	Rename	Move Type	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	7	Coater Challan No	Keep	Coater Challan No	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	8	Product (calc)	Keep	Product ID (snapshot)	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	9	Part / Description (calc)	Keep	Part ID (snapshot)	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	10	Batch (calc)	Keep	Batch ID (snapshot)	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	11	Qty NOS	Keep	Qty NOS	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	12	Wt kg (calc)	Keep	Wt kg	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	13	What Happened	Rename	Defect Description	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	14	Whose Fault	Keep	Whose Fault	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	15	Decision	Keep	Decision	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	16	Re-sent on JWC	Rename	Re-sent on JWC Line ID	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	17	Debit to Coater Rs	Keep	Debit to Coater Rs	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	18	Remarks	Keep	Remarks	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	19	Status (calc)	Rename	Case Status	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	20	Days Open (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	21	Check (calc)	Move to report	Pad control	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	22	Sent on JWC kg (calc)	Move to report	Open JWC Balance	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	23	Month (calc)	Remove	-	
3 Dispatch & Challan	3.6 Coater Return / tblCoaterReturn	24	Mail Coater (calc)	Remove	-	Mail feature removed (scope override 3).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	1	Count Date	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	2	Item	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	3	Qty	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	4	Wt kg	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	5	Counted By	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	6	Source	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.8 FG Count / tblFGCount	7	Month (calc)	Remove	-	FG physical count removed (scope override 5).
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	1	Line ID (calc)	Rename	DC Line ID	Challan No-nn.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	2	Challan No	Keep	Challan No (snapshot)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	3	Date (calc)	Keep	DC Date (snapshot)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	4	Customer (calc)	Keep	Customer ID (snapshot)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	5	Dispatch From (calc)	Keep	Dispatch From (snapshot)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	6	Against JWC No	Remove	-	Derivable from JWC Line ID.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	7	JWC Line	Rename	JWC Line ID	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	8	Product	Rename	Product ID	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	9	Part	Rename	Part ID	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	10	Batch	Keep	Batch ID	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	11	Dispatch Type	Keep	Dispatch Type	Blank on all 626 lines today.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	12	Description	Keep	Description (snapshot)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	13	Qty	Keep	Qty	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	14	Unit (calc)	Rename	Dispatch UOM (snapshot)	33 lines show ‘0’ today.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	15	UOM override	Combine	Customer-Product UOM table	Override moves to master data.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	16	Challan Qty (calc)	Keep	Challan Qty	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	17	Wt/set (calc)	Rename	Wt per Unit kg (snapshot)	Live M_Product lookup removed.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	18	Total Wt kg	Combine	Wt kg (override)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	19	Wt kg (calc)	Keep	Wt kg	

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	20	SqFt (calc)	Keep	Sq ft	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	21	Rate Rs/kg	Combine	Rate Rs (override with reason)	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	22	Rate used (calc)	Rename	Rate Rs (snapshot)	Live sale-rate lookup removed.
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	23	Value Rs (calc)	Keep	Value Rs	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	24	Check (calc)	Move to report	Pad control	
3 Dispatch & Challan	3.10 DC Lines / tblDCLine	25	Month (calc)	Remove	-	
3 Dispatch & Challan	M_Customer / tblCustomer	1	Customer	Keep	Customer	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	2	Full Name	Keep	Full Name	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	3	Address	Keep	Address	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	4	City	Keep	City	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	5	State	Keep	State	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	6	State Code	Keep	State Code	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	7	GSTIN	Keep	GSTIN	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	8	Group	Keep	Group	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	9	Contact	Keep	Contact	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	10	Email	Keep	Email (optional plain reference)	No hyperlink; no mail use.
3 Dispatch & Challan	M_Customer / tblCustomer	11	Tally Ledger	Keep	Tally Ledger	Master / reference field.
3 Dispatch & Challan	M_Customer / tblCustomer	12	Dispatch UOM	Keep	Dispatch UOM (mandatory)	8 of 15 customers blank today.
3 Dispatch & Challan	M_Vendor / tblVendor	1	Vendor	Keep	Vendor	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	2	Short	Keep	Short	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	3	Address	Keep	Address	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	4	GSTIN	Keep	GSTIN	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	5	Email	Keep	Email (optional plain reference)	No hyperlink; no mail use.
3 Dispatch & Challan	M_Vendor / tblVendor	6	State Code	Keep	State Code	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	7	Basis	Combine	tblCoaterRate (effective-dated)	
3 Dispatch & Challan	M_Vendor / tblVendor	8	Rate Rs/SqFt	Combine	tblCoaterRate	
3 Dispatch & Challan	M_Vendor / tblVendor	9	Rate Rs/Kg	Combine	tblCoaterRate	
3 Dispatch & Challan	M_Vendor / tblVendor	10	Status	Keep	Status	Master / reference field.
3 Dispatch & Challan	M_Vendor / tblVendor	11	Note	Keep	Note	Master / reference field.
3 Dispatch & Challan	M_Product / tblProduct	1	Product	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	2	Customer	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	3	Category	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	4	HSN	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.

Workbook	Sheet / table	#	Current column	Action	Final column	Note
3 Dispatch & Challan	M_Product / tblProduct	5	Unit	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	6	Set Wt kg	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	7	Sale Rate Rs/kg	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	8	Class	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Product / tblProduct	9	Dispatched kg (calc)	Combine	X_Product + tblCustPrice	Single product master + Dispatch commercial table.
3 Dispatch & Challan	M_Worker / tblWorker	1	Name	Combine	X_People (import)	Single people master.
3 Dispatch & Challan	M_Worker / tblWorker	2	Role	Combine	X_People (import)	Single people master.
3 Dispatch & Challan	M_Worker / tblWorker	3	Active	Combine	X_People (import)	Single people master.
3 Dispatch & Challan	M_Contacts / tblContact	1	Name	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	M_Contacts / tblContact	2	Team	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	M_Contacts / tblContact	3	Groups	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	M_Contacts / tblContact	4	Email	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	M_Contacts / tblContact	5	Phone	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	M_Contacts / tblContact	6	Note	Remove	-	Mail contacts removed (scope override 3).
3 Dispatch & Challan	3.9 DC Header / tblDCHead	1	Challan No (calc)	Keep	Challan No	Permanent sequential.
3 Dispatch & Challan	3.9 DC Header / tblDCHead	2	Date	Rename	DC Date	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	3	Customer	Rename	Customer ID + Name	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	4	Dispatch From	Keep	Dispatch From	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	5	Vehicle No	Keep	Vehicle No	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	6	Driver	Keep	Driver	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	7	LR No	Keep	LR No	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	8	Invoice No	Keep	Invoice No	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	9	e-Way Bill	Keep	e-Way Bill No	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	10	Status	Rename	DC Status	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	11	Remarks	Keep	Remarks	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	12	Check (calc)	Move to report	Pad control	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	13	Lines (calc)	Move to report	DC totals view	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	14	Qty (calc)	Move to report	DC totals view	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	15	Wt kg (calc)	Move to report	DC totals view	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	16	Value Rs (calc)	Move to report	DC totals view	
3 Dispatch & Challan	3.9 DC Header / tblDCHead	17	Month (calc)	Remove	-	

Appendix C - Formula and compatibility notes

Current function / feature	Minimum Excel	Target replacement
XLOOKUP, XMATCH	2021 / 365	INDEX + MATCH (one MATCH helper per key)
LET	2021 / 365	Helper columns in the pad
FILTER, SORT, SORTBY, UNIQUE, SEQUENCE, spill references (#, ANCHORARRAY)	2021 / 365	Power Query outputs, PivotTables, validation lists from Tables
TEXTJOIN	2019	Separate flag columns; one *Issues* text built with &
MAXIFS / MINIFS	2019	Power Query group-by, or sorted lookup
ENCODEURL + HYPERLINK (Gmail compose)	2013	**Removed** (mail)
TODAY, NOW, CELL, OFFSET, INDIRECT	any (volatile)	Typed As-on date; INDEX ranges; typed folder path
ROW()-based IDs, running COUNTIF IDs	any	IDs posted as values
Power Query (Get & Transform)	2016 (built in)	Kept, controlled refresh
ISFORMULA (integrity check)	2013	Used in pad headers

Appendix D - Terms

Term	Meaning
Entry pad	Small entry area (≈200 rows) with all formulas and checks; rows are posted, then cleared
Register	Append-only Excel Table of posted transactions, values only
Balance	Calculated position for open items (lots, batches, JWC lines, FG, WIP)
Snapshot	Value copied at posting so later master edits do not change history
Extract (X_)	Values-only import of another book's register or balance
Refresh Log	One row per extract per refresh with status and control totals
Batch	One RM ISSUE movement; its Move ID is the Batch ID
Hard stop / warning	Entry blocked / entry allowed with an authorised explanation
Authorised stock adjustment	Approved correction movement with reason and reference; not a count
FY	Financial year (April-March)

Appendix E - Test checklist

ID	Test	Volumes run	Median time	Errors	Result
T-01	Open workbook	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-02	Save workbook	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-03	Recalculate after a normal entry	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-04	Filter / search	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-05	Post a transaction	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested

ID	Test	Volumes run	Median time	Errors	Result
T-06	Stock / batch / dispatch checks	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-07	Refresh MIS extracts	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-08	Open / update dashboard	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-09	Close and reopen after save	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-10	Missing / renamed / moved source file	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-11	Stale source	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-12	Duplicate import	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-13	Stale balance guard	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-14	Reversal	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-15	Period lock	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-16	Closed-period change detection	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested
T-17	Excel version compatibility	[ ] 10k [ ] 25k [ ] 50k [ ] 100k [ ] 400k+			Not yet tested